Ion xromatografiya tizimi

Foydalanish bo'yicha qo'llanma

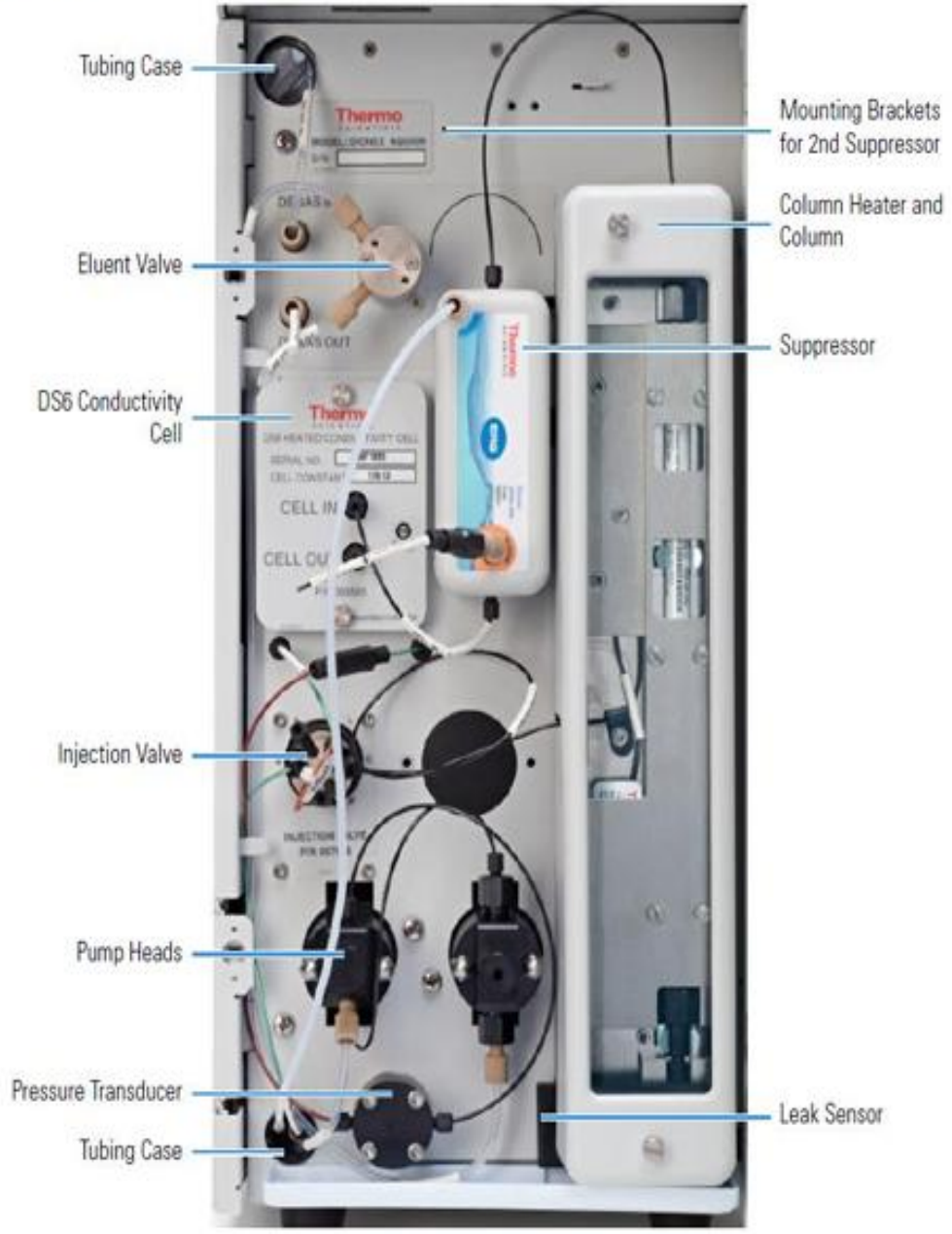


Ishlab chiqaruvchi	**Thermo Fisher Scientific**
Mahsulot nomi	**Aquion IC tizimi** **AS-DV avtosampler bilan**

I. AQUION qurilmaiga kirish

1. Qurilma komponentlari

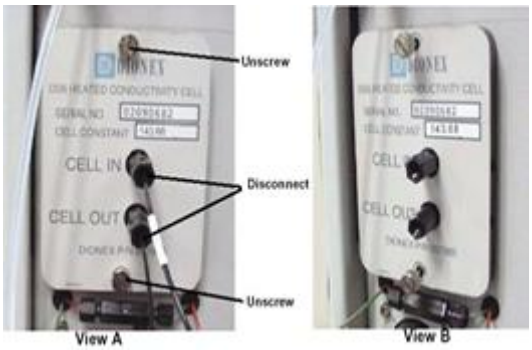
Figure 5. Dionex Aquion component panel



2. Old panel

AQUION qurilmasini faqat maxsus dasturiy ta'minot orqali boshqarish mumkin. Qurilmaning old paneli tizim holatini, jumladan, aloqa holatini, nasosni yoqish/o'chirish holatini va bostiruvchini yoqish/o'chirish holatini ko'rsatadigan LED ko'rsatkichlari bilan jihozlangan.

3. O'tkazuvchanlik datchiklari(yacheykalar)



Past hajmli, yuqori samaradorlikka ega termostatlangan hujayra yuqori sezgirlikni ta’minlaydi va hatto aniq harorat nazorati qiyin bo‘lgan muhitlarda ham barqaror ishlashni kafolatlaydi. Raqamli boshqaruv avtomatik diapazonni tanlash funksiyasi bilan keng dinamik diapazonni ta’minlab, bitta namuna yuborilishi orqali asosiy va kichik komponentlarni bir vaqtning o‘zida aniqlash imkonini beradi.

4. Supressor



Keng turdagi qo‘llanmalar talablariga javob beradigan barcha Dionex suppressorlari ushbu tizim bilan ishlatilishi mumkin. Reagentsiz (Reagent-Free) suppressiya kislotali yoki asosiy regenerant eritmalarni tayyorlash zaruratini yo‘qotadi, natijada past shovqin, yuqori sezgirlik va texnik xizmat ko‘rsatish talablarining kamayishini ta’minlaydi.

5. Kolonka isitgichi (Column Heater)



Haroratni boshqarish ma'lum ilovalarda kolonkaning barqarorligini va natijalar takrorlanuvchanligini yaxshilaydi.

Past dispersiyali issiqlik almashgich eluent(eritma)ni kolonkaga kirishidan oldin isitadi, bu esa belgilangan kolonka haroratining aniq ta’minlanishiga yordam beradi. Shaffof qopqoq issiqlik yo‘qotmasdan kolonka ulanishlarini vizual tekshirish imkonini beradi.

6. Oqish sensori va tomchi patnisi (Leak Sensor va Drip Tray)

Ichki o‘rnatilgan optik oqish detektori va tomchi patnisi oqish yuz bergan holatda tizimning avtomatik to‘xtatilishini ta’minlaydi hamda tezkor tuzatish choralarini ko‘rish imkonini beradi.

7. USB aloqasi

Plug-and-play USB aloqa texnologiyasi USB orqali boshqariladigan qurilmalar bilan tez va ishonchli ulanishni ta’minlaydi.

Avtomatik aniqlash va sozlash (auto-configuration) funksiyasi tizimni oson va foydalanuvchi uchun qulay tarzda sozlash imkonini beradi.

8. Eluent klapani(eritma klapani)



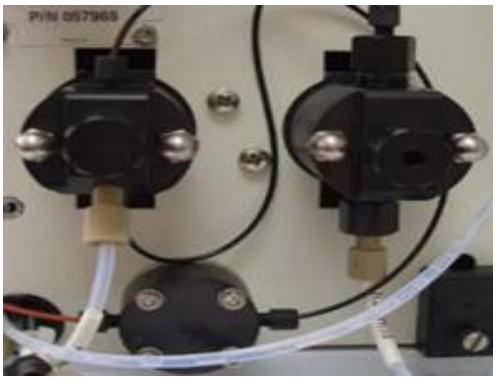
Elektron boshqariladigan klapan nasosning yuqori qismida eritma oqimini

to’xtatish imkonini beradi, bu esa texnik xizmat ko'rsatishni osonlashtiradi.
Nasos o‘chirilganda klapan avtomatik yopilib, teskari oqimning oldini oladi va nasos yoqilganda avtomatik ravishda ochiladi.

9. Vakuumli gazsizlantirish

Eluentni vakuumli gazsizlantirish uni ifloslanish va parchalanishdan himoya qiladi, hamda tahlil natijalarining takrorlanuvchanligini ta’minlaydi.
Gazsizlantirish jarayoni avtomatik ravishda boshqariladi va faqat gazsizlantirish zarur bo'lganda faollashadi.

10. Nasos



Ikki porshenli nasos konstruksiyasi eritmani minimal drift bilan barqaror yetkazib beradi va deyarli pulsatsiyasiz oqimni ta’minlaydi, natijada barqaror bazaviy chiziq va past aniqlash chegaralari yuzaga keladi.
PEEK polimer oqim yo‘li metall bilan bog‘liq ifloslanish va korroziyaning oldini oladi.

11. Injektor



Kichik kechikish hajmiga ega, elektr orqali boshqariladigan Rheodyne klapani PEEK oqim yo‘li bilan birgalikda namunaning barqaror va ishonchli yuborilishini ta’minlaydi.

12. Avtomatik namuna oluvchi (AS-DV)(Autosampler)



AS-DV avtomatik namuna oluvchi (autosampler) tasodifiy kirish (random access) rejimini qo‘llab-quvvatlaydi.

U 0,5 mL va 5 mL hajmdagi flakonlarni sig‘dira oladi, hamda qo‘shimcha modullar namunani avtomatik oldindan tayyorlash imkonini beradi.

Ⅱ. Tahlilga tayyorgarlik

1. Tahlilga tayyorgarlik

1) Anion tahlili

$$100mg \times \frac{Molecular\ or\ atomic\ weight\ of\ the\ anion + (Na^{+}\ or\ \ H_2O)}{Molecular\ or\ atomic\ weight\ of\ the\ anion}$$

·Izoh: Analitik darajadagi (eng yuqori tozalikdagi) reagentlardan foydalanish kerak.

Standart eritmalarni tayyorlash uchun hisoblash usuli
1000 ppm anion standart eritmasini tayyorlash uchun kerakli ionning konsentratsiyasi 1000 mg/L bo'lishi kerak. Na^+ kabi reagentlar alohida holda mavjud bo‘lmaganligi sababli, anion tuz hosil qilgan birikmalar (masalan, Cl^- → NaCl ko‘rinishida) ishlatilishi kerak. Shuning uchun 100 mL standart eritma tayyorlashda nazariy jihatdan 100 mg kerakli ion talab etiladi. Biroq amalda massa qarshi ion (masalan, Na^+ yoki K^+)ni o‘z ichiga olgan tuzning molekulyar massasiga asoslanib qayta hisoblanishi kerak. Shu sababli standart eritmalar quyida keltirilgan hisoblash asosida tayyorlanadi.

(1) Standart eritma tayyorlash

1000 mg/L (1000 ppm) boshlang‘ich (stok) eritma tayyorlanishi va keyinchalik zaruratga qarab suyultirilishi kerak. Juda yuqori tozalik darajasiga ega reagentlardan foydalanish lozim.

Quyidagilar 1000 ppm boshlang‘ich eritmalarni tayyorlash uchun zarur bo‘lgan reagent massalariga misollardir:

·F^-: 2.2100 g NaF / 1 L

·Br^-: 1,2876 g NaBr / 1 L

·NO_2^- : 1.4900 g $NaNO_2$ / 1 L

·NO_3^- : 1.3710 g $NaNO_3$ / 1 L

·BrO_3^- : 1,3057 g $KBrO_3$ / 1 L

·PO_4^{3-} : 1,4330 g KH_2PO_4 / 1 L

·ClO_3^- : 1,2753 g $NaClO_3$ / 1 L

·Cl^- : 1,6485 g NaCl / 1 L

·SO_4^{2-} : 1,4790 g Na_2SO_4 / 1 L

·ClO_2^- : 1,3410 g $NaClO_2$ / 1 L

Dionex 7-anion kombinatsiyalangan standart eritmasi uchun qism raqami (P/N) 56933 ga teng.

Ehtiyot choralari
Standart stok eritmalar 4 °C da olti (6) oygacha saqlanishi kerak.
NO_2^- va HPO_4^{2-} standartlari ishlatishdan oldin har kuni yangi suyultirilishi kerak.
Boshqa barcha standartlar suyultirilishi va haftada kamida bir marta ishlatilishi kerak.

(2) Eluent(eritma)

Kolonka turi	**Eluent(eritma) konsentratsiyasi**	**Eluent(eritma) tayyorlash**	**Oqim tezligi**
IonPac AS4A-SC, AS4A	1,8 mM Na_2CO_3 + 1,7 mM $NaHCO_3$	0,159 g Na_2CO_3 + 0,143 g $NaHCO_3$ / L	2,0 ml/min
IonPac AS12A	2,7 mM Na_2CO_3 + 0,3 mM $NaHCO_3$	0,286 g Na_2CO_3 + 0,025 g $NaHCO_3$ / L	1,5 ml/min
IonPac AS14	3,5 mM Na_2CO_3 + 1,0 mM $NaHCO_3$	0,371 g Na_2CO_3 + 0,084 g $NaHCO_3$ / L	1,2 ml/min
IonPac AS14A	3,5 mM Na_2CO_3 + 1,0 mM $NaHCO_3$	0,849 g Na_2CO_3 + 0,084 g $NaHCO_3$ / L	1,0 ml/min
IonPac AS11	100 mM NaOH	5,25 ml 50% (v/v) NaOH eritmasini 1 l ga suyultiring	1,0 ml/min
IonPac AS22	4,5 mM Na_2CO_3 + 1,4 mM $NaHCO_3$	0,478 g Na_2CO_3 + 0,117 g $NaHCO_3$ / L	1,2 ml/min
IonPac AS23	4,5 mM Na_2CO_3 + 0,8 mM $NaHCO_3$	0,478 g Na_2CO_3 + 0,066 g $NaHCO_3$ / L	1,0 ml/min

(3) Regenerant

(Faqat AMMS supressoridan foydalanilganda talab qilinadi)

50 mN H_2SO_4 → 5.6 mL Konsentrlangan H_2SO_4/4L (98% sulfat kislota)

(4) Supressor

- AERS

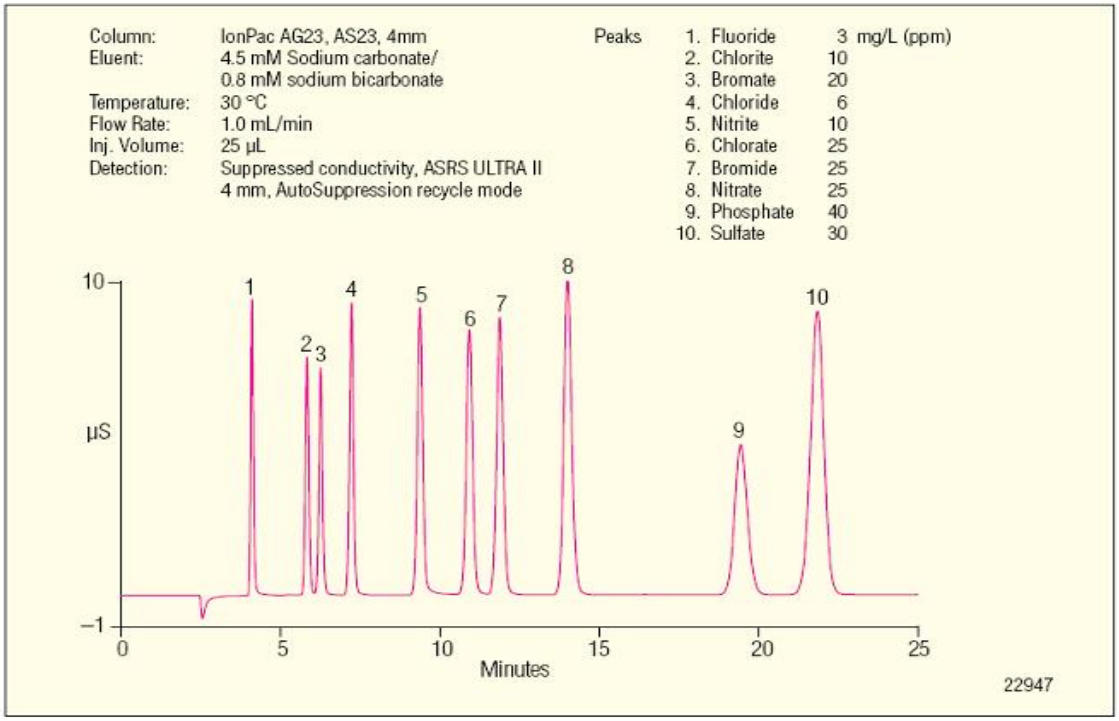
AERS oqimi ishlatilayotgan kolonkaga qarab, quyida ko‘rsatilganidek sozlanishi kerak.

Kolonka turi	ERS oqimi	Kolonka turi	ERS oqimi
IonPac AS4A-SC, AS4A	27 mA	IonPac AS11	248 mA
IonPac AS14	20 mA	IonPac AS22	31 mA
IonPac AS14A	24 mA	IonPac AS23	25 mA

- AMMS

50 mN sulfat kislota (H_2SO_4) regeneranti 2–6 psi azot (N_2) gaz bosimi ostida yetkazib berilishi kerak.

Regenerant oqim tezligi 2–5 mL/min darajada saqlanishi kerak, bu chiqindi liniyasidan har ikki soniyada taxminan uch tomchi chiqishiga mos keladi.



<Tahlil namunasi>

2) Kation tahlili

(1) Standart

Stok eritmalari ACS (Amerika Kimyo Jamiyati) reaktiv darajasidagi kimyoviy moddalar yordamida tayyorlanishi va ishlatishdan oldin kerak bo'lganda suyultirilishi kerak.

Standart eritmalar xona haroratida polietilen yoki polipropilen idishlarda saqlanishi kerak.

*Olti kationli kombinatsiyalangan standart (P/N 46070)

1000 ppm kation standartlarini tayyorlash

Quyidagilar alohida kationlarning 1000 ppm boshlang‘ich eritmalarini tayyorlash uchun zarur bo‘lgan reagentlar va ularning miqdorlaridir:

·K^+: 1.9067 g KCl / 1 L

·NH_4^+ : 2.9654 g NH_4Cl / 1 L

·Na^+: 2.5420 g NaCl / 1 L

·Ca^{2+}: 3,5580 g $CaCl_2 \cdot 2H_2O$ / 1 L

·Mg^{2+} : 8.3650 g $MgCl_2 \cdot 6H_2O$ / 1 L

(2) Eluent(eritma)

20 mM MSA (Metansulfon kislotasi) tayyorlash

MSA odatda standart IonPac CS12 ustun sharoitida 20 mM konsentratsiyada qo'llaniladi.

Quyidagi hisoblash Fluka / Aldrich MSA (M861-4) ga asoslangan.

Metansulfon kislotasi (MSA):

·Molekulyar og'irlik: 96.10

·Zichlik: 1,319 g/ml

·Konsentratsiya: 70% eritma

1 litr 20 mM MSA tayyorlash uchun:

· Kerakli massa:

$20 \times 96.10 \times 10^{-3} = 1.922 g$

· Hajm birligiga o'tkazish:

$$1.922 \div 1.319 = 1.457\,\text{mL}$$

·70% eritma uchun tuzatish

$$1.457 \div 0.70 = 2.08\,\text{mL}$$

Shuning uchun **2,08 mL 70% li metansulfon kislota** qo'shiladi va hajm deionizatsiyalangan suv bilan 1 L gacha yetkaziladi.

21 mM sulfat kislota tayyorlash

Sulfat kislota odatda standart IonPac CS12A kolonkasi sharoitida 21 mM konsentratsiyada qo'llaniladi.

Quyidagi hisoblash 1 L eluent(eritma) tayyorlashga asoslangan.

(Eslatma: Ion xromatografiyasi uchun maxsus mo'ljallangan reagent darajasidagi sulfat kislota ishlatilishi kerak.

Oddiy laboratoriya darajasidagi sulfat kislota qabul qilinmaydi.)

Sulfat kislotasi (H_2SO_4):

·Molekulyar og'irligi: 98.08

·Zichlik: 1,84 g/ml

·Konsentratsiya: 95–97% eritma (o'rtacha 98%)

1 litr 21 mM sulfat kislota tayyorlash uchun:

· Kerakli massa:

$$21 \times 98.08 \times 10^{-3} \div 2 = 1.0295\,g$$

· Hajm birligiga o’tkazish:

$$1.0295 \div 1.84 = 0.5595\,\text{mL}$$

·98% eritma uchun tuzatish:

$$0.5595 \div 0.98 = 0.57\,\text{mL}$$

Shuning uchun **0,57 mL 98% li sulfat kislota** qo‘shiladi va hajm deionizatsiyalangan suv bilan 1 L gacha yetkaziladi.

(3) Regenerant (Faqat CMMS supressoridan foydalanilganda qo'llaniladi)

100 mM TBAOH (Tetrabutilamoniy gidroksid)

Reaktiv: Fluka, P/N 86880

→ 55% li TBAOH eritmasidan 200 mL olinib, hajm 4 L gacha suyultirib tayyorlanadi.

(4) Kolonka

- IonPac CS12

·Eluent(eritma): 20 mM MSA (Metansulfon kislotasi)

·Oqim tezligi: 1,0 ml/min

- IonPac CS12A

·Eluent(eritma): 20 mM MSA (Metansulfon kislotasi)

·Oqim tezligi: 1,0 ml/min

- IonPac CS14

·Eluent(eritma):

MSA (metansulfon kislota) izokratik elyutsiya sharoitida yoki MSA va atsetonitril (ACN) yordamida gradient elyutsiya sharoitida qo‘llaniladi.

(Bu usul CS12/CS12A kolonkalarida ajratiladigan odatiy kationlar hamda alifatik aminlarni ajratish uchun mos keladi.)

·Oqim tezligi: 1,0 ml/min

- IonPac CS15

·Eluent(eritma): 10 mM H_2SO_4 (Sulfat kislotasi) + 9% asetonitril

·Oqim tezligi: 1,2 ml/min

(5) Supressor

- CERS

ERS oqimi ishlatilayotgan kolonkaga qarab, quyida ko‘rsatilganidek sozlanishi kerak.

Kolonka turi	ERS oqimi (mA)	Kolonka turi	ERS oqimi (mA)
IonPac CS12	59	IonPac CS12A	59
IonPac CS14	59	IonPac CS15	56

- CMMS

100 mM TBAOH (tetrabutilammoniy gidroksid) regeneranti 2–6 psi bosimdagi azot (N_2) gazi ostida berilishi kerak.

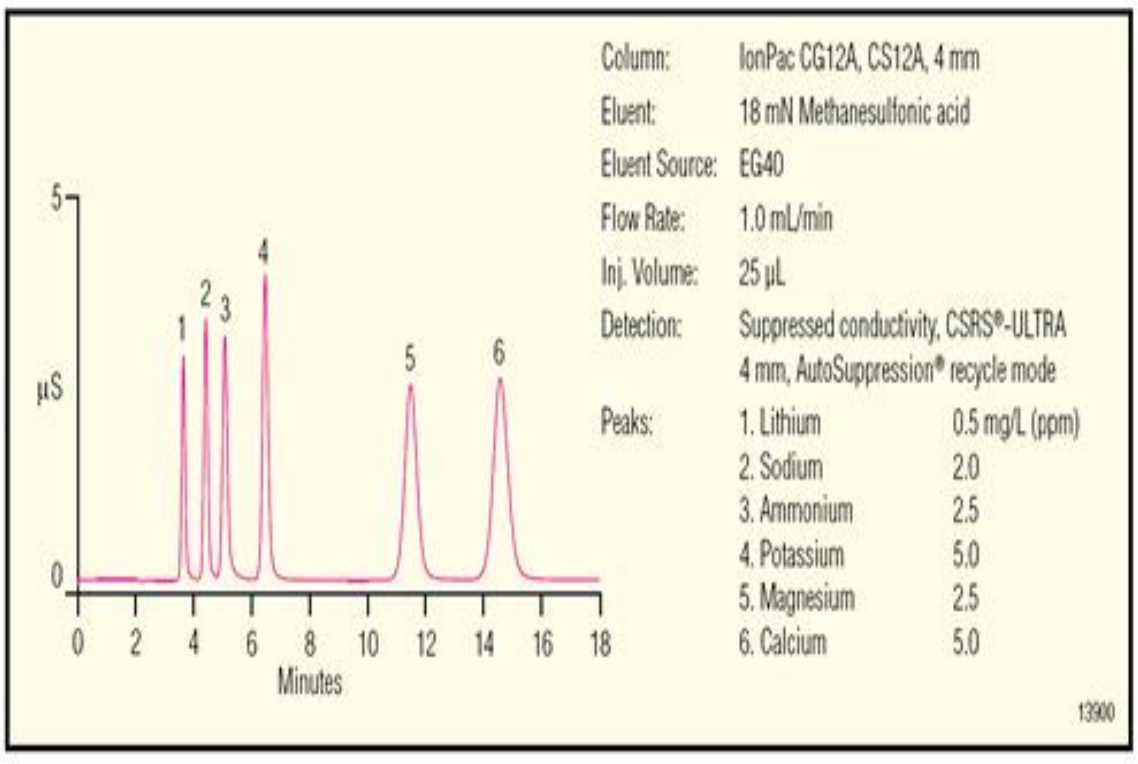
·100 mM TBAOH tayyorlash:

55% li TBAOH eritmasidan (Fluka, P/N 86880) 200 mL olinib, hajm 4 L gacha suyultiriladi.

· Foydalanish shartlari:

Oqim tezligi: 2–5 ml/min

Chiqindi liniyasi: Har ikki soniyada taxminan uch tomchi

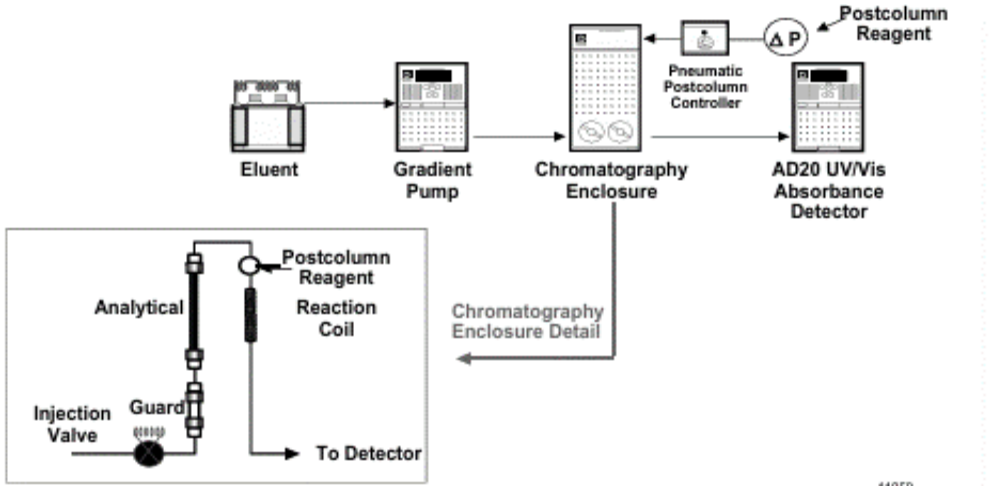


<Tahlil namunasi>

3) O'tish metallini tahlil qilish

(1) Tizim konfiguratsiyasi va oqim yo'li

Ion xromatografiya tizimining o' tish metallari uchun konfiguratsiyasi



2) Analitik shartlar

(1) Standart

Atom-absorbsiya spektrometriyasi (AAS) uchun mo‘ljallangan, savdoda mavjud 1000 ppm standart eritmalardan foydalanish mumkin.

Ushbu standartlar kislotali eritmada taqdim etilganligi sababli, ular IC (ion xromatografiya) standartlari sifatida ham mos keladi.

(2) Eluent(eritma)

PDCA buferidan foydalanilganda (ikki va uch valentli temir ionlari, Fe^{2+} va Fe^{3+} ni tahlil qilish uchun):

- 6 mM PDCA (Piridin-2,6-dikarbon kislotasi), 99%
- 50 mM sirka kislotasi (99%)
- 50 mM natriy asetat trigidrat

→ 1 g PDCA + 6,8 g natriy asetat + 2,9 ml sirka kislotasi/1L

Oksalat buferidan foydalanganda (Pb^{2+} tahlili uchun)

- 50 mM oksalat kislotasi digidrati
- 95 mM Litiy gidroksid monogidrat (kristal)

→ 6,3 g oksalat kislotasi + 4,0 g litiy gidroksid/1L

Oksalat buferi ishlatilganda, kadmiy (Cd) va marganets (Mn) bir vaqtning o'zida eritiladi.

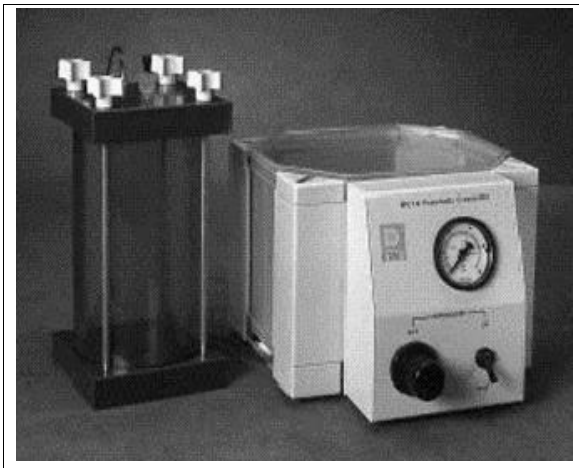
(3) Postcolumn reaktivi

- 0,4 mM PAR (4-(2-piridilazo)rezorsinol mononatriy tuzi monogidrat)
- 1M Sirka kislotasi (99%)
- 3M ammoniy gidroksid (28%)

→ 0.1 g PAR + 57㎖ Sirka kislotasi + 200㎖ Ammoniy gidroksid/1L

(4) O'tish metallini tahlil qilish uchun Dionex reaktivlari

- MetPac PDCA Eluent Konsentrati (5×), 1000ml gacha suyultirilgan(P/N 46088)
- MetPac Oksalat kislotasi Eluent Konsentrati (10×), 500 ml gacha suyultirilgan (P/N 46091)
- MetPac PAR Postcolumn erituvchisi, 1000 ml gacha suyultiriladi (P/N 46094)
- PDCA: 20 g (P/N 39671)
- PAR: 5 g (P/N 39672)



Qo'shimcha eslatmalar

PAR: 4-(2-piridilazo)rezorsinol

Oqim tezligi: 0,7 ml/min

Ish bosimi: 50–60 psi da azot gazi bilan bosim o'tkaziladi

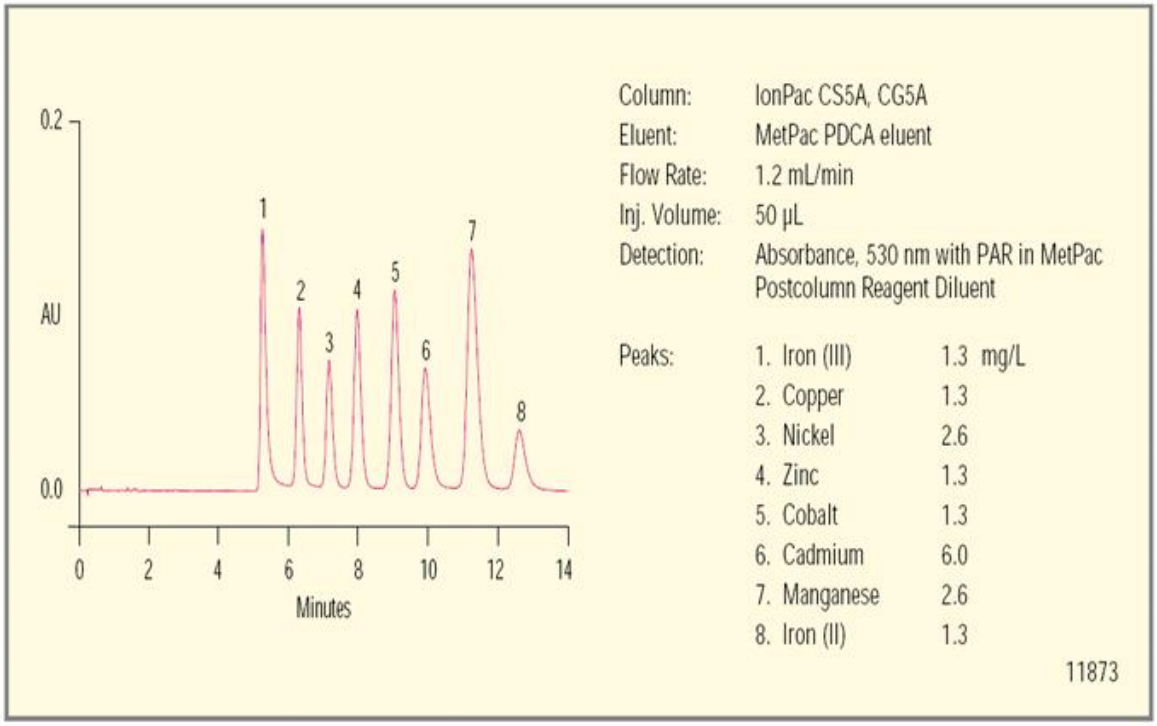
(rasmda ko'rsatilganidek, PC10 postcolumn reagent yetkazib berish tizimidan foydalanganda)

(5) Kolonka

- IonPac CS5, IonPac CS5A
- Oqim tezligi: 1,0 ml/min

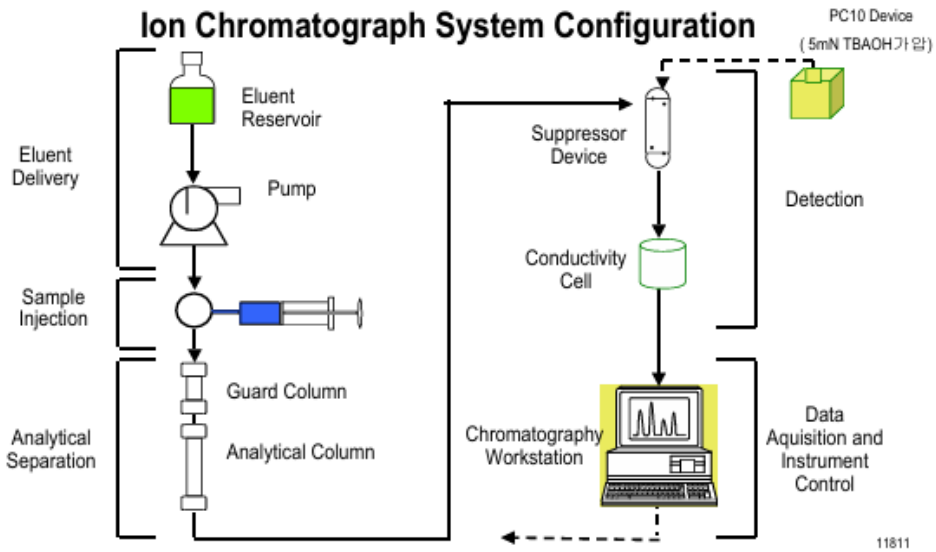
(6) Detektor (Yutilish detektori)

- Yorug'lik manbai: Ko'rinadigan diapazon
- To'lqin uzunligi: 520 nm yoki 530 nm



<Tahlil namunasi>

4) Organik kislota tahlili



(1) Tizim konfiguratsiyasi va oqim yo'li

(2) Analitik shartlar

- Kolonka: IonPac ICE-AS6

·Eluent(eritma): 0,4 mM geptaftorbutirik kislota (Fluka reaktivi)

→ 1 L uchun 52,35 mkL (99% geptaflorobutirik kislota ishlatilganda)

·Oqim tezligi: 1,0 ml/min

- Supressor

·ICE mikromembrana supressori (ICE-AMMS)

·5.0 mN TBAOH (Tetrabutilamoniy gidroksid)

·2–6 psi azot (N_2) gaz bosimi ostida yetkazib beriladi

·Regenerant oqim tezligi: 2–5 ml/min

·Oqim tezligi 2–5 ml/min da saqlanishi kerak, bu chiqindi liniyasidan har ikki soniyada taxminan uch tomchi tushirish tezligiga to'g'ri keladi.

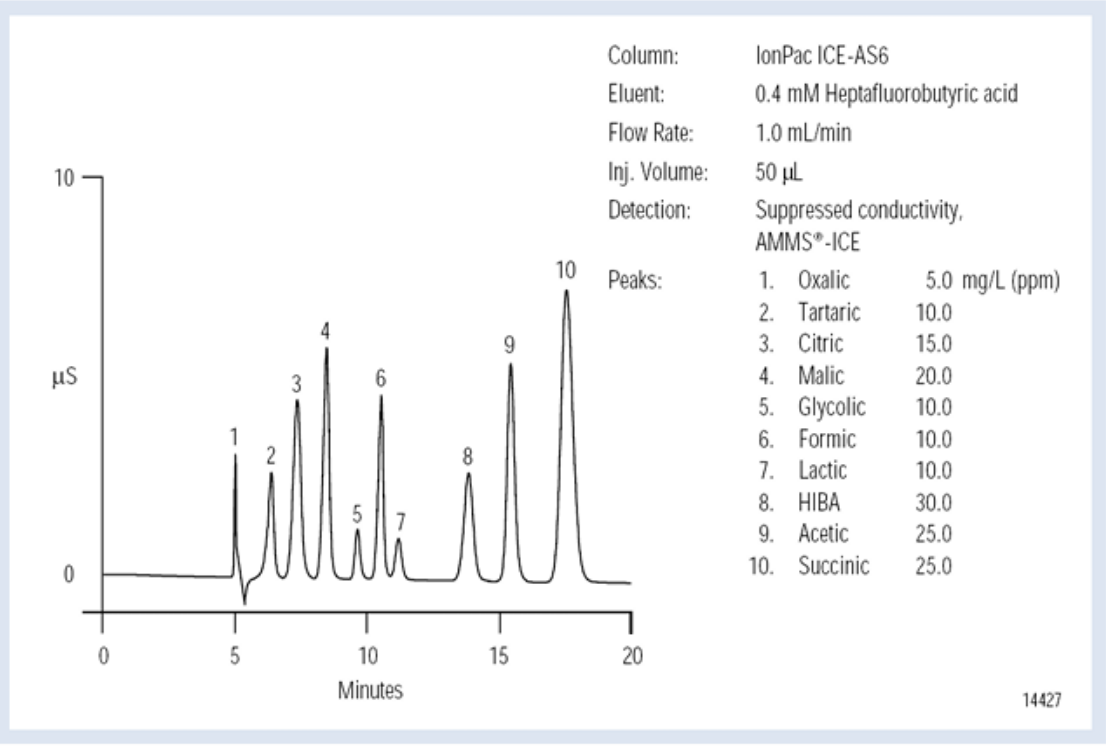
·5.0 mM TBAOH tayyorlash:

55% li TBAOH eritmasidan (Fluka, P/N 86880) 10 mL olinib, deionizatsiyalangan suv bilan hajm 4 L gacha suyultirib tayyorlanadi.

-Detektor

·O'tkazuvchanlik detektori

·Fon o'tkazuvchanligi: 23–27 µS



<Tahlil namunasi>

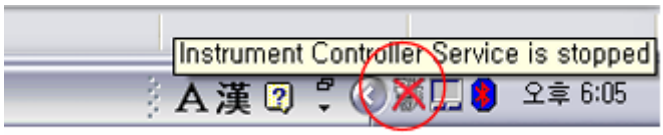
Ⅲ. AQUION tizimini tez barqarorlashtirish

1. AQUION tizimini yoqish

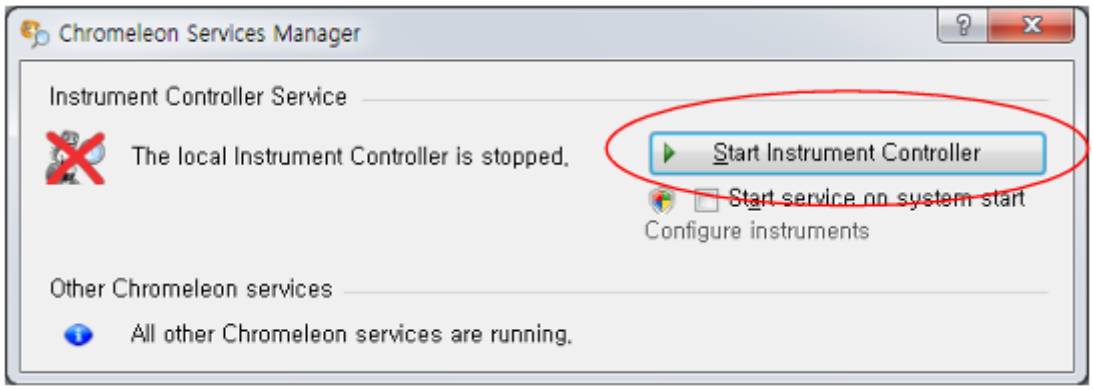
AQUION tizimi 220 V AC quvvat manbaidan ishlaydi. Elektr shovqinini yo'q qilish uchun to'g'ri yerga ulash kerak. Barqaror kuchlanish ta'minotini ta'minlash uchun tizim barqarorligini oshirish uchun avtomatik kuchlanish regulyatoridan (AVR) foydalanish tavsiya etiladi.



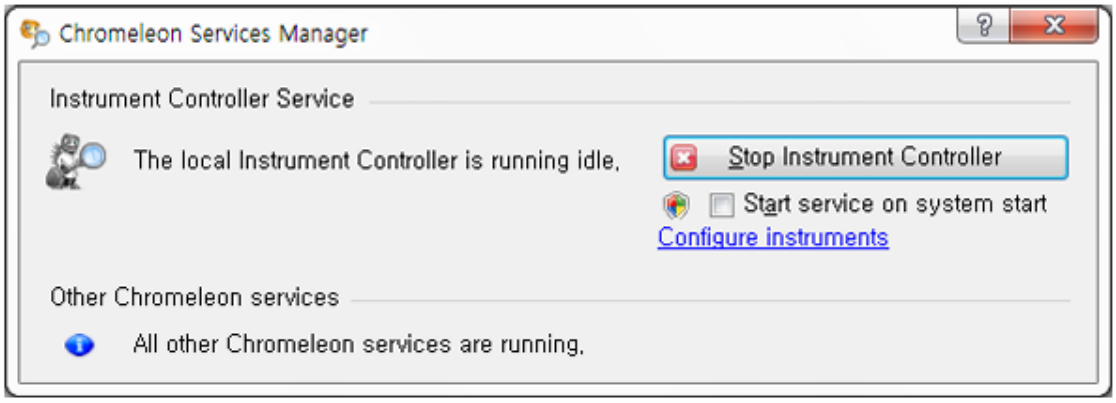
2. Kompyuterni ishga tushiring va monitor ekranining pastki qismida joylashgan Instrument Controller Service tugmasini bosing.



3. Xizmat oynasi faol bo‘lganda, davom etish uchun ustiga bosing.



4. Ekranda ko'rsatilganidek, muvaffaqiyatli faollashtirish ko'rsatilganda, Yopish tugmasini bosing.



5. Dasturiy ta'minotni ishga tushirish uchun ish stolidagi Chromeleon7 belgisini ikki marta bosing.

Chromeleon7 konsolning asosiy ekranidan iborat. Konsol ekranining pastki chap qismida joylashgan Navigator Qurilmalar va Ma'lumotlarga bo'lingan.

1) Instrument — tizimni boshqarish uchun ishlatiladi.

2) Data — ma’lumotlarni qayta ishlash uchun ishlatiladi.

AQUION tizimini boshqarish uchun Instrument ni tanlang, keyin Pump_ECD yorlig'ini tanlang.

6. Prime (to‘ldirish/primlash)

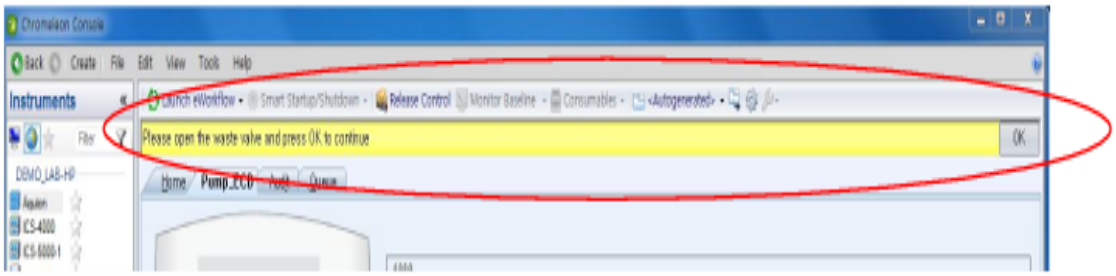
Primlash eluent idishi bilan nasos orasidagi havo pufakchalarini olib tashlash uchun amalga oshiriladi.



1) Rasmda ko‘rsatilganidek, primlashga tayyorgarlik uchun ikkinchi nasos

boshchasining chiqindi klapanini 2–3 marta aylantirib oching.

2) Chromeleon 7 dasturining Instrument panelida Prime tugmasini bosing. Ekranning yuqori qismida sariq ogohlantirish xabari paydo bo‘ladi. Primlashni boshlash uchun OK tugmasini bosing.



3) Primlash jarayonida chiqindi liniyasini kuzatib, barcha havo pufakchalari chiqarib yuborilganini tasdiqlang. Primlash tugagach (odatda 5 daqiqa ichida), primlashni to‘xtatish uchun Pump Off tugmasini bosing. So‘ng chiqindi klapanini soat miliga teskari yo‘nalishda burab yopib, primlash jarayonini yakunlang.

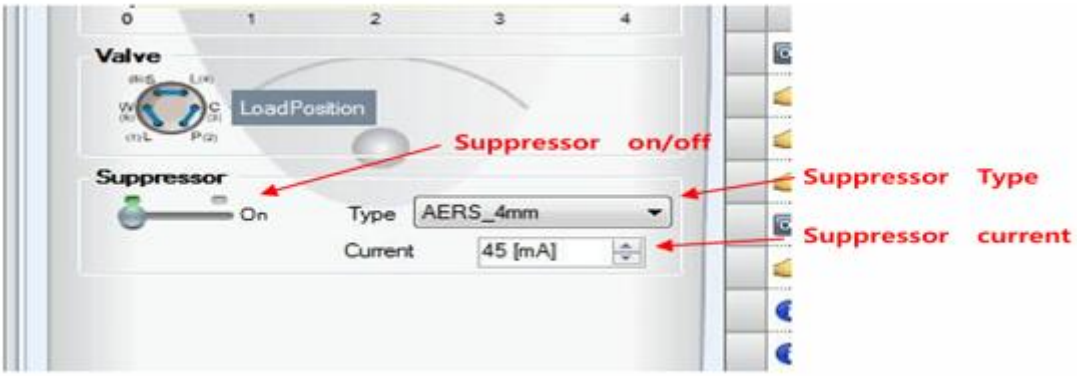
4) Nasosni yoqishdan oldin kerakli oqim tezligini kiriting, so‘ng Pump On tugmasini bosing. Oqim tezligi kolonkaga bog‘liq bo‘lgani uchun, mos qiymatlarni tanlashda II-1 bo‘lim — Tahlilga tayyorgarlik bo'limiga qarang.

7. Suppressorni sozlash

Nasosni yoqib, bosim qiymati barqarorlashganini tasdiqlagandan so‘ng, suppressor turini va suppressor oqimini sozlang.

Suppressorni sozlashning batafsil qiymatlari II-1 bo‘lim — Tahlilga tayyorgarlik da keltirilgan.

Suppressorning to‘g‘ri elektrolitik ishlashi regeneratsiya chiqish liniyasida gaz pufakchalari hosil bo‘lishi orqali tasdiqlanishi mumkin.



8. Tahlilning boshlanishi

Bosim barqaror saqlanib, stabil bazaviy chiziq (fon) hosil bo‘lgandan so‘ng, tahlilni boshlash mumkin.

Batafsil tahlil jarayonlari uchun Chromeleon 7 dasturining ishlash bo'limiga qarang.

9. Tizimning o'chirilishi

Tahlil yakunlangandan so‘ng tizimni quyidagi tartibda o‘chiring:

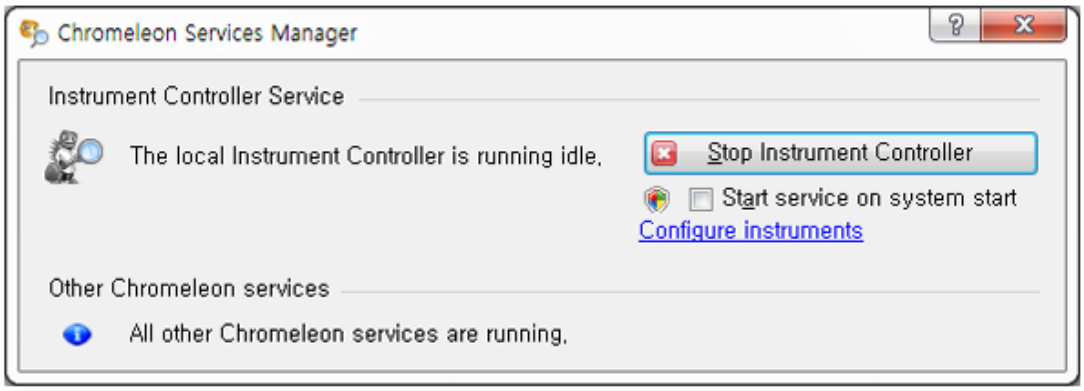
(1) Suppressorni o‘chiring.

(2) Nasosni o‘chiring.

Ushbu bosqichlar bajarilgach, tizimni o‘chirish yakunlanadi. So‘ng Chromeleon 7 dasturining Console oynasini yoping.

10. Chromeleon xizmatlarini tugatish

Kompyuter ekranining pastki o‘ng burchagida joylashgan Chromeleon Services Manager ikonkasini bosib, xizmatlar menejerini oching. So‘ng Stop tugmasini bosib, Chromeleon xizmatlarini to‘xtating..



11. Quvvatni o'chirish

Nihoyat, qurilmaning quvvatini o'chiring va kompyuterni o'chiring.

IV.CHROMELEON 7.2 Dasturiy ta'minotning ishlashi

1. Dionex Chromeleon dasturining afzalliklari

1) Chromeleon dasturi Windows Explorer uslubidagi brauzer interfeysi orqali saqlangan ma’lumotlarni oson ko‘rish imkonini beradi va shu bilan birga qurilmani to‘liq boshqarishni ta’minlaydi.

2) Instrument Method Files va Processing Method Files fayllarini oson yaratish mumkin, shuningdek ma’lumotlarni qayta ishlash (reprocessing) qo‘llab-quvvatlanadi.

3)Dastur intuitiv va o‘rganish uchun qulay tarzda ishlab chiqilgan bo‘lib, har qanday foydalanuvchi uni samarali ishlata oladi.

2. Chromeleon dasturiy ta'minotining komponentlari

Chromeleon dasturi quyidagi komponentlardan iborat:

1) Server menejeri

2) Server konfiguratsiyasi (Server menejeri ichida)

3) Chromeleon 7

4) Service menejeri

Qurilmaning seriya raqamini taniydi va uni kompyuterga ulaydi.

5) Qurilmani sozlash(Configure Instrument)

Dastur dastlab o‘rnatilgandan so‘ng asosiy qurilma ma’lumotlarini sozlash uchun ishlatiladi.

6) Chromeleon belgisi

Qurilmalarning ishlashi va ma'lumotlarni qayta ishlash uchun ishlatiladigan asosiy dasturni ishga tushiradi.

3. Xizmat ko'rsatish menejeri(Service Manager)

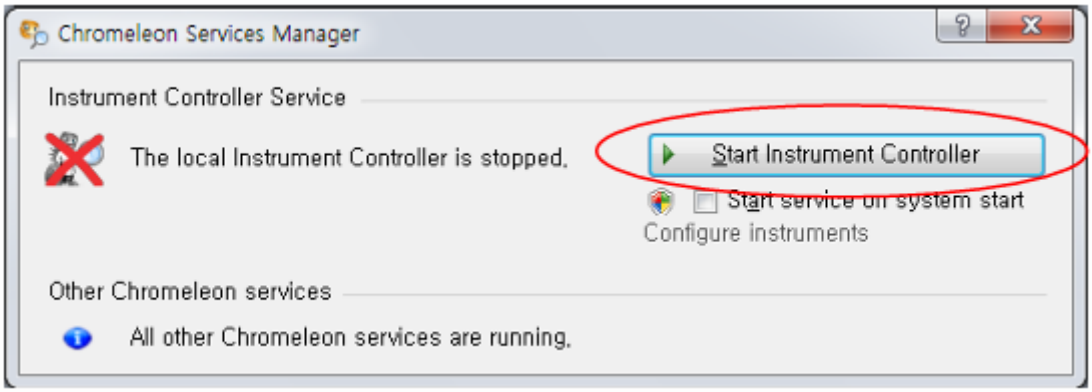


1) Ish stolida Start → All Programs → Service Manager ni tanlang.

Shu bilan bir qatorda, ekranning pastki o'ng burchagida joylashgan Xizmat menejeri(Service Manager) belgisini ikki marta bosing.



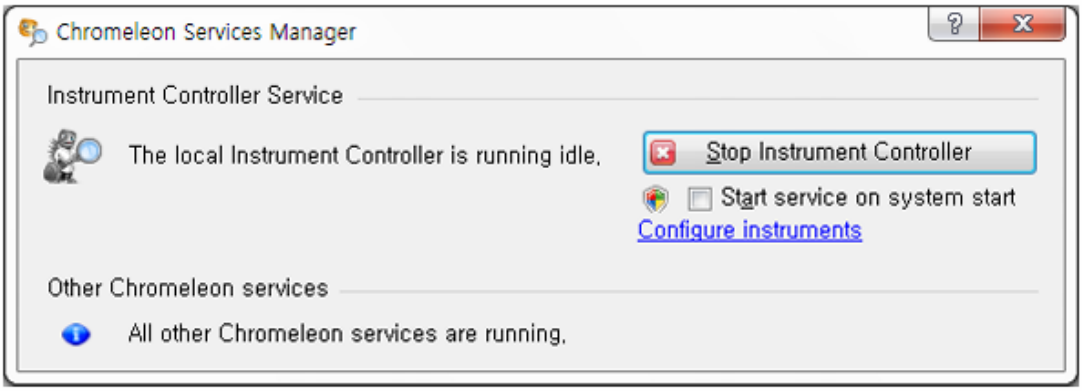
2) Xizmat menejeri oynasi paydo bo'lganda, davom etish uchun bosing.





Bosish

3) Quyida ko'rsatilgandek “Mahalliy qurilmalar boshqaruvchisi ishlayapti…” (The local Instrument Controller is running…) xabari paydo bo'lganda, oynani yoping.

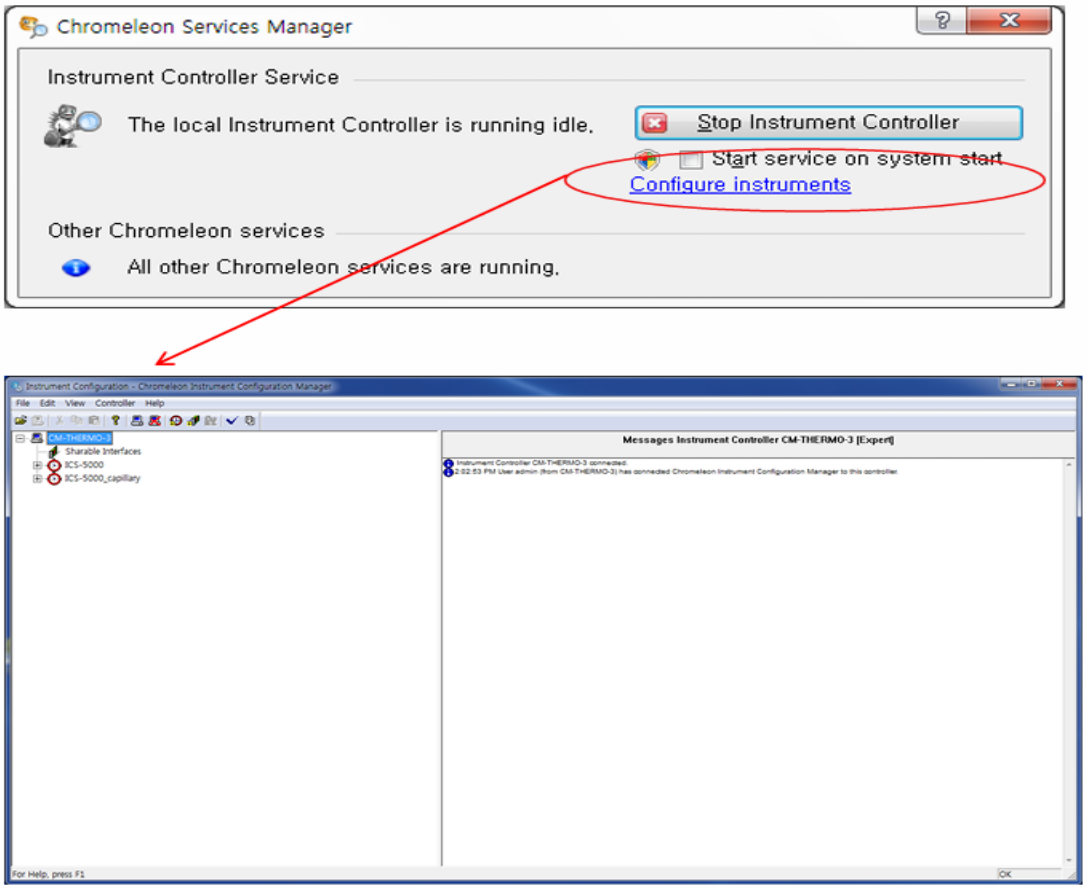


·Maslahat:

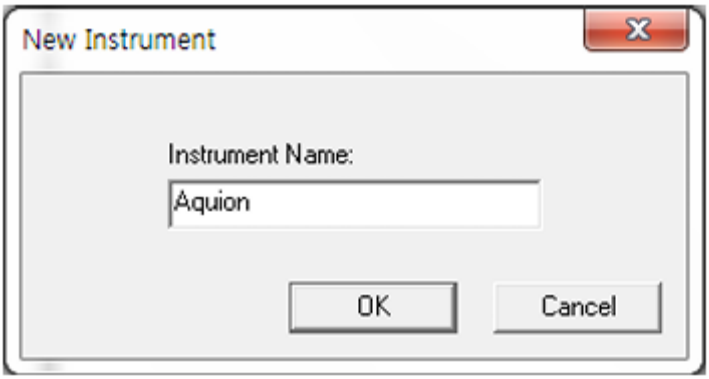
Agar ushbu jarayon kompyuter ishga tushganda avtomatik boshlanishini xohlasangiz, Start Service on System Start bandini belgilang.

4. Chromeleon server konfiguratsiyasi (Chromeleon Server Configuration)

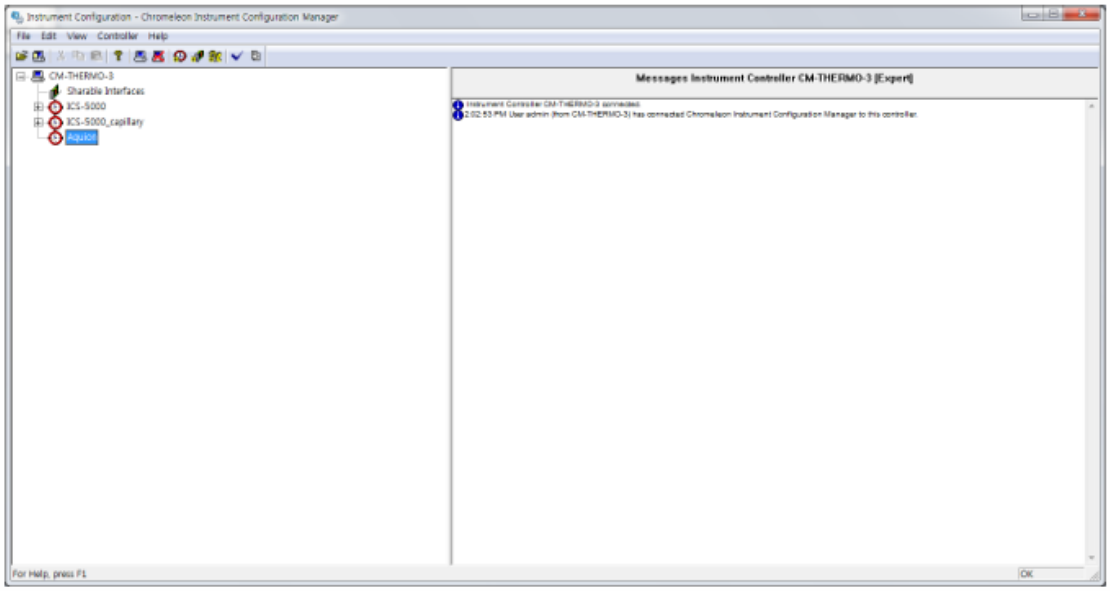
1) Configure Instrument tugmasini bosib, konfiguratsiya oynasini oching.



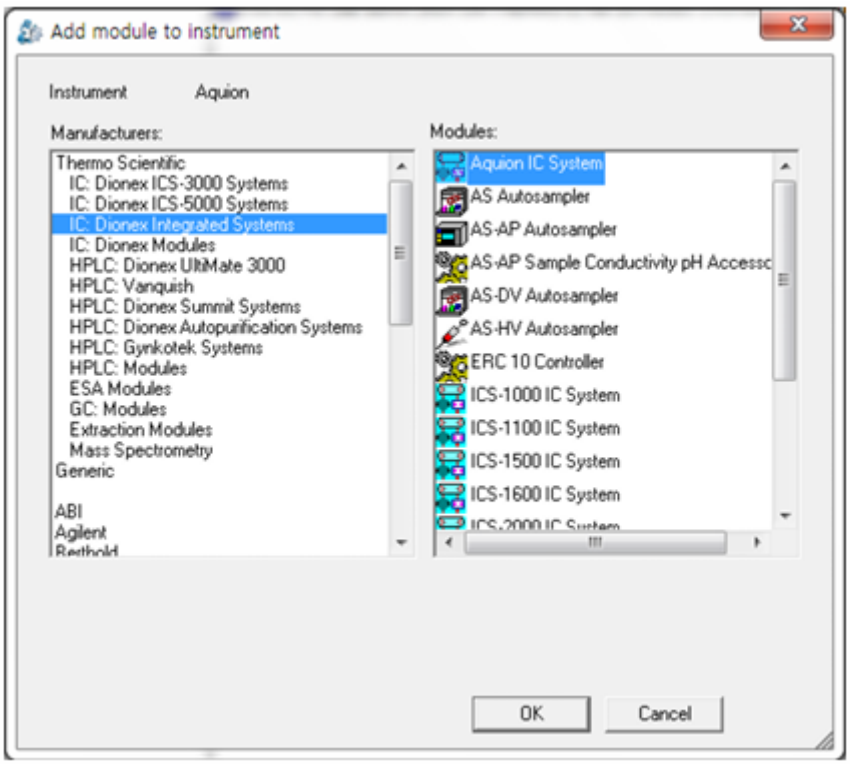
2) Qurilma qo'shish( ) tugmasini bosing, ko'rsatilgandek nom bering va keyin OK tugmasini bosing.



3) Konfiguratsiya rasmda ko‘rsatilganidek aks etayotganini tasdiqlang.
Soat shaklidagi belgi Timebase ni bildiradi, ya’ni bir xil vaqt oralig‘ida birgalikda ishlaydigan qurilmalar guruhini anglatadi.



4) Qurilma qo'shish( ) tugmasini bosing, ko'rsatilgandek kerakli qurilma modelini tanlang va keyin OK tugmasini bosing.

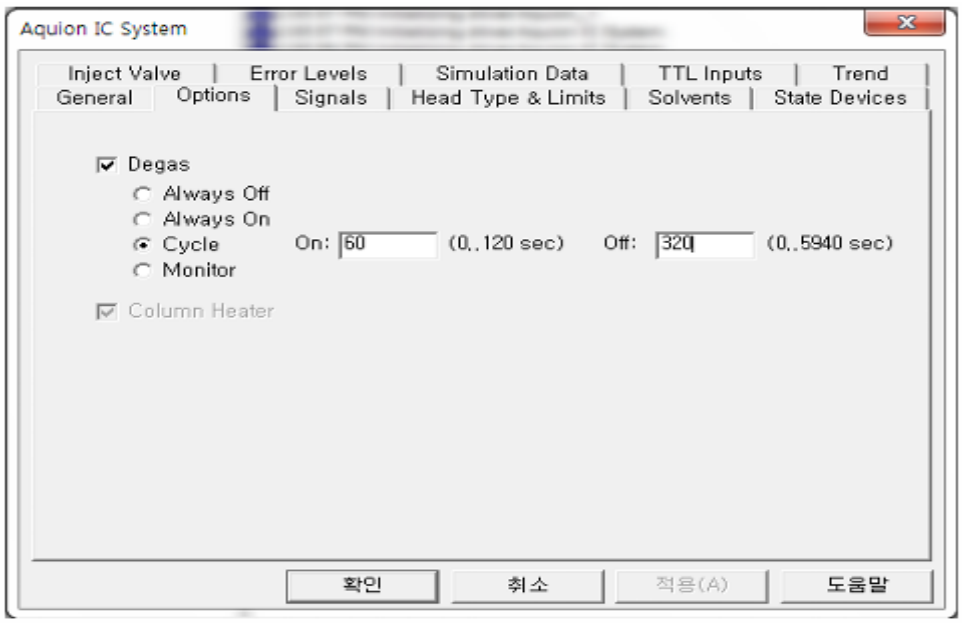


5) Quyidagi ekran ko‘rsatiladi. Ko‘rsatilgan modullar ro‘yxatidan Serial Number (S/N) ni tanlang.

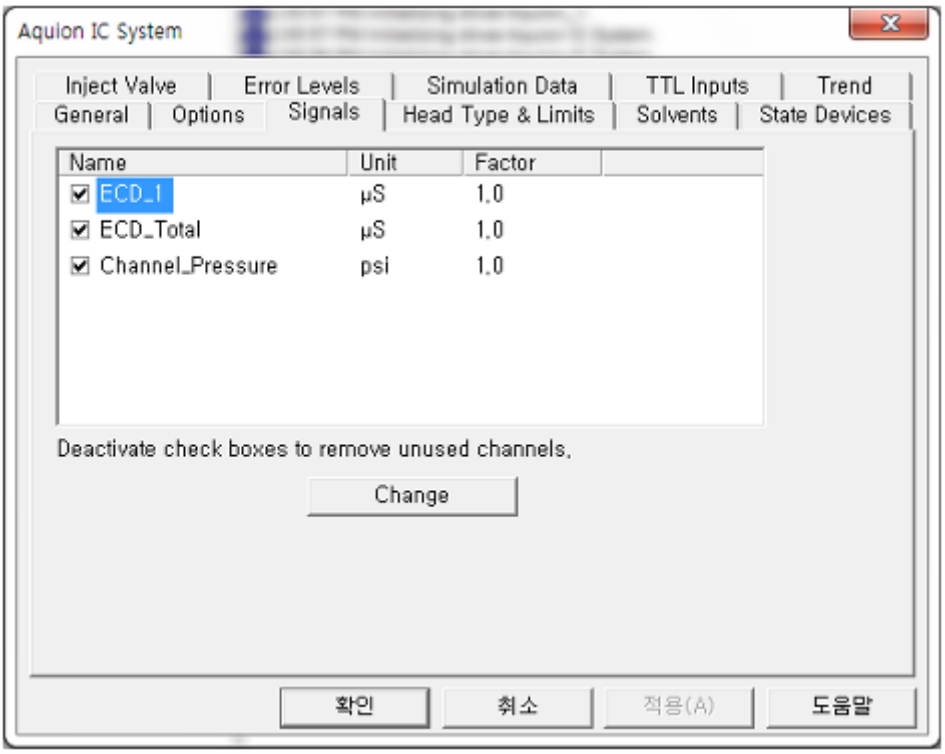
Module Serial No maydonida ochiluvchi ro‘yxat (drop-down) tugmasini bosib, kompyuterga ulangan qurilmalarning seriya raqamlarini ko‘ring va mos keladigan seriya raqamini tanlang.



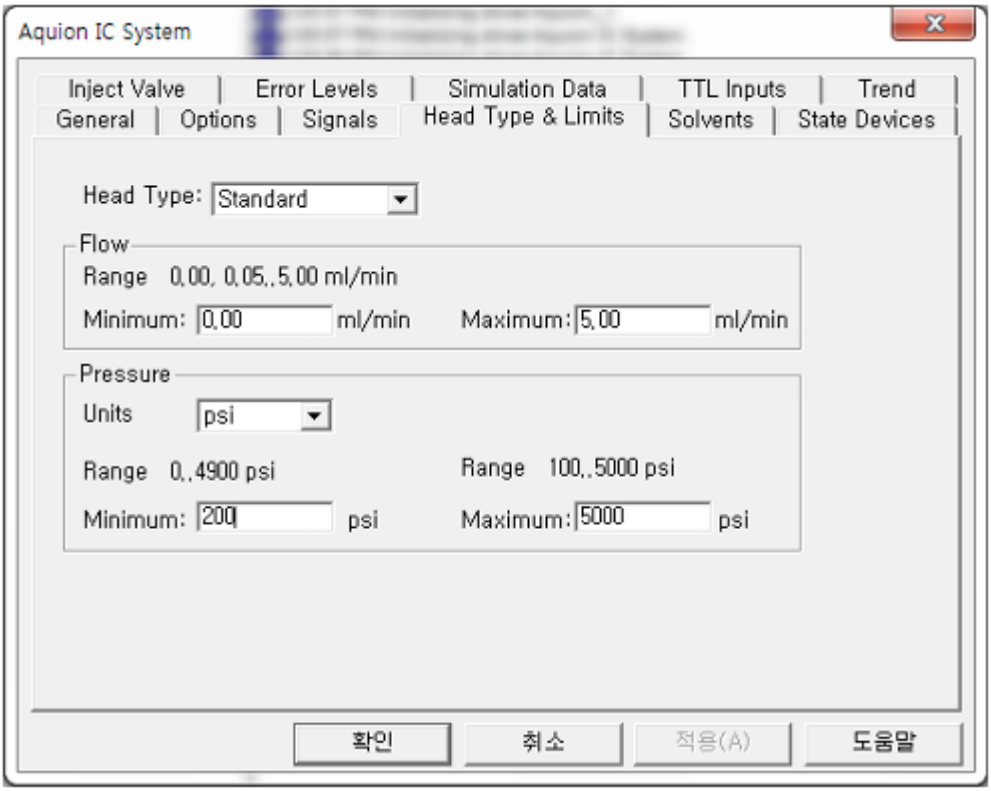
6) Options bo‘limida tizimda hozirda o‘rnatilgan opsiyalarni tanlang. Agar degasser o‘rnatilgan bo‘lsa, sozlamalarni rasmda ko‘rsatilganidek moslang.



7) Signallar sozlamalari ko'rsatilgandek sozlanganligini tasdiqlang.

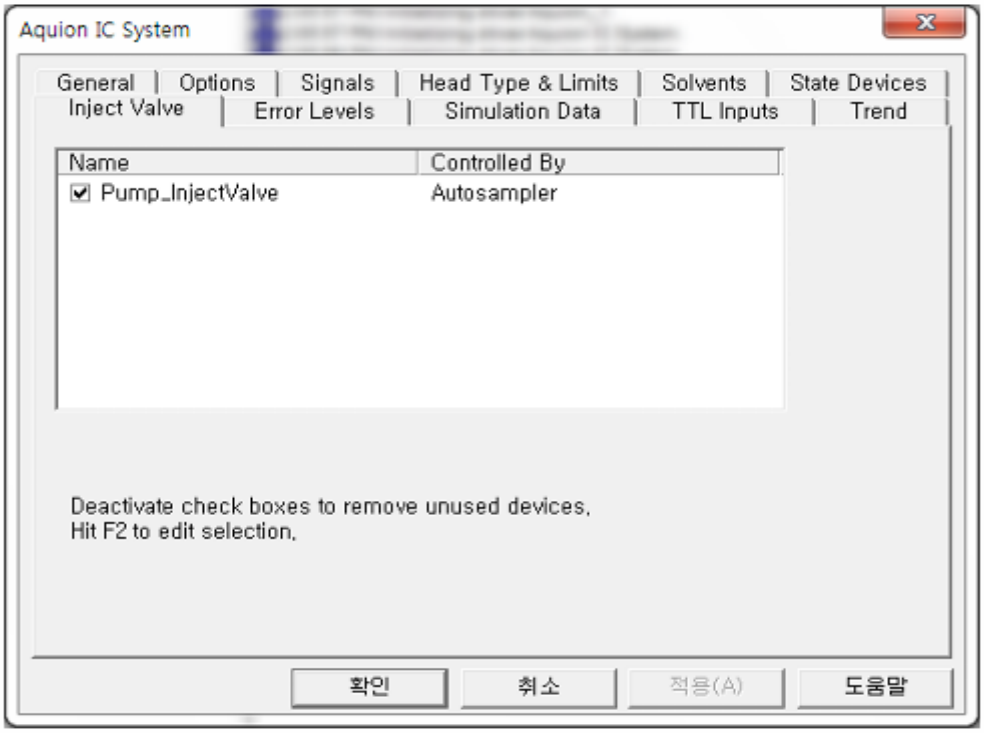


8) Ko'rsatilganidek, Head Type & Limits sozlamalarini sozlang.

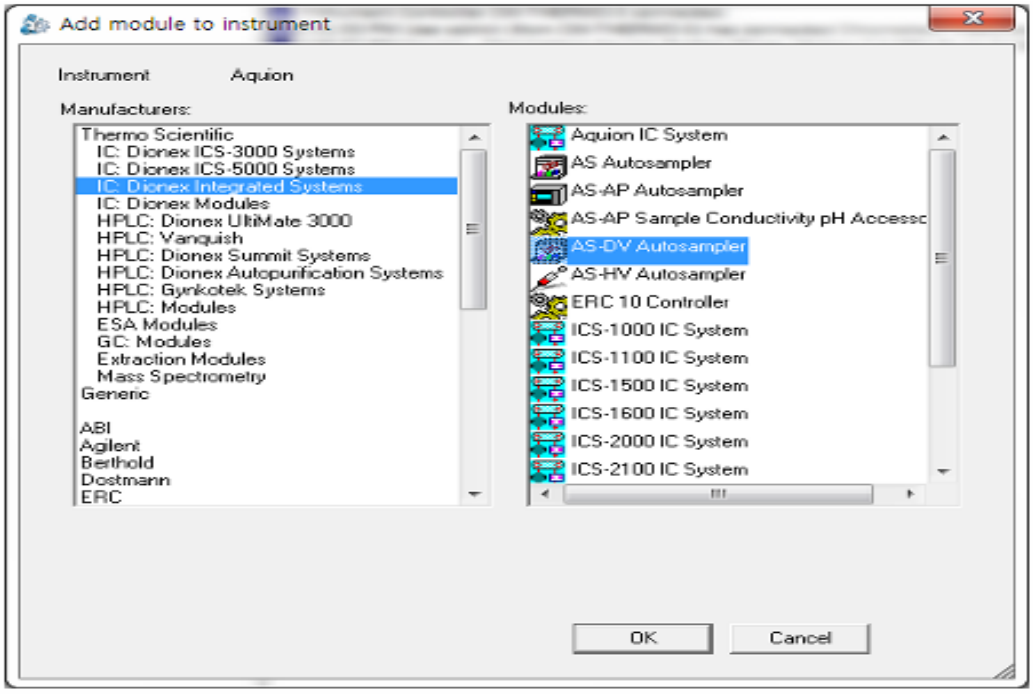


9) Injection Valve sozlamalarini rasmda ko‘rsatilganidek tanlang.

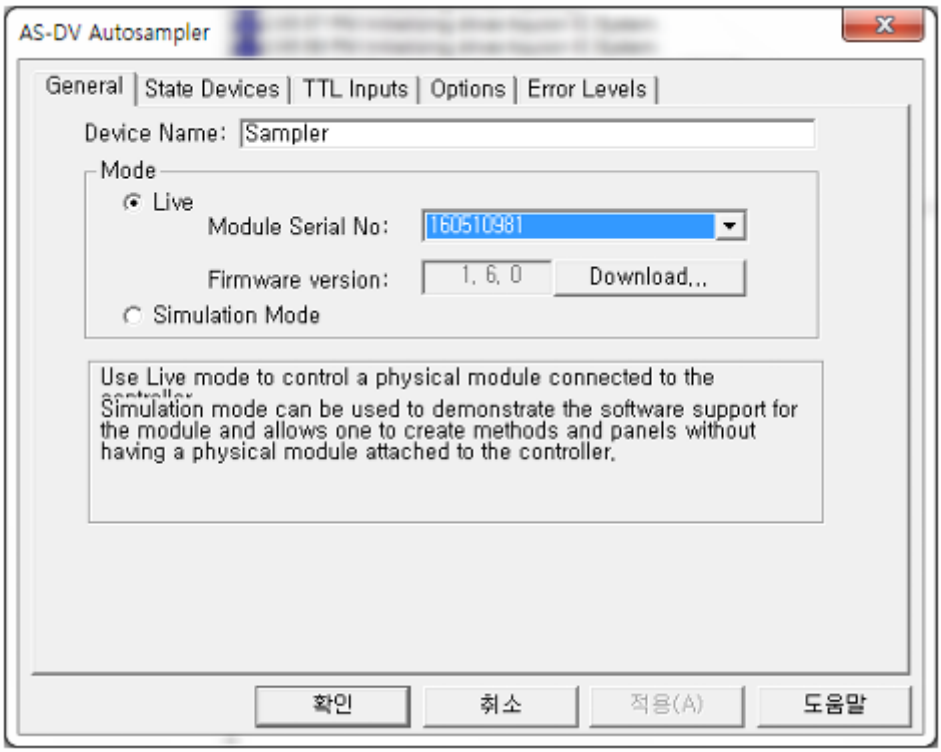
Barcha sozlamalar yakunlangach, konfiguratsiyani tugatish uchun pastdagi OK tugmasini bosing.



10) AS-HV Sampler yoki AS-DV Sampler qo‘shilganda, yuqorida bayon etilgan tartibga muvofiq xuddi shu amallarni bajaring.

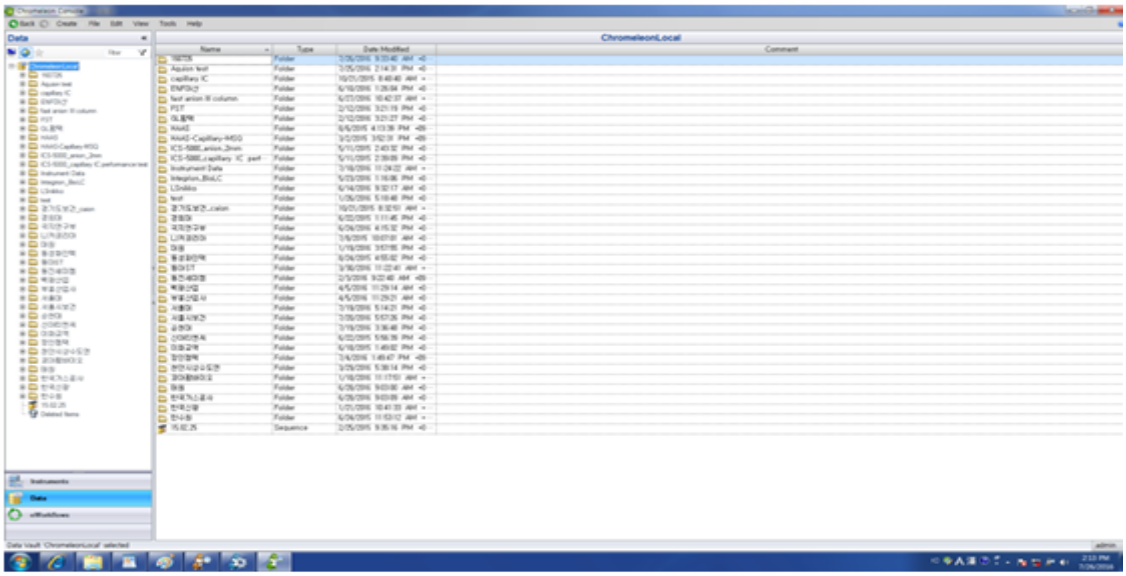


11) Modulning seriya raqamini tanlang, so'ngra konfiguratsiyani yakunlash uchun OK tugmasini bosing. Barcha sozlamalarni tugatgandan so'ng, konfiguratsiyani saqlash va chiqish uchun Yopish tugmasini bosing.



5. Chromeleonni ishga tushirish

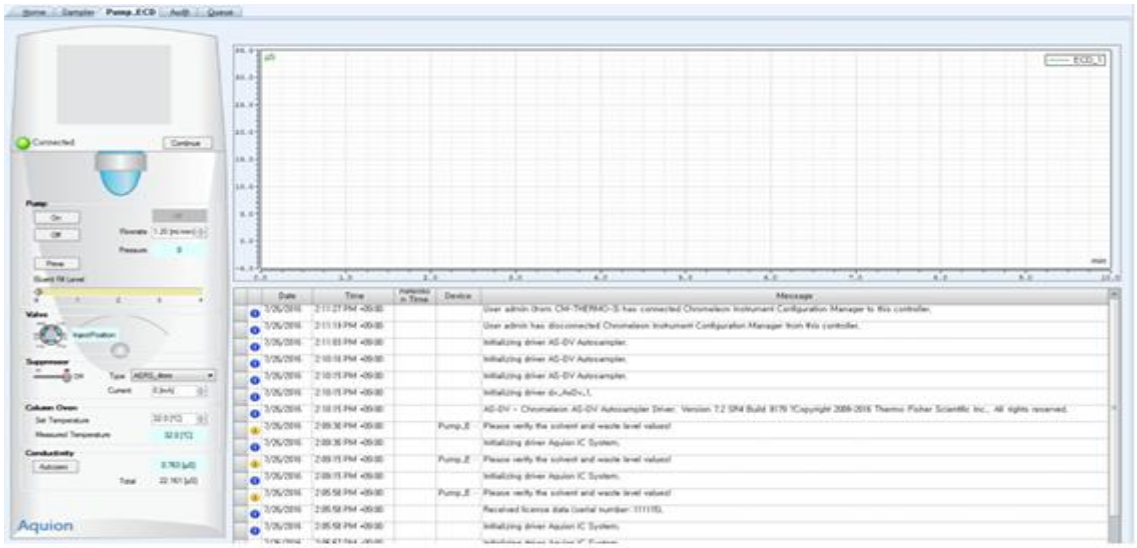
1) Chromeleon konsolini ochish uchun ish stolidagi Chromeleon 7 belgisini ikki marta bosing.

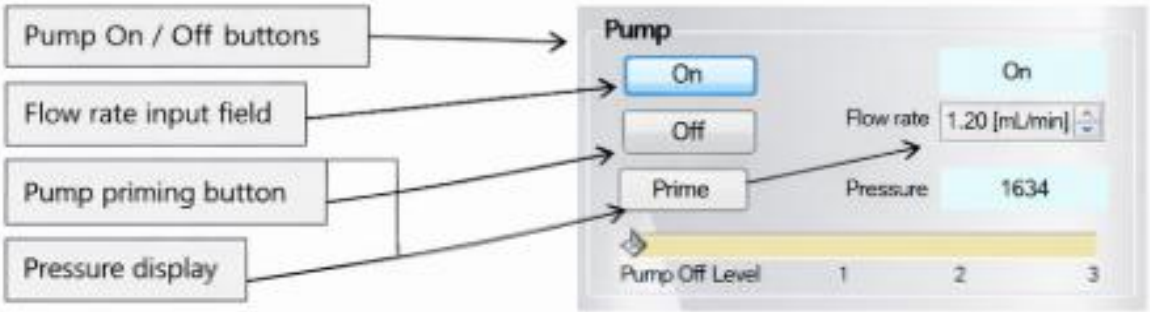


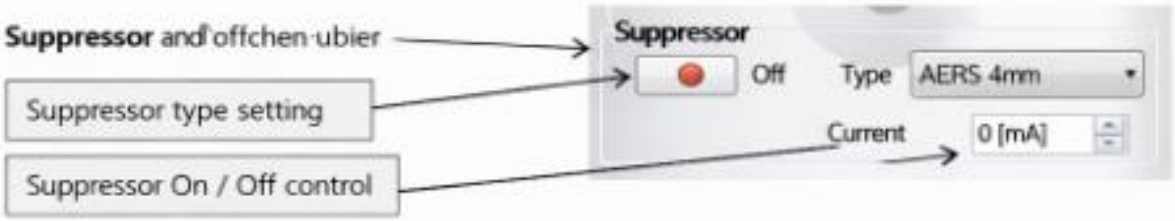
2) Ekranning pastki chap burchagida joylashgan Qurilmalar(Instruments) tugmasini bosing.

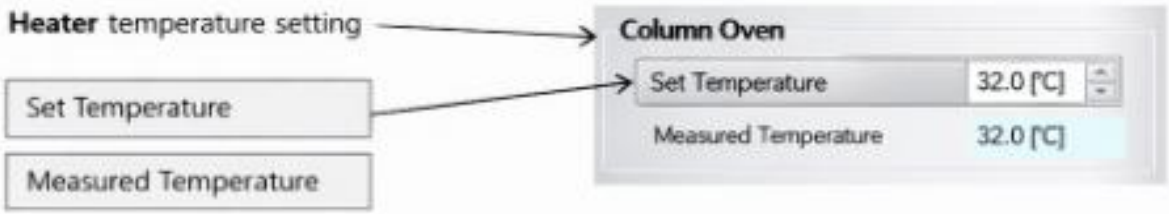


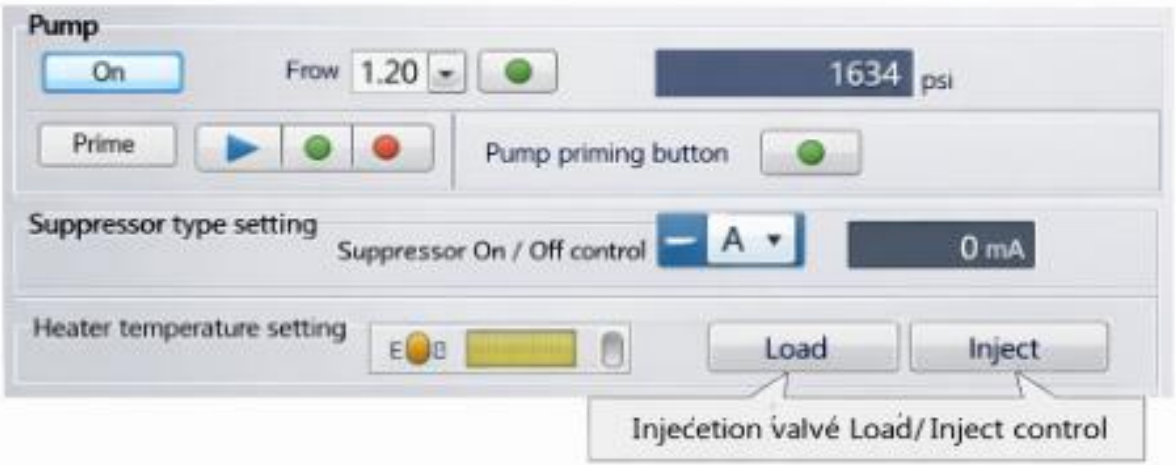
3) Qurilma tanlanganda, tizim boshqaruv paneli quyida ko'rsatilgandek paydo bo'ladi.



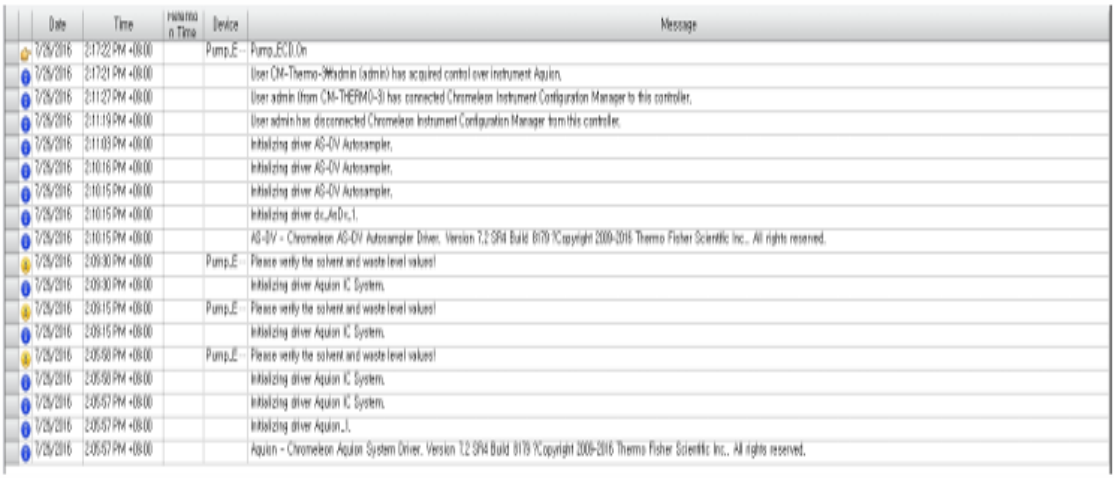

Pump On / Off buttons
Flow rate input field
Pump priming button
Pressure display
Pump
On
Off
Prime
On
Flow rate
1.20 [mL/min]
Pressure
1634
Pump Off Level
1
2
3


Suppressor and offchen ubier
Suppressor type setting
Suppressor On / Off control
Suppressor
Off
Type
AERS 4mm
Current
0 [mA]


Heater temperature setting
Set Temperature
Measured Temperature
Column Oven
Set Temperature
32.0 [°C]
Measured Temperature
32.0 [°C]

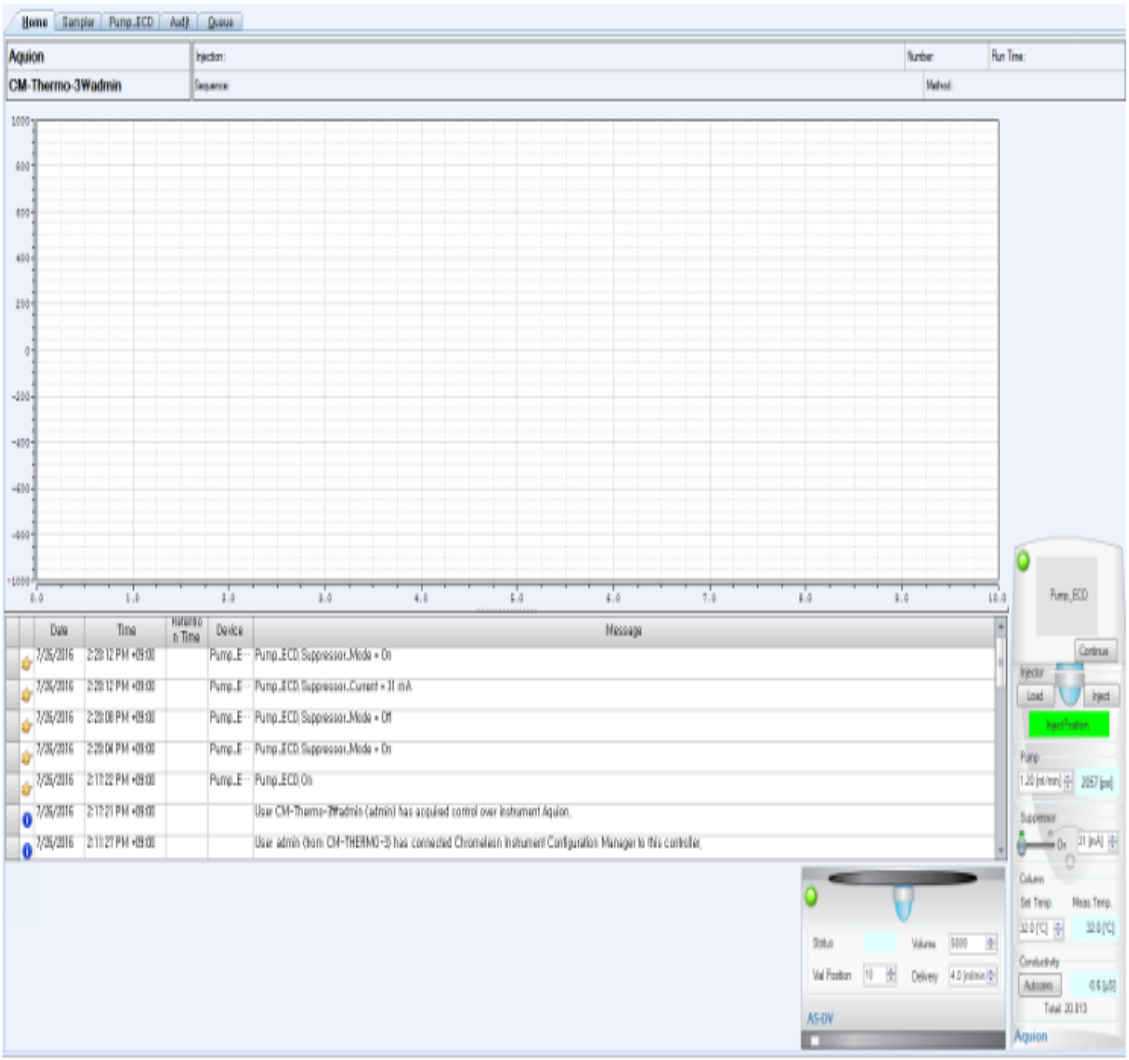

Pump
On
Frow
1.20
1634
psi
Prime
Pump priming button
Suppressor type setting
Suppressor On / Off control
A
0 mA
Heater temperature setting
Load
Inject
Injection valve Load/Inject control

Qurilma buyruqlari va tizim holatini ko'rsatuvchi Audit Trail bo'limi



Date	Time	Retention Time	Device	Message
7/26/2016	2:17:22 PM +08:00		Pump_E...	Pump_ECD On
7/26/2016	2:17:21 PM +08:00			User CM-Thermo-3\Wadmin (admin) has acquired control over instrument Aquion.
7/26/2016	2:11:27 PM +08:00			User admin (from CM-THERMO-3) has connected Chromeleon Instrument Configuration Manager to this controller.
7/26/2016	2:11:19 PM +08:00			User admin has disconnected Chromeleon Instrument Configuration Manager from this controller.
7/26/2016	2:11:03 PM +08:00			Initializing driver AS-DV Autosampler.
7/26/2016	2:10:16 PM +08:00			Initializing driver AS-DV Autosampler.
7/26/2016	2:10:15 PM +08:00			Initializing driver AS-DV Autosampler.
7/26/2016	2:10:15 PM +08:00			Initializing driver dx_AxD=_1.
7/26/2016	2:10:15 PM +08:00			AS-DV - Chromeleon AS-DV Autosampler Driver. Version 7.2 SP4 Build 8179 ?Copyright 2009-2016 Thermo Fisher Scientific Inc.. All rights reserved.
7/26/2016	2:09:30 PM +08:00		Pump_E...	Please verify the solvent and waste level values!
7/26/2016	2:09:30 PM +08:00			Initializing driver Aquion IC System.
7/26/2016	2:09:15 PM +08:00		Pump_E...	Please verify the solvent and waste level values!
7/26/2016	2:09:15 PM +08:00			Initializing driver Aquion IC System.
7/26/2016	2:05:58 PM +08:00		Pump_E...	Please verify the solvent and waste level values!
7/26/2016	2:05:58 PM +08:00			Initializing driver Aquion IC System.
7/26/2016	2:05:57 PM +08:00			Initializing driver Aquion IC System.
7/26/2016	2:05:57 PM +08:00			Initializing driver Aquion_1.
7/26/2016	2:05:57 PM +08:00			Aquion - Chromeleon Aquion System Driver. Version 7.2 SP4 Build 8179 ?Copyright 2006-2016 Thermo Fisher Scientific Inc.. All rights reserved.

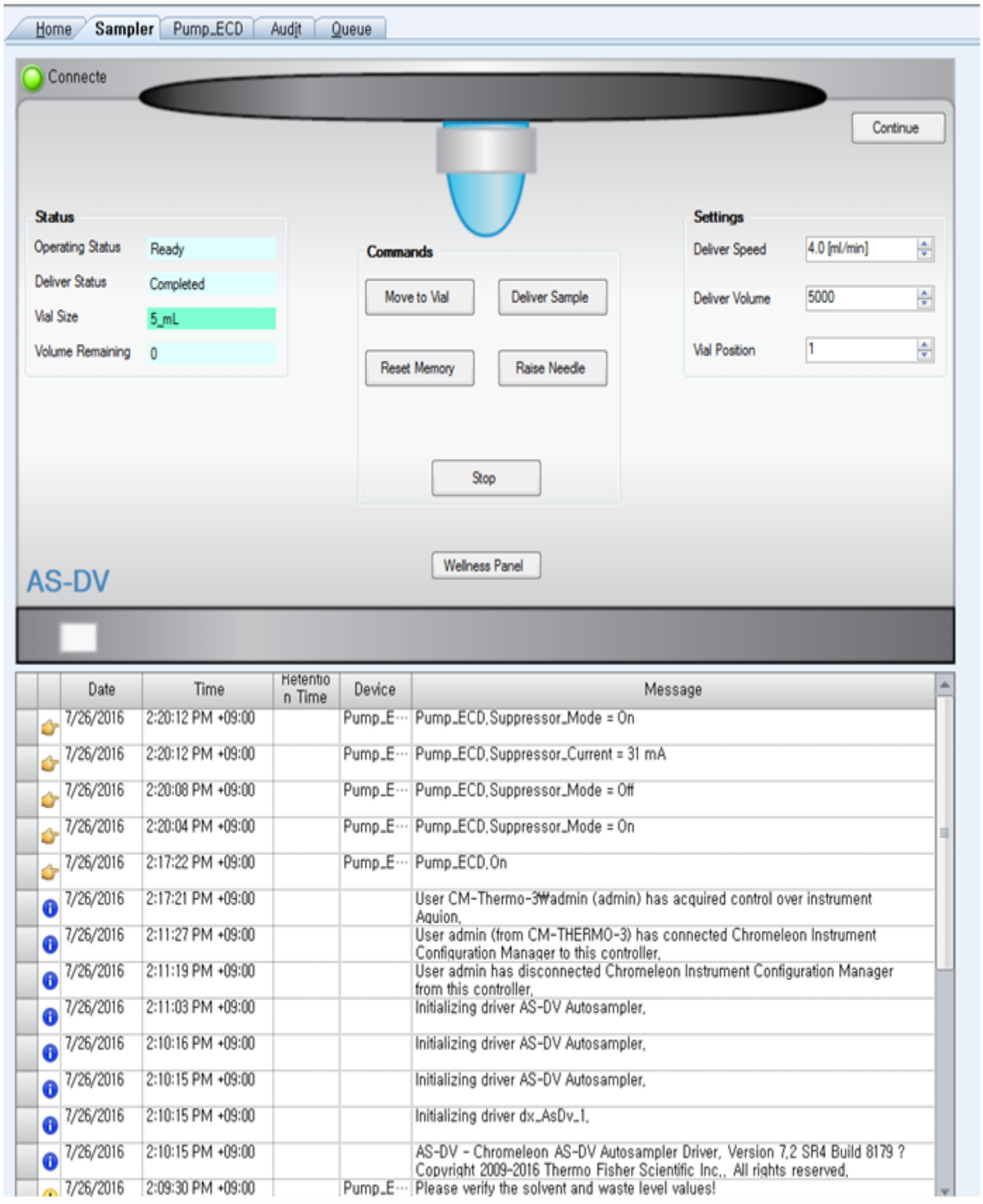
Xromatografik ma'lumotlarni ko'rsatuvchi xromatogramma displey maydoni

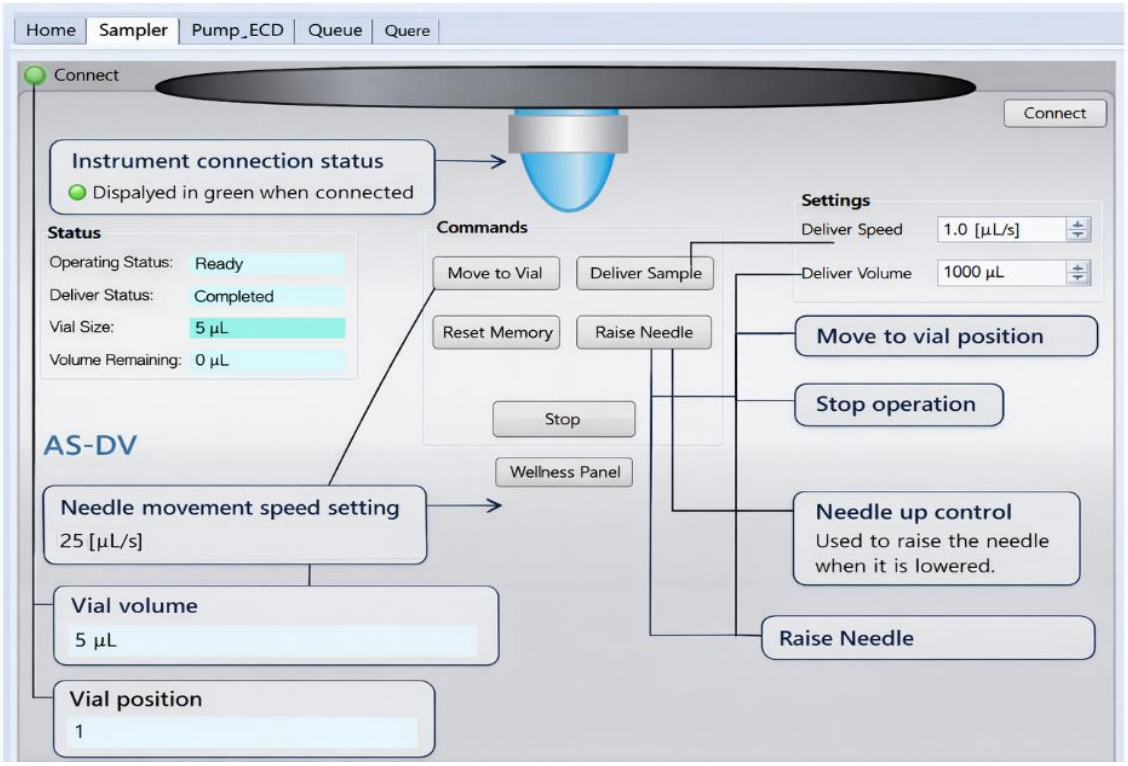


AS-DV Sampler boshqaruv paneli

AS-DV sampler boshqaruv paneli samplerning joriy ishlash holatini ko‘rsatadi va uni boshqarish imkonini beradi.

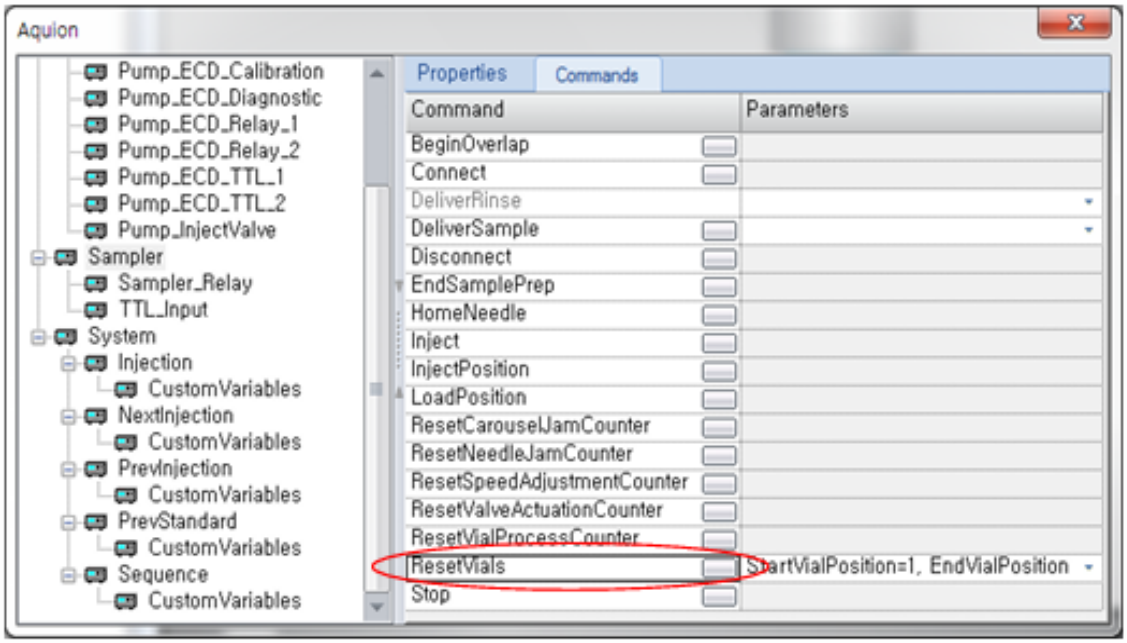
Panel orqali foydalanuvchi ish jarayonida flakon joylashuvi, igna (needle) holati va samplerning umumiy holatini kuzatishi mumkin.





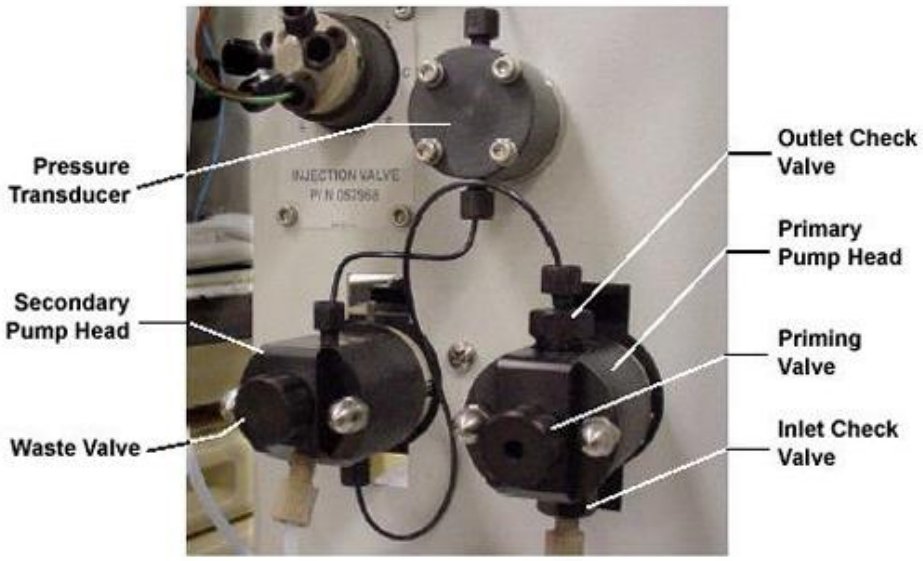
AS-DV xotira funksiyasi bilan jihozlangan. Barcha namunalar tahlil qilingandan so‘ng, xotirani tiklash uchun Reset All tugmasini bosish kerak.

Muqobil ravishda, qurilmani boshqarish oynasida F8 tugmasini bosish orqali ham xotirani tiklash mumkin.



4) Prime (to‘ldirish/primlash)

Primlash eluent idishi bilan nasos orasidagi havo pufakchalarini olib tashlash uchun amalga oshiriladi..



(1) 1) Rasmda ko‘rsatilganidek, primlashga tayyorgarlik uchun ikkinchi nasos boshchasining chiqindi klapanini 2–3 marta aylantirib oching.

2) Chromeleon 7 dasturining Instrument panelida Prime tugmasini bosing. Ekranning yuqori qismida sariq ogohlantirish xabari paydo bo‘ladi. Primlashni boshlash uchun OK tugmasini bosing.



(3) 3) Primlash jarayonida chiqindi liniyasini kuzatib, barcha havo pufakchalari chiqarib yuborilganini tasdiqlang. Primlash tugagach (odatda 5 daqiqa ichida), primlashni to‘xtatish uchun Pump Off tugmasini bosing. So‘ng chiqindi klapanini soat miliga teskari yo‘nalishda burab yopib, primlash jarayonini yakunlang.

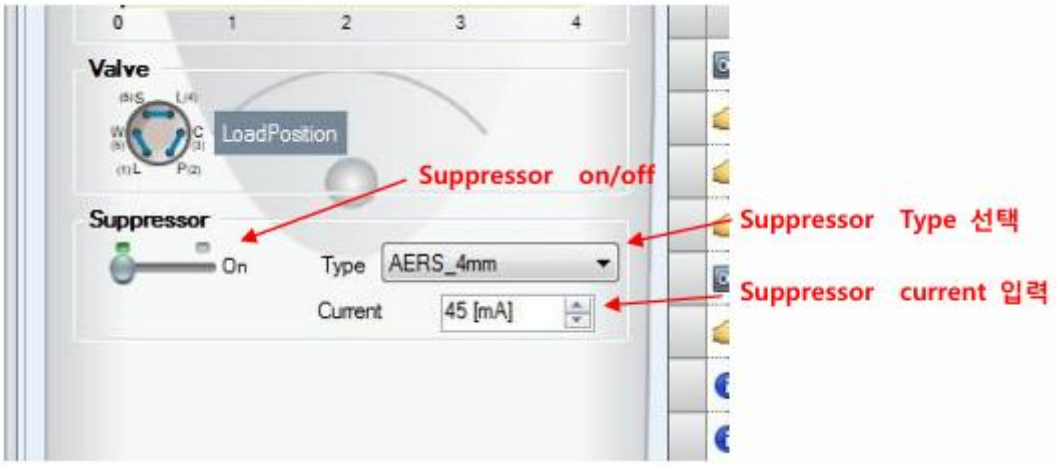
4) Nasosni yoqishdan oldin kerakli oqim tezligini kiriting, so‘ng On tugmasini bosing. Oqim tezligi kolonkaga bog‘liq bo‘lgani uchun, mos qiymatlarni tanlashda II-1 bo‘lim — Tahlilga tayyorgarlik bo'limiga qarang.

5) Nasosni yoqib, bosim qiymati barqarorlashganini tasdiqlagandan so‘ng,

suppressor turini va suppressor oqimini sozlang.

Suppressorni sozlashning batafsil qiymatlari II-1 bo‘lim — Tahlilga tayyorgarlik da keltirilgan.

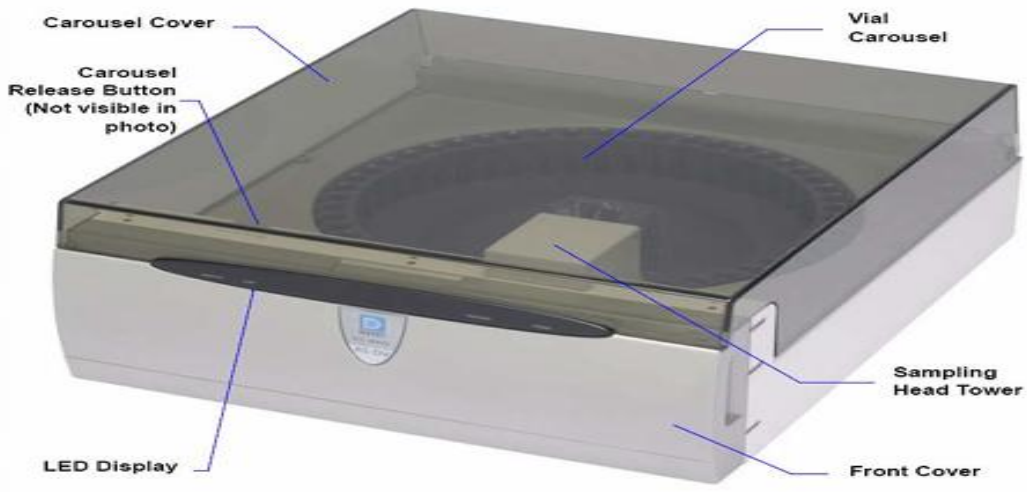
Suppressorning normal ishlashi regeneratsiya chiqish liniyasida gaz pufakchalari hosil bo‘lishi orqali tasdiqlanadi, bu elektrolitik regeneratsiya jarayoni to‘g‘ri ishlayotganini ko‘rsatadi.



6) Namunalarni ketma-ket AS-DV ga joylashtiring.

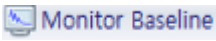
AS-DV Carousel Release tugmasi bosilganda, namuna karuseli oson harakatlantirilishi mumkin.

Agar Rinse opsiyasi yoqilgan bo‘lsa, namunalar har bir flakon joyiga joylashtirilmaydi; aksincha, chayish funksiyasi ishlatilganda faqat belgilangan flakonlarni to‘ldirish kerak.

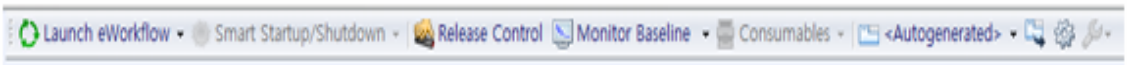


Namunalar tayyorlangandan so‘ng, Carousel Release tugmasini yana bir bor bosishni unutmang. Agar bu tugma bosilmasa, tahlil boshlanmaydi.



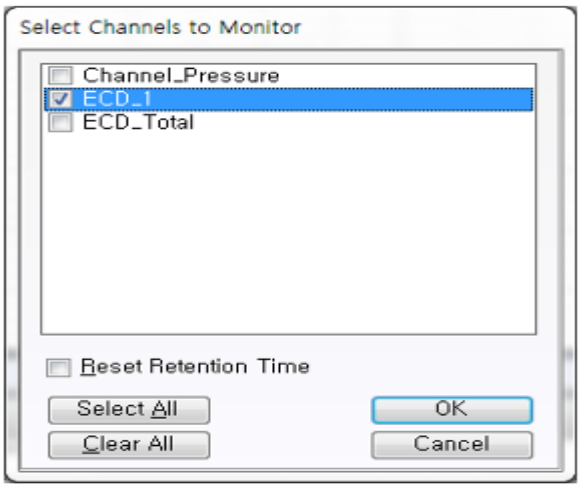
7) Bazoviychiziqni ko'rish uchun Monitor Baseline(

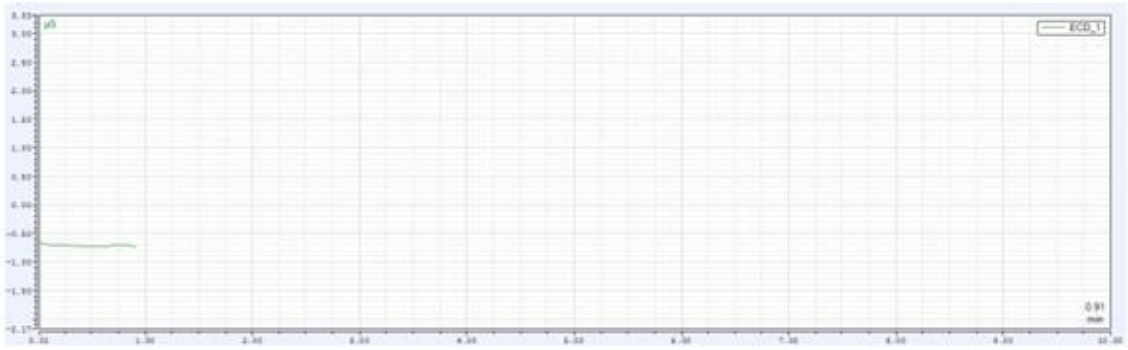
) tugmasini



bosing.

Quyidagi ekran paydo bo'lganda, ECD_1 tugmasini bosing, so'ngra bazoviy chiziqni ko'rsatish uchun OK tugmasini bosing.



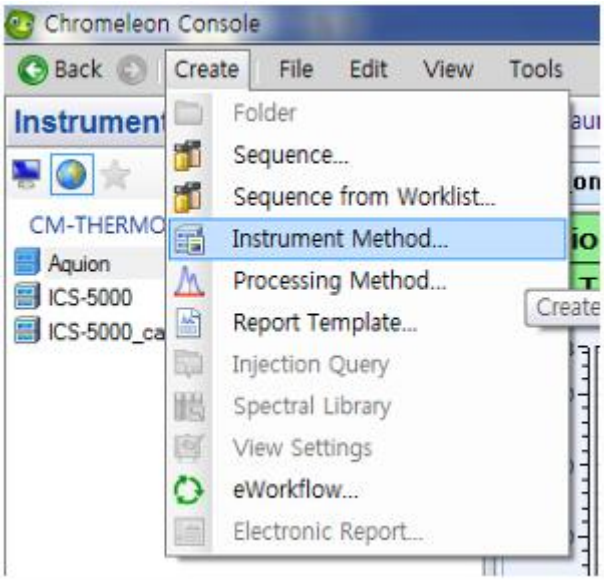


Bazoviy chiziqni ko'rsatishni to'xtatish uchun yana bir bor Monitor Baseline(

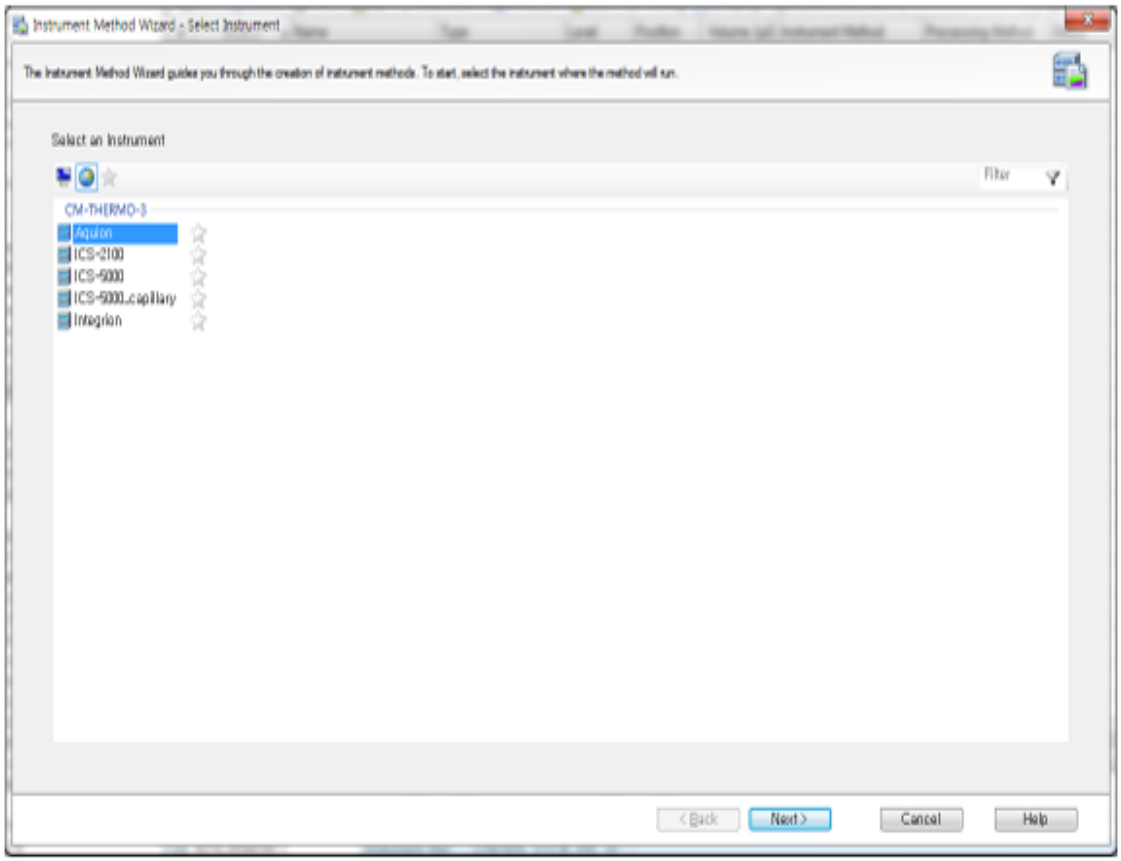
Monitor Baseline) tugmasini bosing.

6. Qurilma usulini (Instrument Method) qanday yaratish kerak

1) Create → Qurilma usuli(Instrument Method) tugmasini bosing.

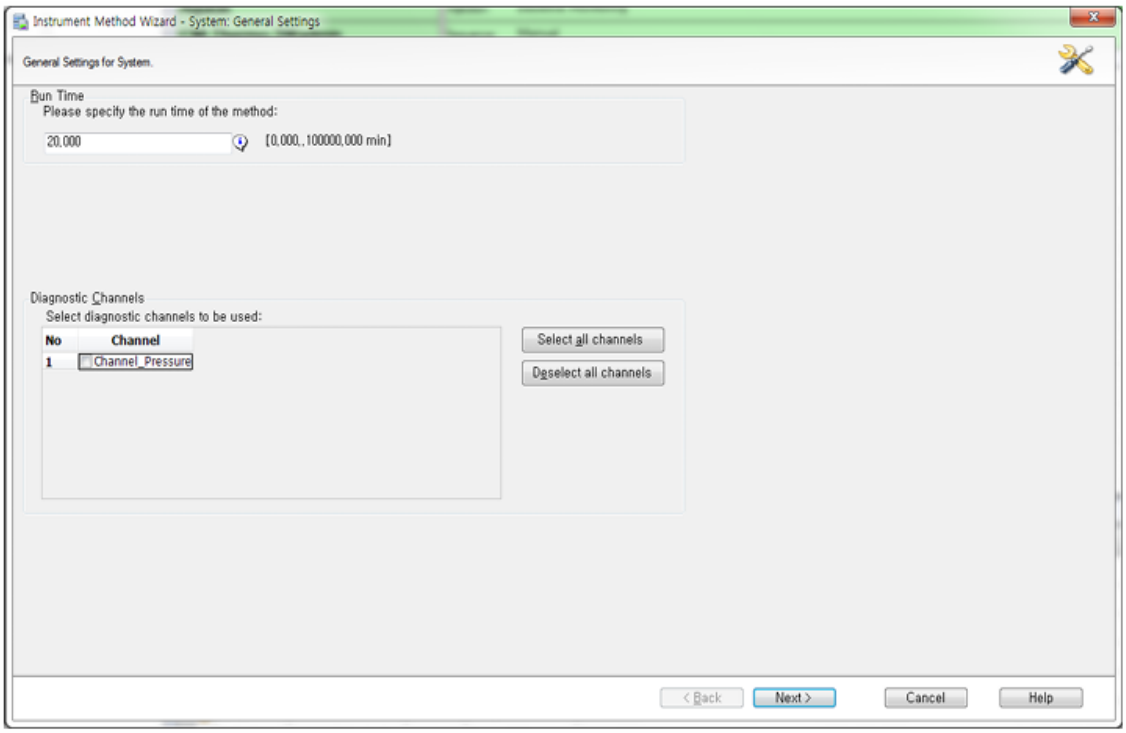


2) AQUION-ni tanlang, so'ngra “Keyingi(Next)” tugmasini bosing.

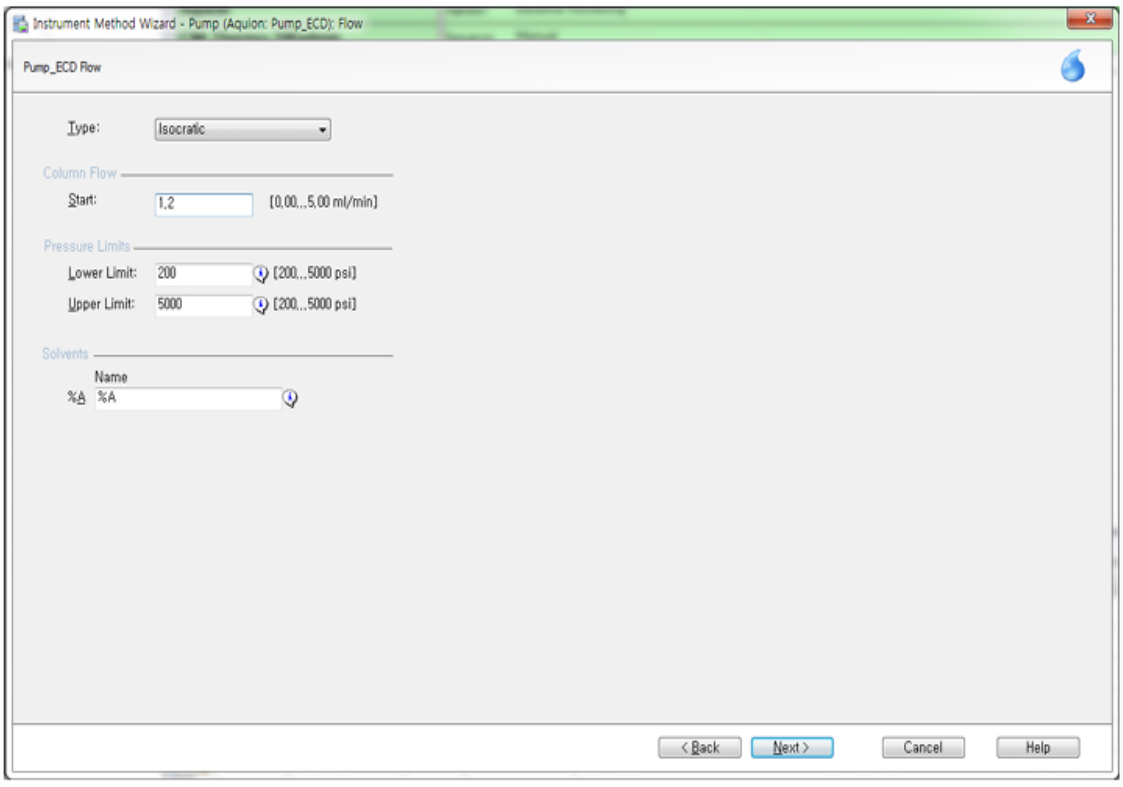


3) Keyingi oynada Run Time maydoniga kerakli tahlil vaqtini kiriting.

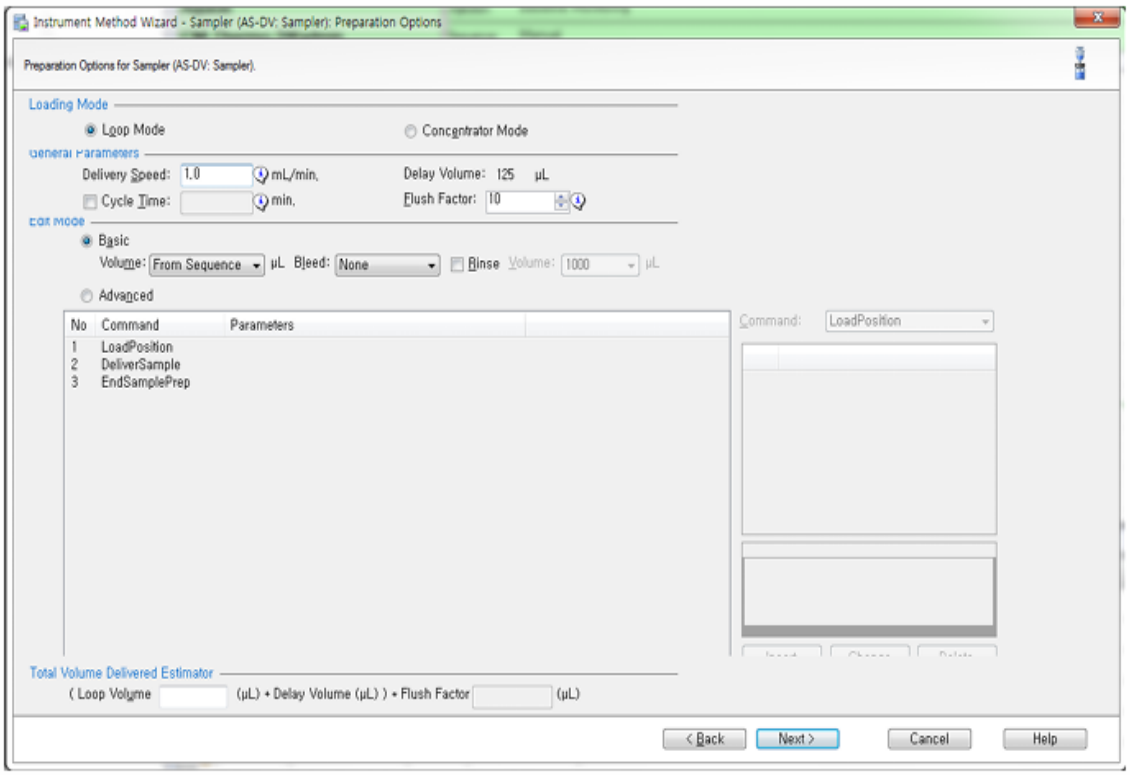
Agar bosim signalini kuzatmoqchi bo‘lsangiz, Channel Pressure opsiyasini belgilang.



4) Column Flow maydoniga oqim tezligini kiriting.



5) Quyidagi oyna Avto Sampler konfiguratsiyasi ekranidir.



(1) Loop Mode rejimida ishlating.

(2) Basic Mode rejimida ishlating.

(3) Rinse Volume ni sozlang (standart hajm: 1000 µL). Kerakli chayish hajmini kiriting.

(4) Flush Factor yuborilgan namuna hajmi necha marta yuvilishini (flush qilinishini) bildiradi.

Yuboriladigan namuna hajmi Delay Volume va Sample Volume ga bog‘liq.

Namuna yuborish hajmi

= (Delay Volume + Sample Volume) × Flush Factor.

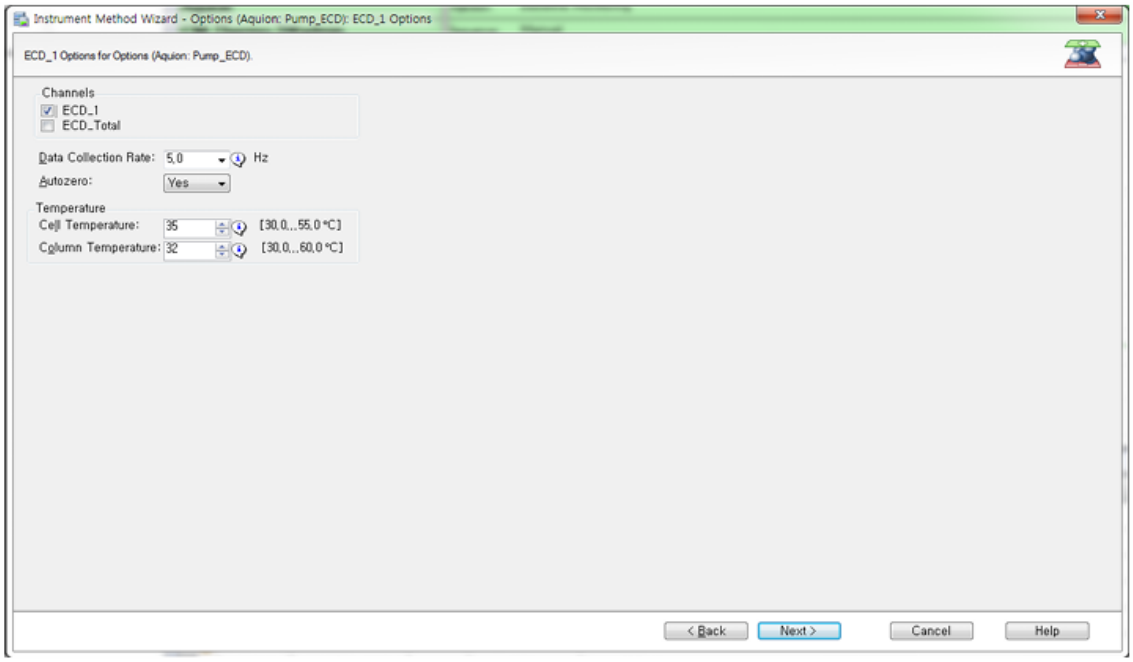
Misol:

1500 µL = (125 µL + 25 µL) × 10

(5) Bleed va Concentration Mode namunalarni konsentratsiyalashda (boyitishda) qo‘llaniladi.

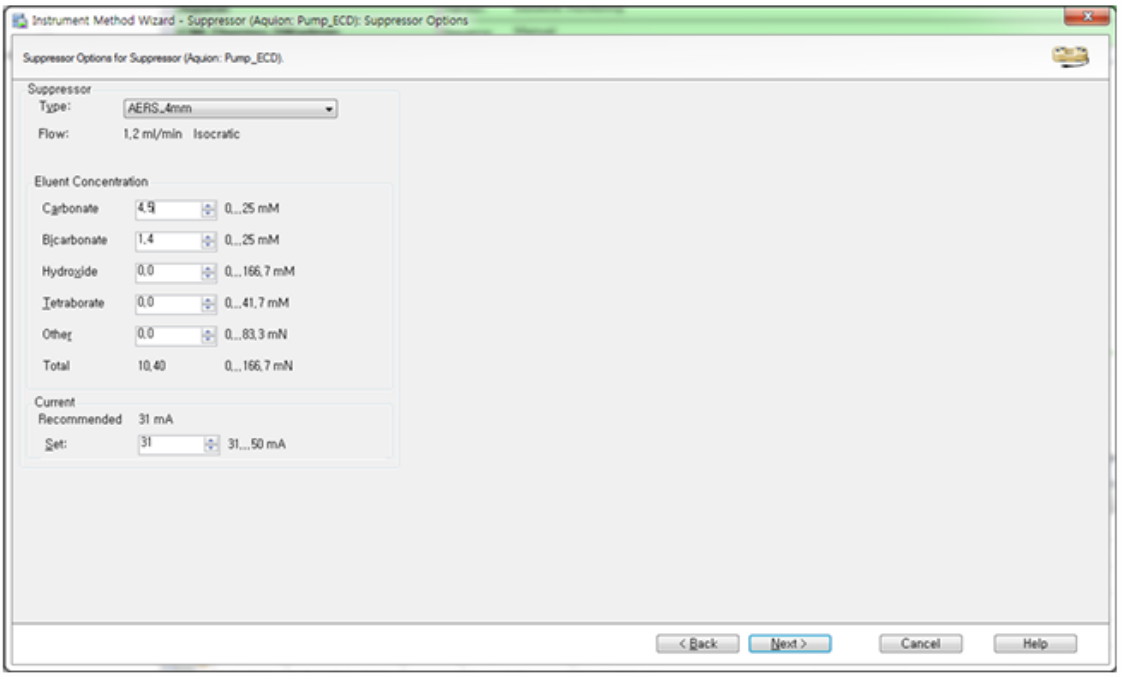
(6) Delivery Speed namunaning yuborilish tezligini belgilaydi. Ushbu tezlikni oshirish yuborish vaqtini qisqartirishi mumkin, biroq namuna yuborish vaqtida bosim oshishi ehtimoli mavjud. Shuning uchun oqim tezligini taxminan 1 mL/min darajada sozlash tavsiya etiladi.

6) Keyingi oynada yacheyka harorati va kolonka haroratini o'rnating.

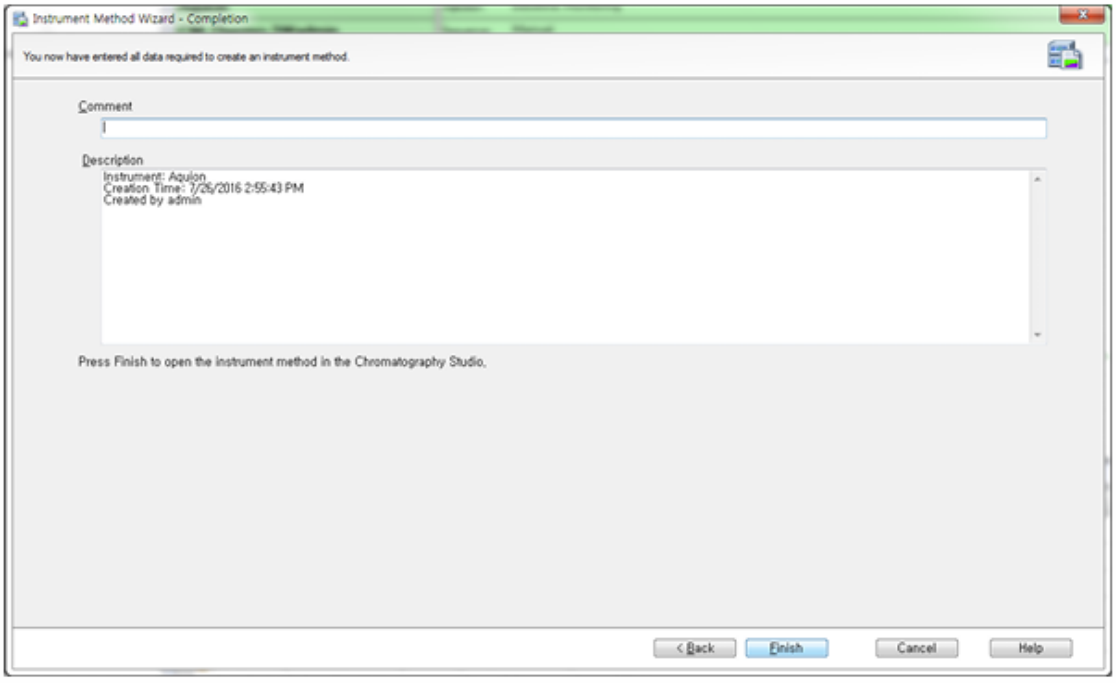


Quyida ko‘rsatilganidek, ECD_1 ni belgilang, so‘ng yacheyka harorati va kolonka haroratini sozlang. Next tugmasini bosing. Yacheyka va kolonka haroratini rasmda ko‘rsatilganidek o‘rnating.

7) Ushbu oynada suppressor sozlamalari amalga oshiriladi. Suppressor Type ni tanlang va karbonat konsentratsiyasini kiriting. Ushbu qiymatlar kiritilgandan so‘ng, tavsiya etilgan oqim qiymati avtomatik ravishda ko‘rsatiladi.

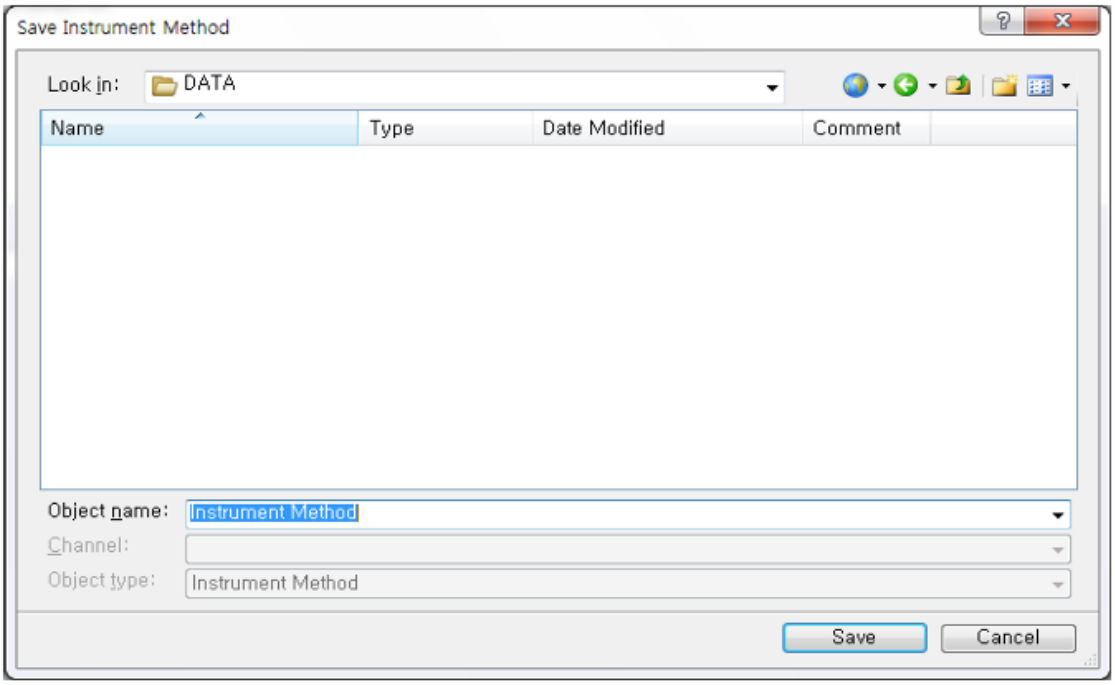


8) Keyingi bosqichda Comment (izoh) kiritish mumkin, biroq bu bosqichni o‘tkazib yuborish ham mumkin.

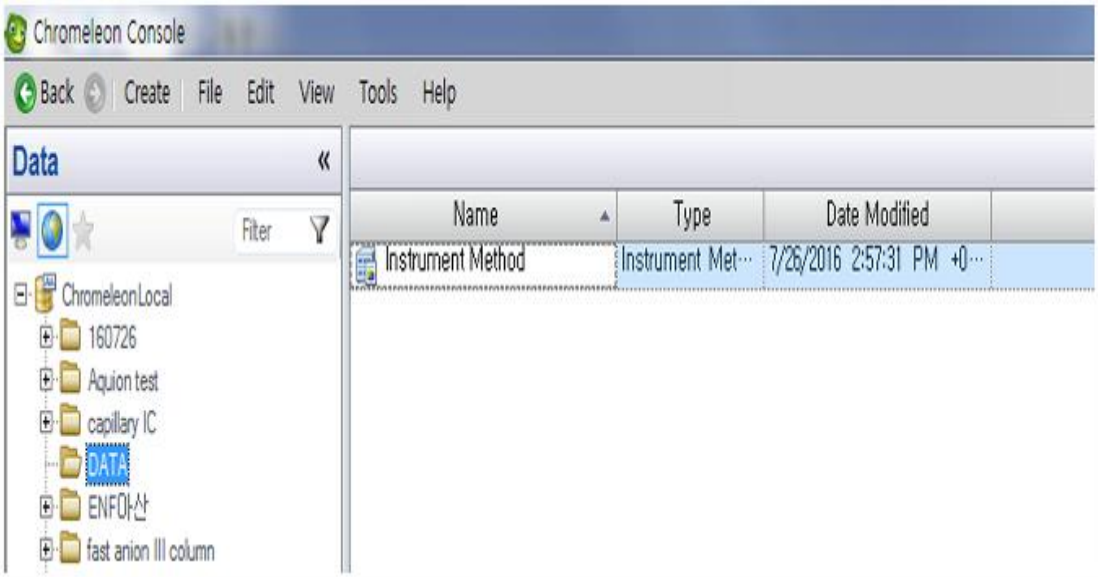


Sozlamalar kiritilgandan so‘ng, Finish tugmasini bosib, Program faylini yaratishni yakunlang.

9) Kerakli joyga dastur faylining nomini kiriting va saqlang.

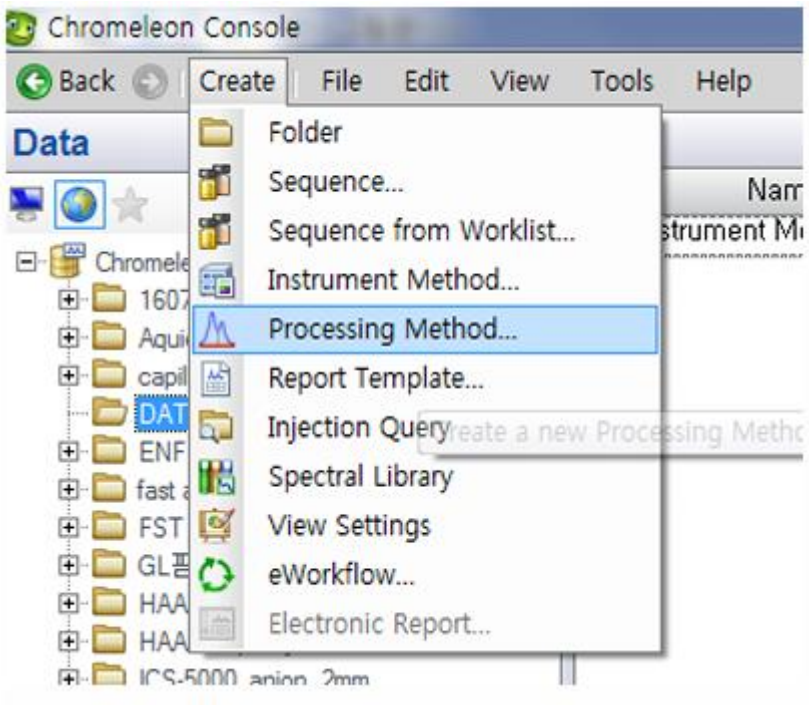


Quyidagi rasmda yaratilgan Instrument Method ko‘rsatilgan.

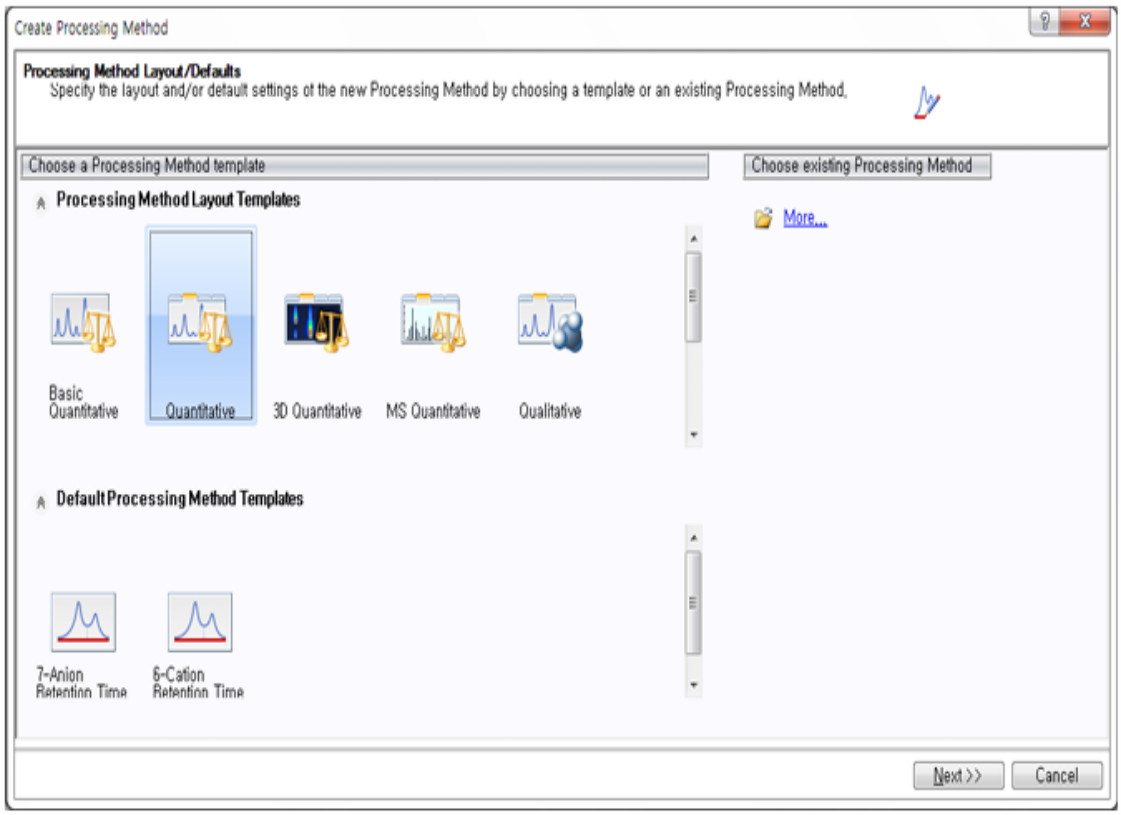


7. Processing Method File yaratish tartibi

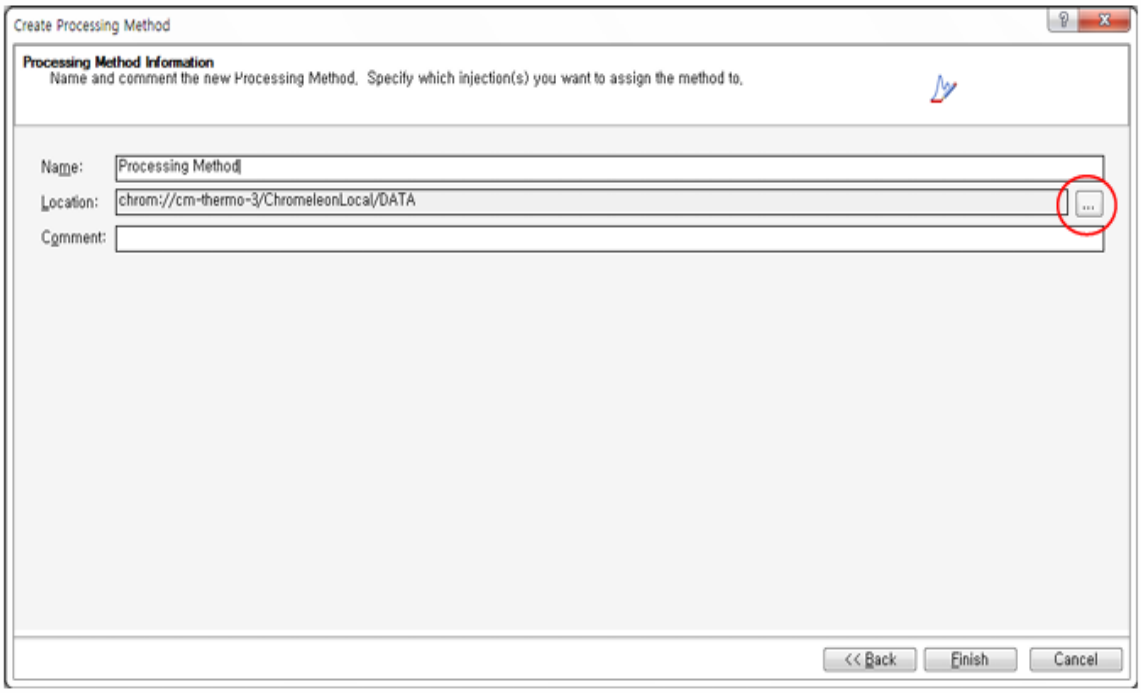
1) Create → Processing Method ni tanlang.

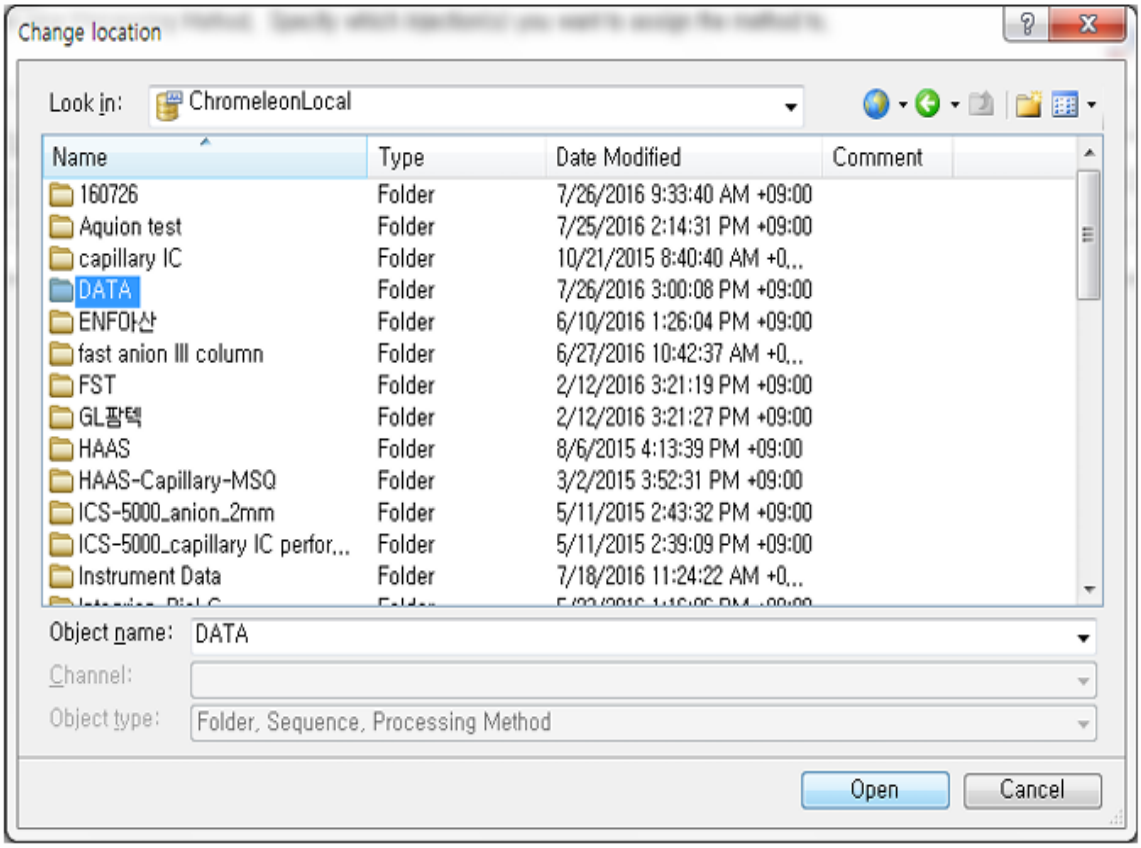


2) Keyingi ekranda Quantitative ni tanlang.



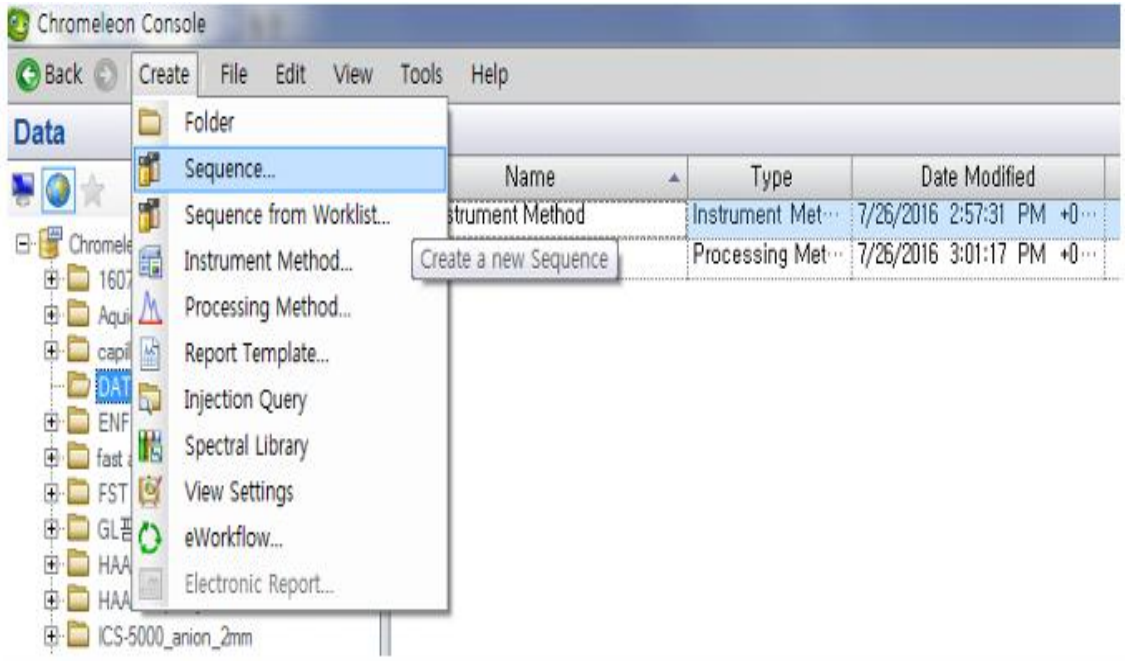
3) Nom kiritilgandan so‘ng, uni kerakli joyga saqlang. Processing Method faylini tahrirlash kalibrlash egri chizig‘i uchun STD o‘lchovi yakunlangandan keyin amalga oshiriladi.



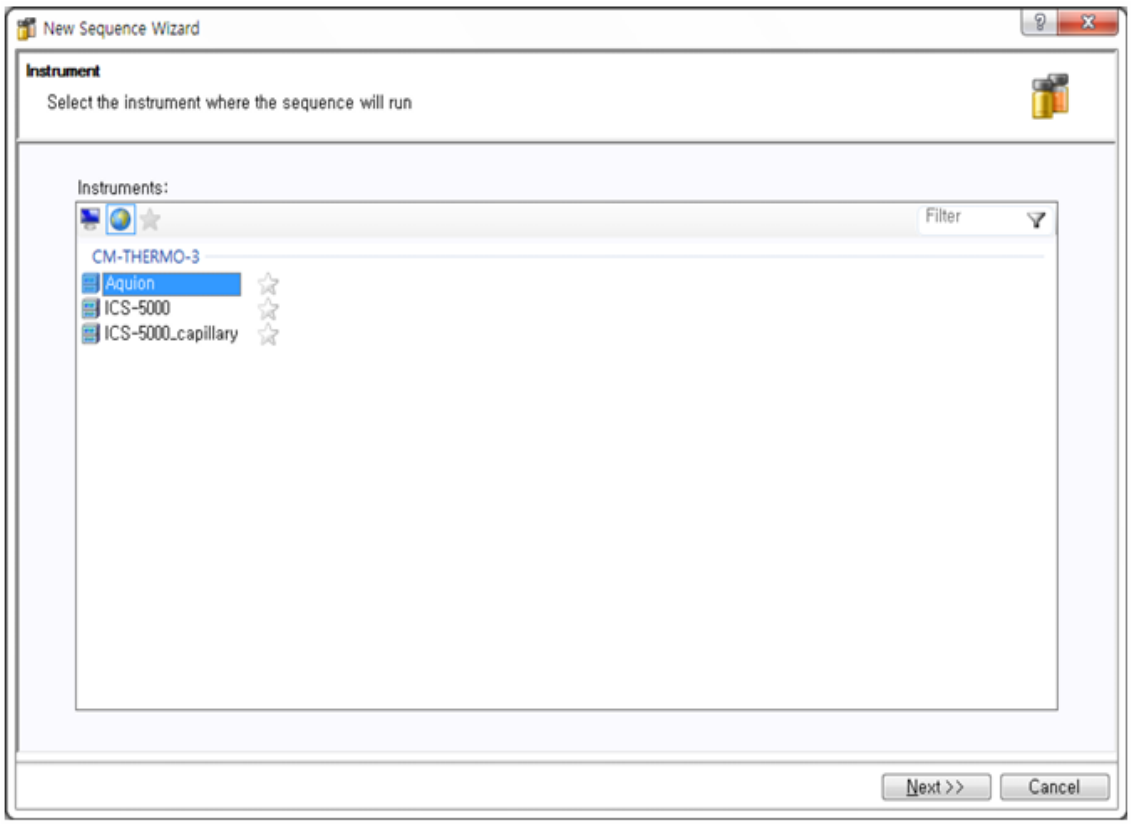


8. Sequence yaratish tartibi

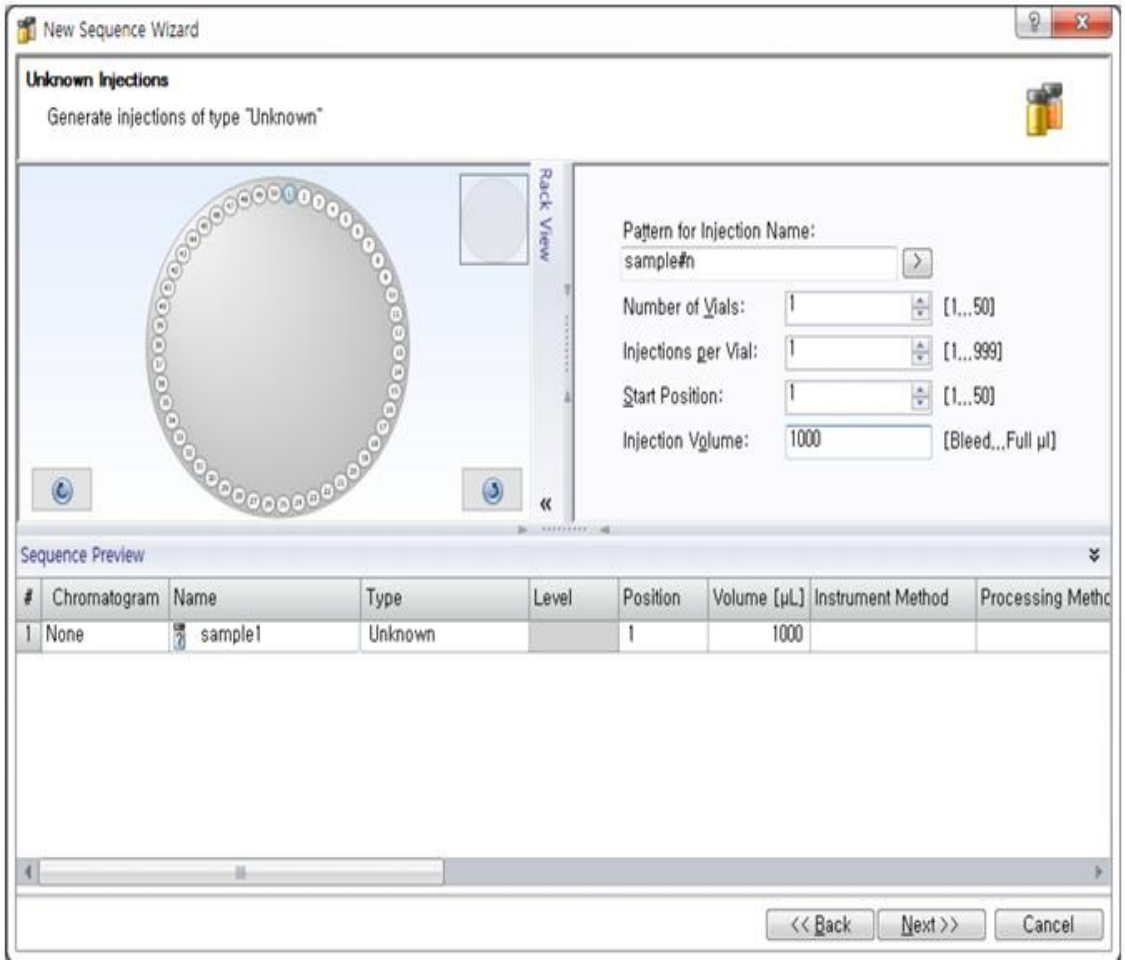
1)Create → Sequence ni tanlang.



2) Kerakli qurilmani tanlang va keyin Next tugmasini tanlang.



3) Namunalar sonini va namuna nomlarini belgilang.



Agar namuna nomini kiritib, yoniga #n qo‘shsangiz, Preview oynasida namunalar quyida ko‘rsatilganidek aks etishini ko‘rishingiz mumkin.

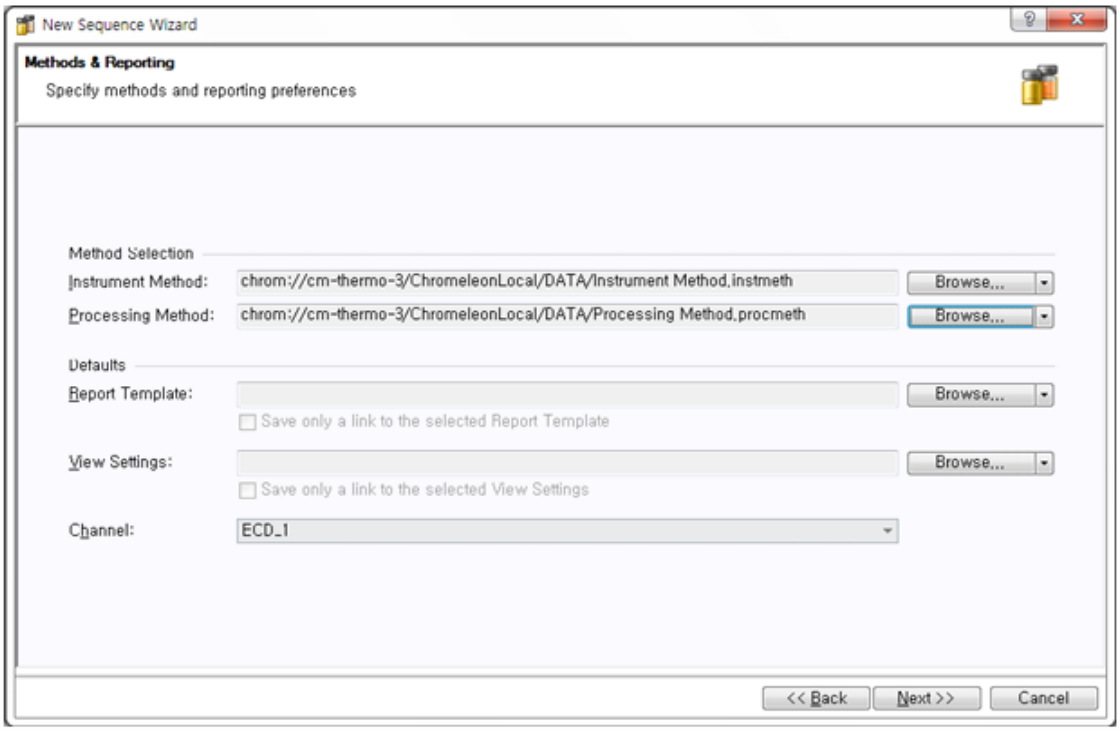
(1) Number of Vials: namunalar soni

(2) Injections per Vial: har bir flakondan yuborishlar soni

(3) Start Position: namuna ketma-ketligi boshlanadigan flakon pozitsiyasi

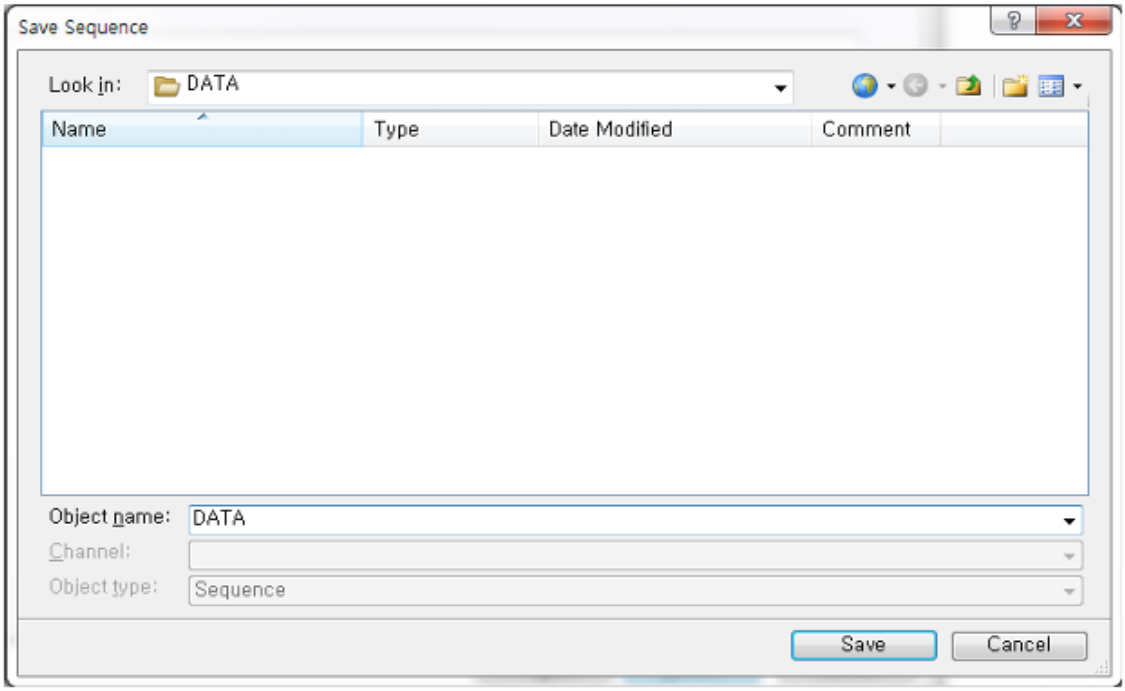
(4) Injection Volume: namuna yuborish hajmi

4) Tahlil uchun ishlatiladigan Processing Method, Instrument Method va Sequence fayllarini tanlang.



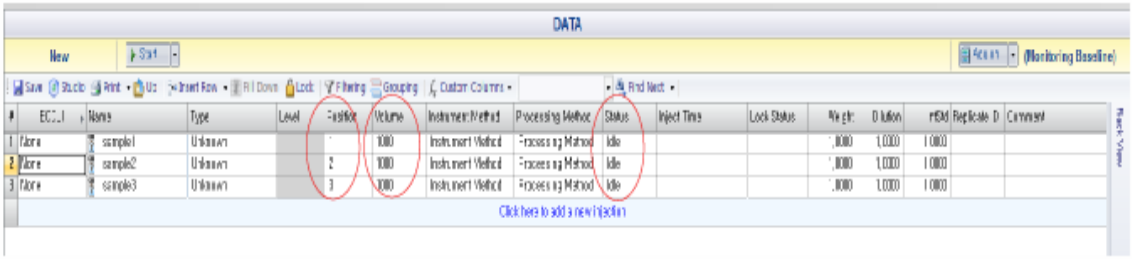
5) Namuna ma'lumotlarini va flakonlarning pozitsiyasini kiriting.

Barcha sozlamalarni bajargandan so'ng, Sequence faylini yaratishni yakunlash uchun Finish tugmasini bosing.



9. Namuna tahlilini boshlash

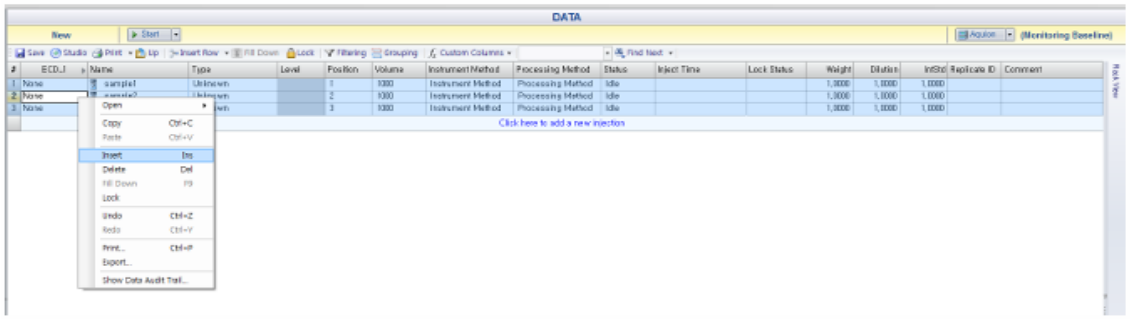
Yaratilgan Sequence faylini tanlang, so'ngra namunaviy tahlilni boshlash uchun Start tugmasini bosing.



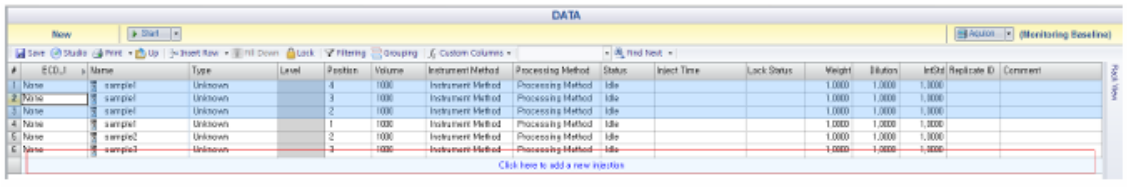
Keyingi oynada namunalar joylashuvi va samplerdagi pozitsiyasi to‘g‘ri o‘rnatilganini tekshiring, shuningdek namuna hajmi ham to‘g‘ri kiritilganini tasdiqlang.

Tahlil faqat Status, Idle holatida bajarilishi mumkin, shuning uchun tahlilni boshlashdan oldin holat Idle ekanligini tekshiring.

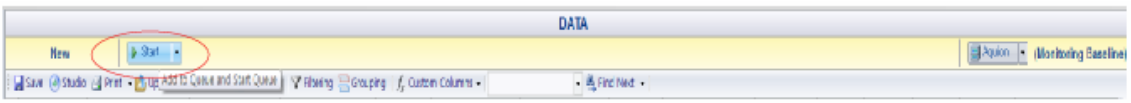
Agar tahlilni vaqtincha to‘xtatib turmoqchi bo‘lsangiz, Finished ni tanlab, saqlang. Namuna qo‘shish uchun rasmda ko‘rsatilganidek namunani tanlang va Insert tugmasini bosib, yangi namuna qo‘shing.



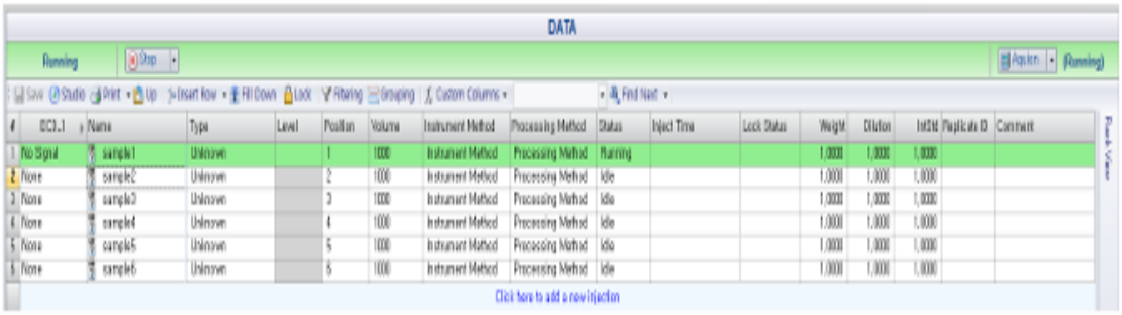
Agar namunalarni tezda qo'shmoqchi bo'lsangiz, quyidagi bo'sh qatorni bosing. Keyin namunalar soni bir vaqtning o'zida bitta qatorga ko'payadi.



Agar siz namunalarni o'lchashni xohlasangiz, "Start" tugmasini bosing.



"Start" tugmasini bosganingizda, namunaviy holat quyida ko'rsatilgandek "Running" ga o'zgaradi.



10. Ma'lumotlarni qayta ishlash

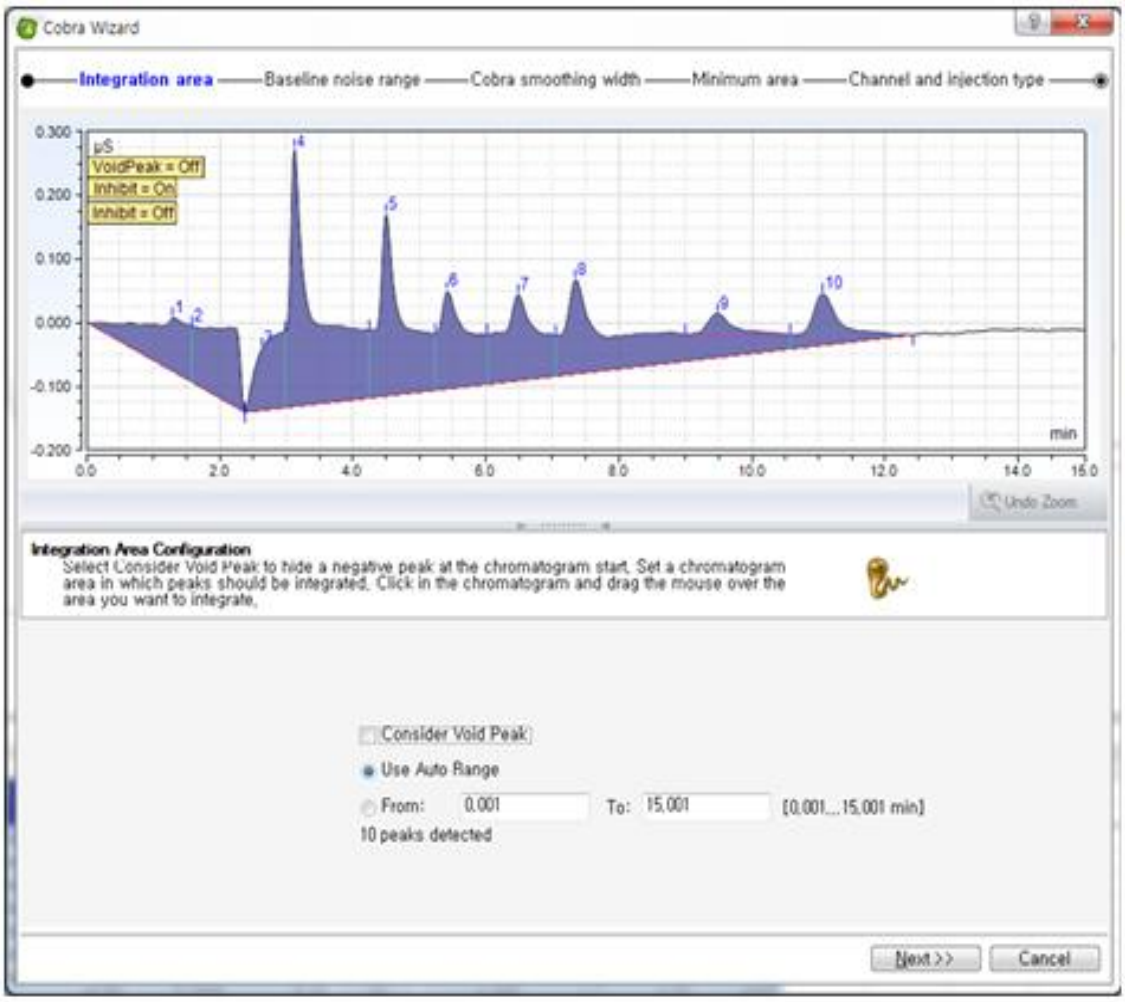
1) Avval tahlil yakunlangan standart ma'lumotlarni ikki marta bosing.



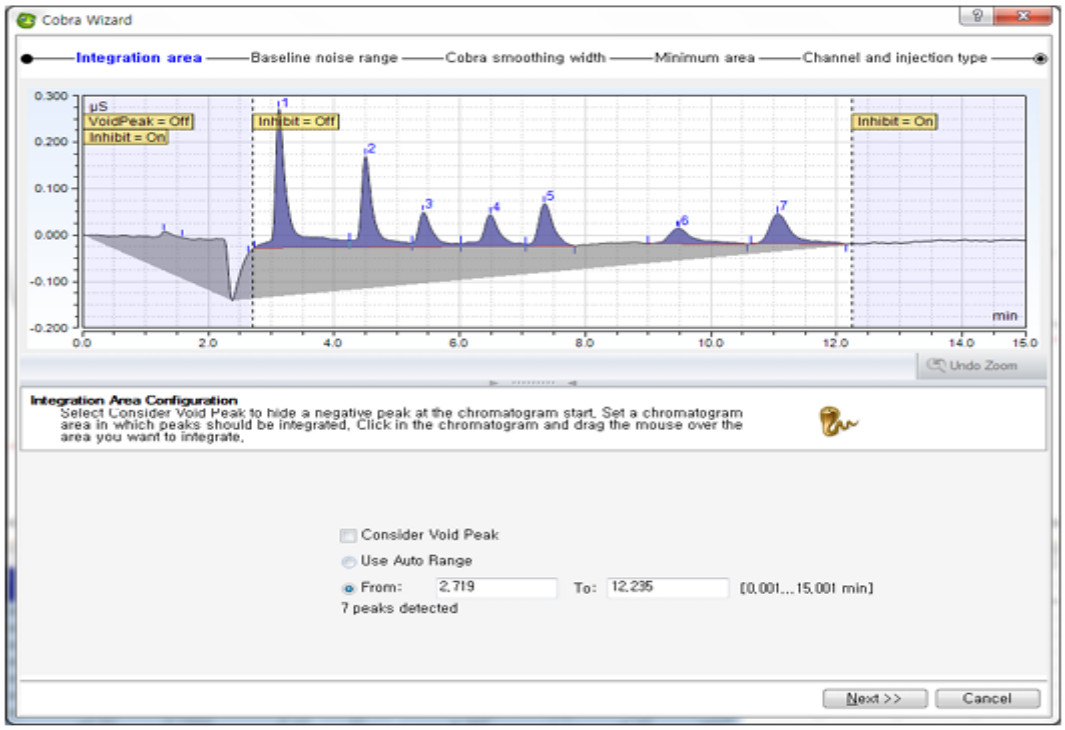
2) Quyida ko'rsatilganidek, Processingni tanlang va keyin [

]ni tanlang.

3) Quyidagi wizard yaratildi.

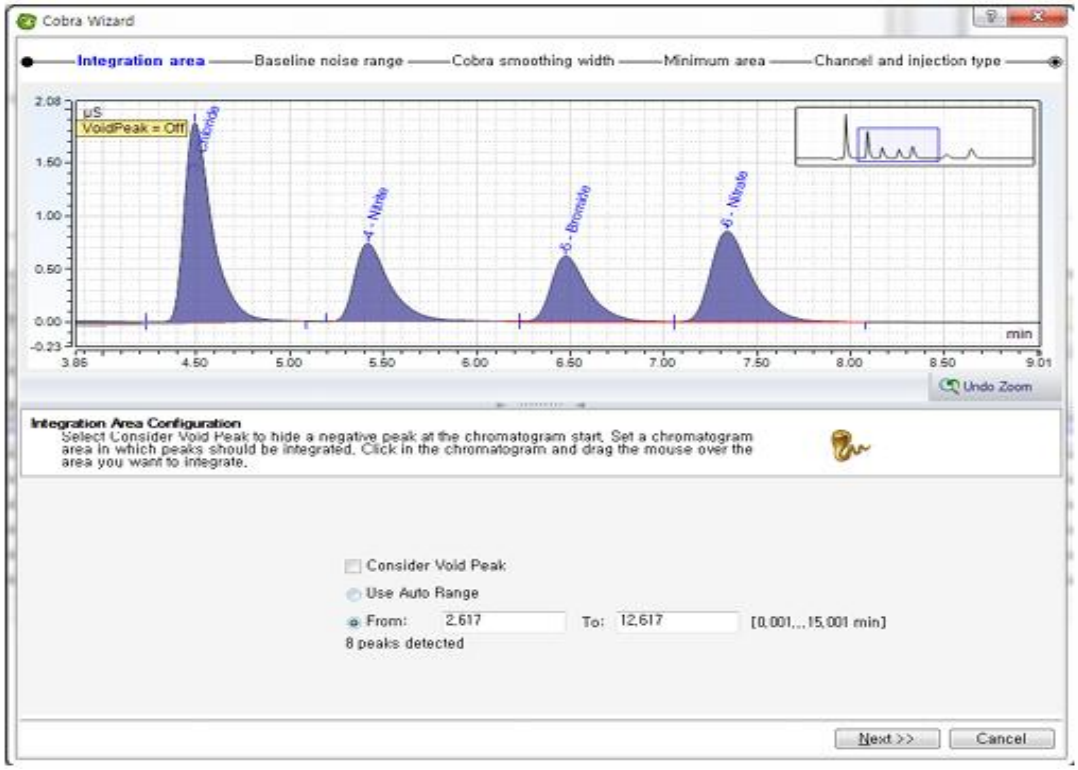


4) Integratsiya qilmoqchi bo‘lgan hududni belgilang.



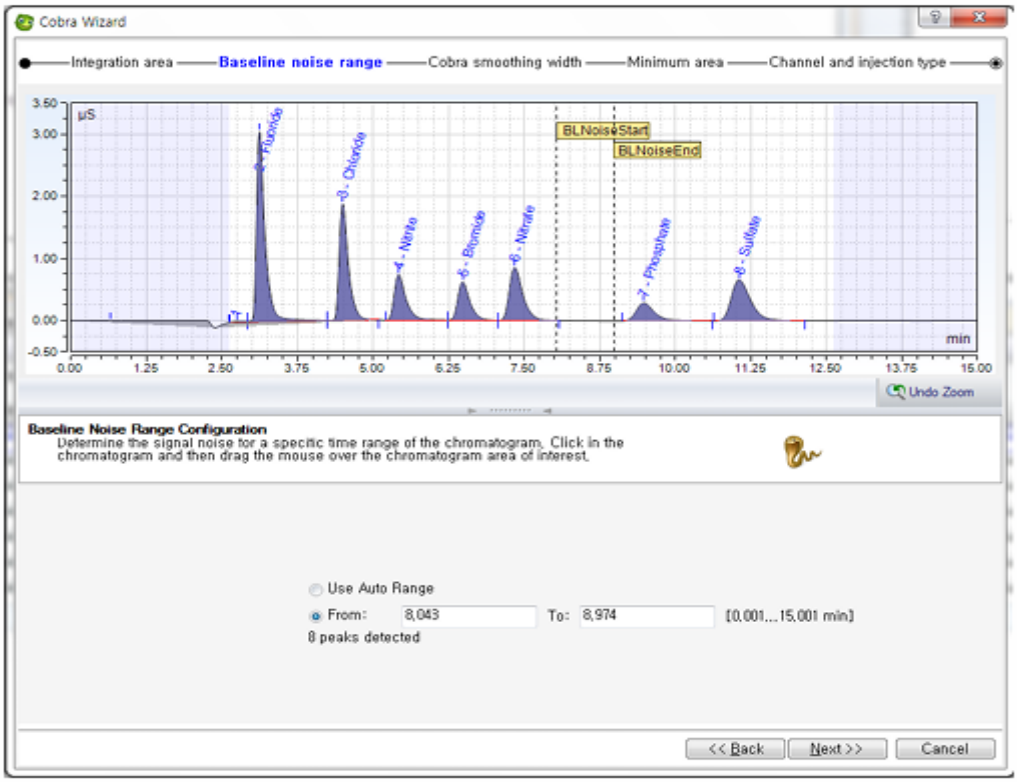
5) Zoom key tugmasidan foydalanib, eng tor peak(cho’qqi)ni torting.

6) Natija quyida ko'rsatilgandek ko'rinadi.

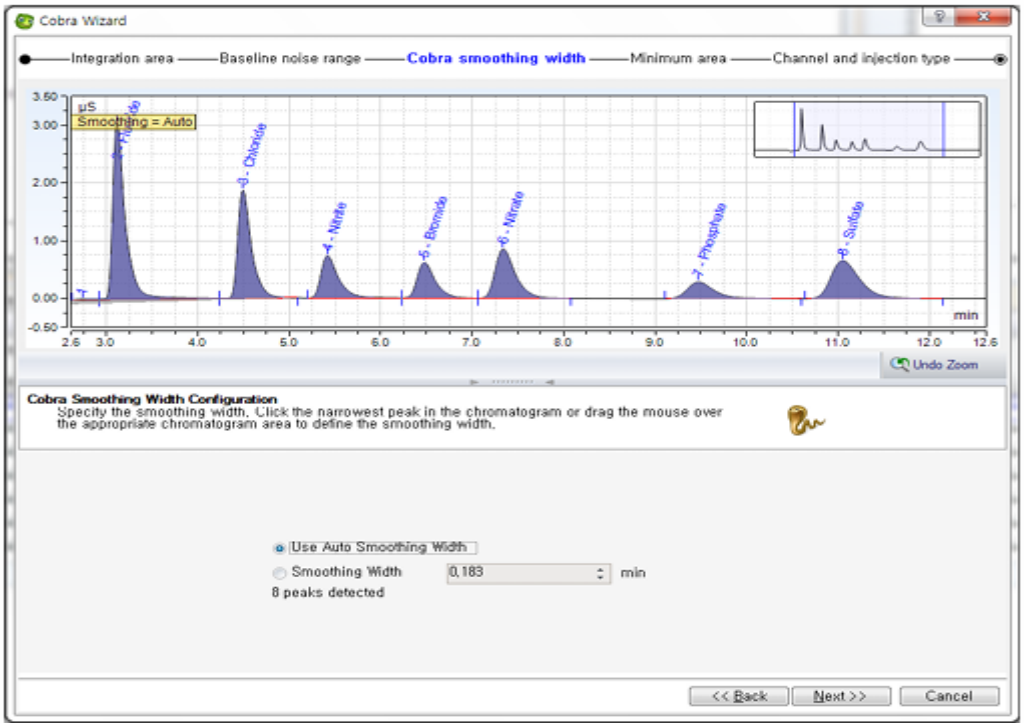


7) Keyingi bosqichga o'tish uchun Next tugmasini bosing.

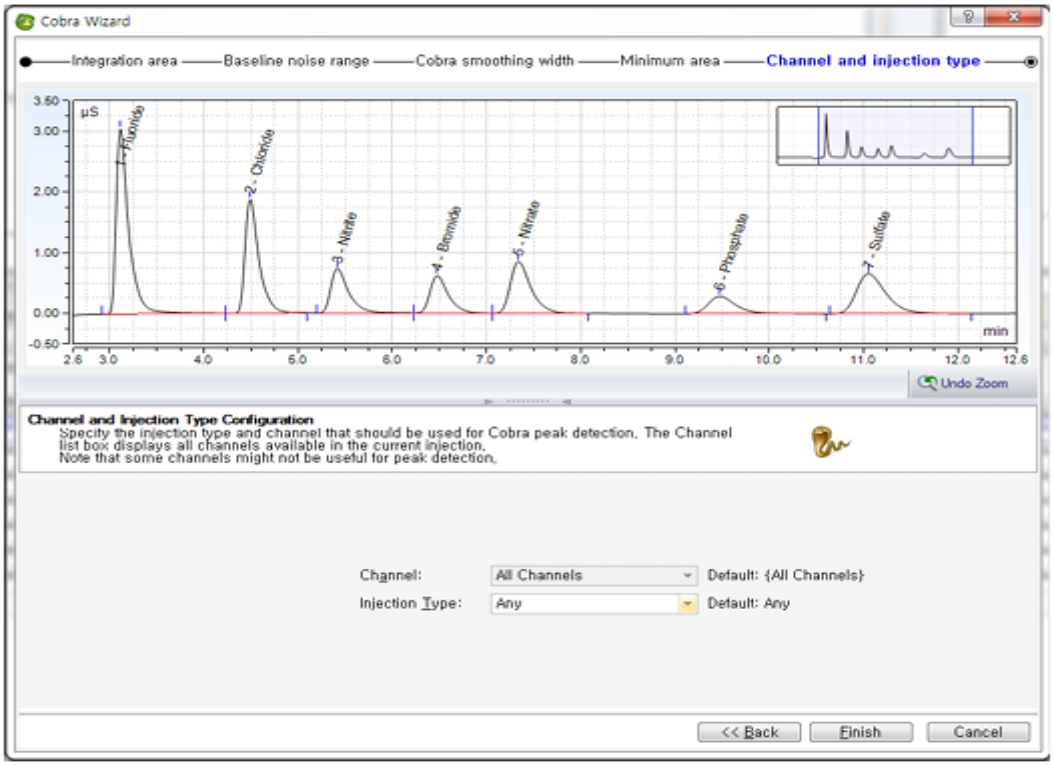
8) Baseline noise ni o'rnating.



9) Keyingi bosqichga o'tish uchun Next tugmasini bosing.

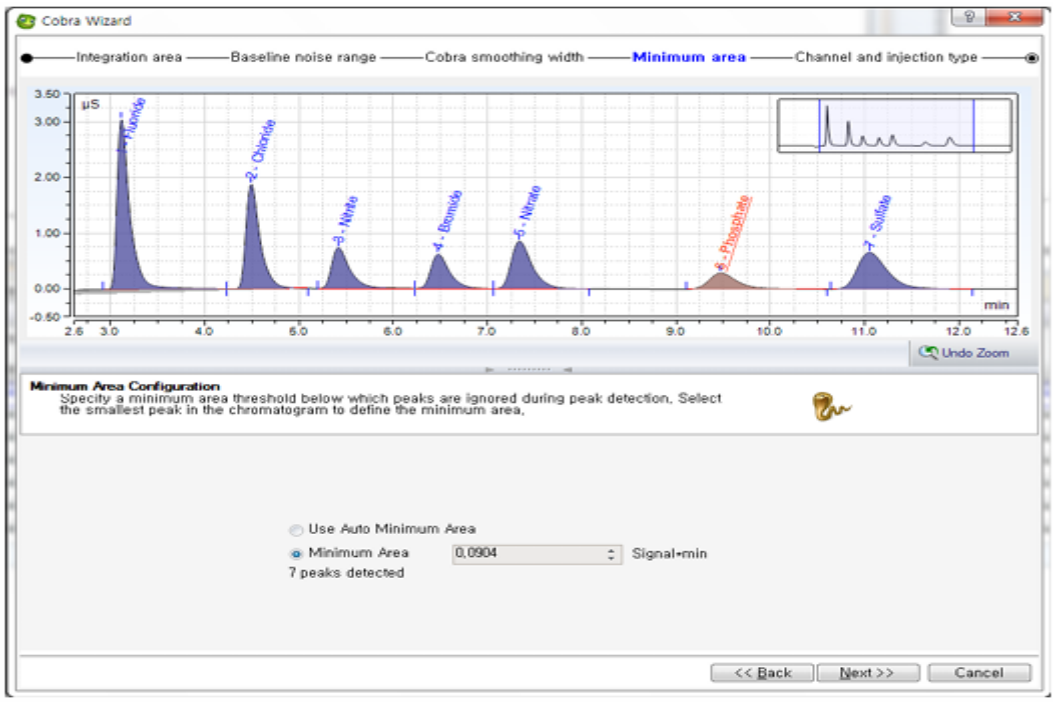


10) Minimal maydonni (minimum area) sozlash uchun eng kichik pik maydonini sichqoncha bilan belgilang.



11) Integratsiya quyida ko'rsatilgandek to'g'ri bajarilganligini tasdiqlaganingizdan so'ng, Next tugmasini bosing.

In'ektsiya turi va kanalini tanlagandan so'ng, jarayonni yakunlash uchun Finish tugmasini bosing.



11. Smart Peak’dan qanday foydalanish kerak

1) Kerakli standart peakni ikki marta bosing.

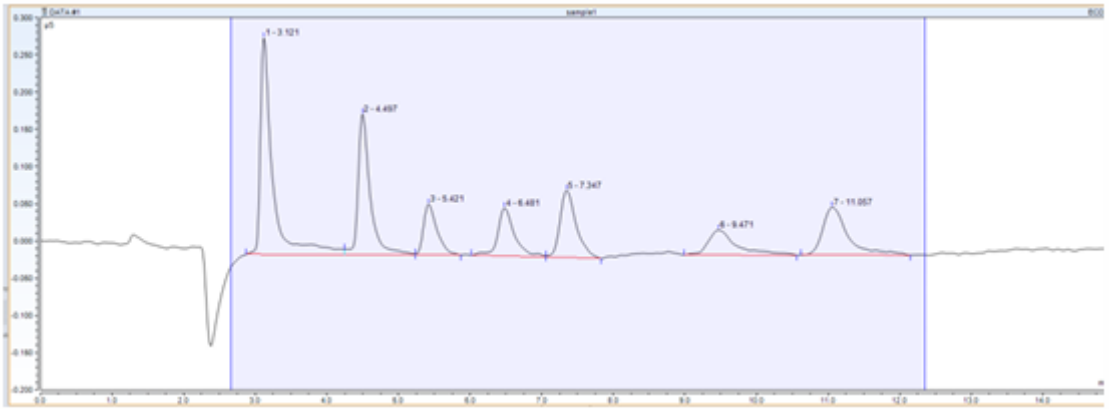
2) Processing yorlig‘ini bosing.



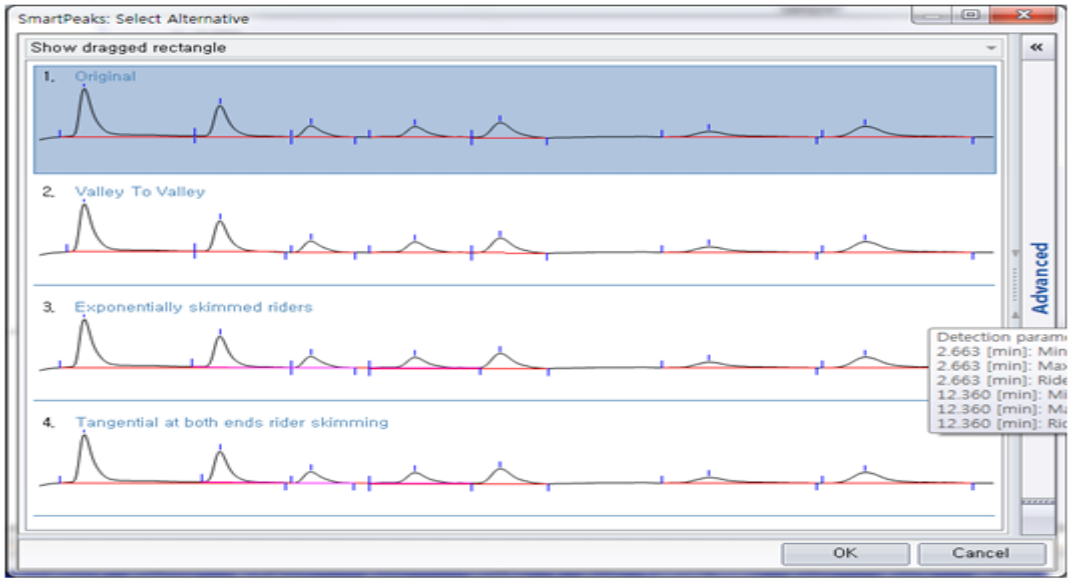
3) Smart Peak tugmasini bosing.(

)

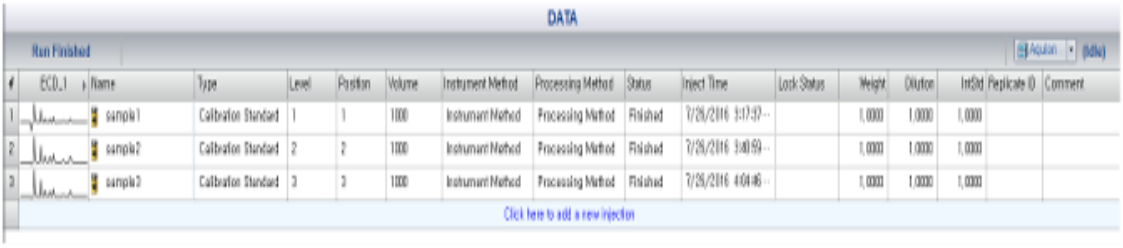
4) Pik joylashgan hududni sichqoncha yordamida belgilang.



5) Turli integratsiya variantlari quyida ko‘rsatilganidek aks ettiriladi.



6) Kerakli pik shakliga mos keladigan integratsiya variantini tanlang, so‘ng OK tugmasini bosing.

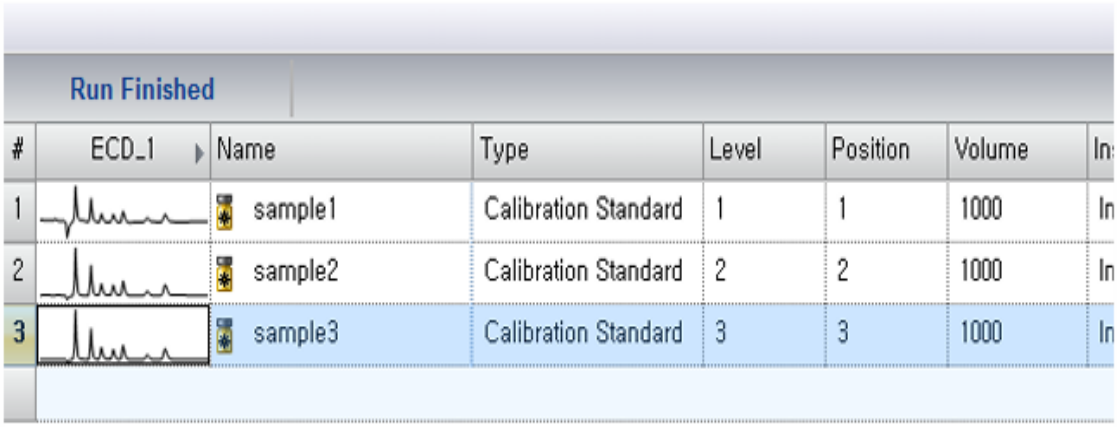


7) Quyida ko‘rsatilganidek, integratsiya avtomatik ravishda qo‘llanganini tasdiqlashingiz mumkin. Keyin quyida ko‘rsatilgan yorliqlarni tahrirlashga o‘ting.

Yorliqlarni tahrirlashdan oldin, Browser oynasida Standard ni qo‘shing.

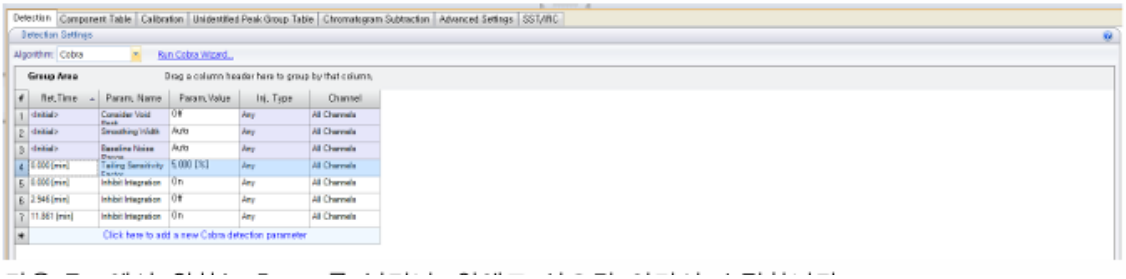


Quyida ko'rsatilgandek standartlarni qo'shing va har bir standartni uning darajasiga qarab o'rnating.



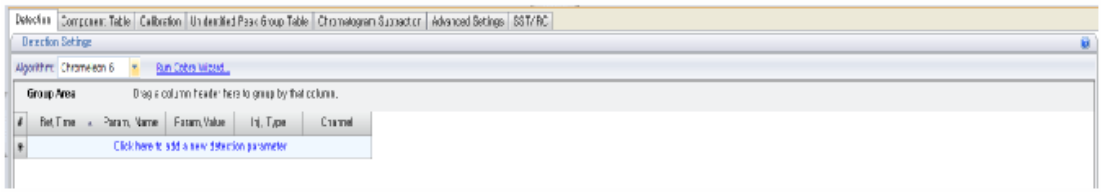
V.Chromeleon ilovalari

1. Aniqlash yorlig'i(Detection Tab)



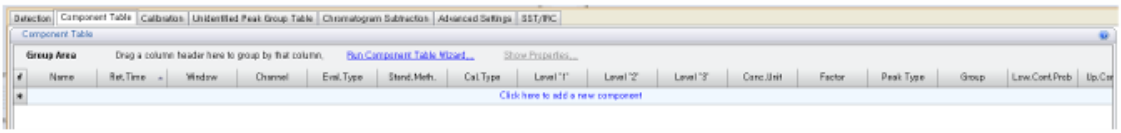
Ushbu yorliqda kerakli faktor qiymatini kiriting yoki zarur bo‘lsa uni o‘chirib tashlang.

Agar siz ilgari Chromeleonning oldingi versiyasidan foydalangan bo'lsangiz, Algoritm uchun Chromeleon 6 ni tanlash Chromeleon 6 bilan bir xil tarzda ishlashga imkon beradi.

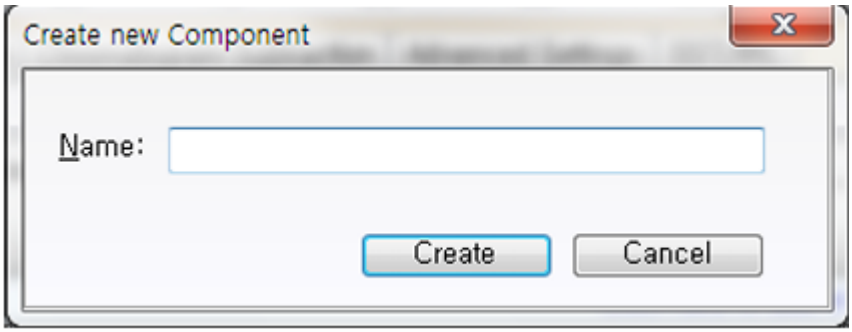


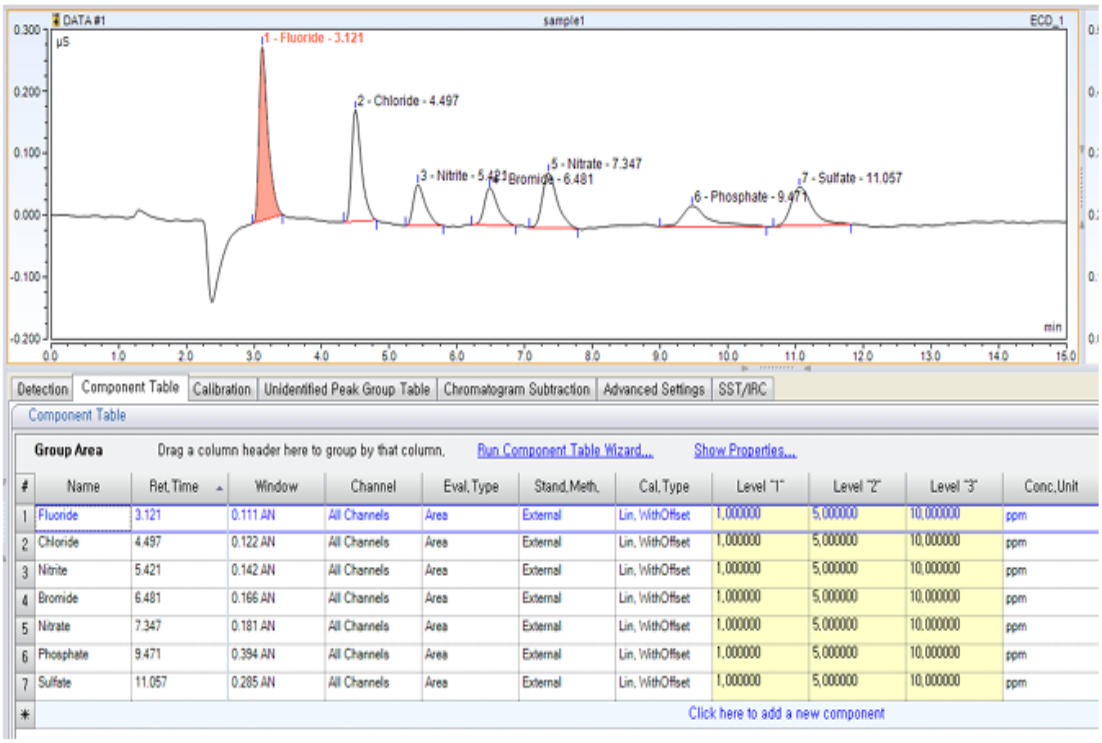
2. Komponentlar jadvali(Component Table) yorlig'i

1) Yangi komponent qo'shish uchun ushbu maydonni bosing.



2) Namuna nomini kiriting.

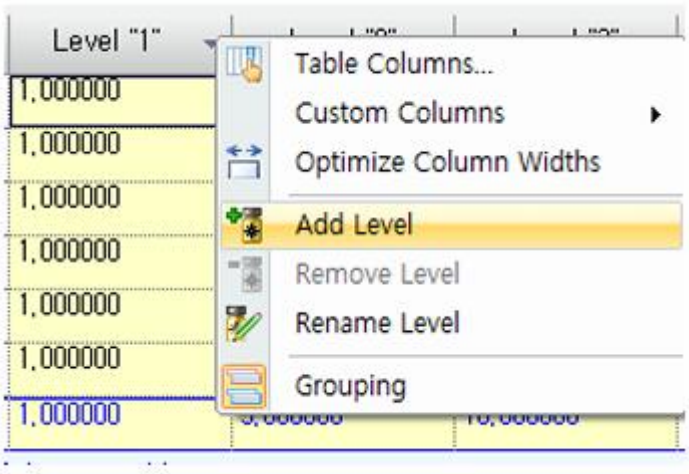




#	Name	Ret.Time	Window	Channel	Eval.Type	Stand.Meth.	Cal.Type	Level "1"	Level "2"	Level "3"	Conc.Unit
1	Fluoride	3.121	0.111 AN	All Channels	Area	External	Lin, WithOffset	1,000000	5,000000	10,000000	ppm
2	Chloride	4.497	0.122 AN	All Channels	Area	External	Lin, WithOffset	1,000000	5,000000	10,000000	ppm
3	Nitrite	5.421	0.142 AN	All Channels	Area	External	Lin, WithOffset	1,000000	5,000000	10,000000	ppm
4	Bromide	6.481	0.166 AN	All Channels	Area	External	Lin, WithOffset	1,000000	5,000000	10,000000	ppm
5	Nitrate	7.347	0.181 AN	All Channels	Area	External	Lin, WithOffset	1,000000	5,000000	10,000000	ppm
6	Phosphate	9.471	0.394 AN	All Channels	Area	External	Lin, WithOffset	1,000000	5,000000	10,000000	ppm
7	Sulfate	11.057	0.285 AN	All Channels	Area	External	Lin, WithOffset	1,000000	5,000000	10,000000	ppm

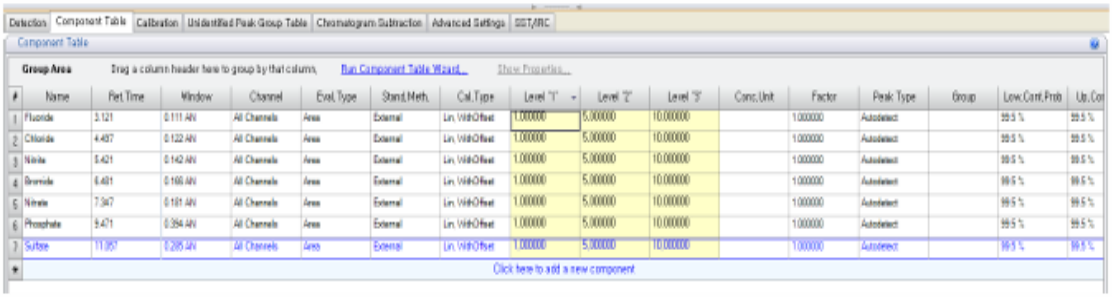
Fluoride ni tanlaganingizdan so‘ng, yuqorida ko‘rsatilgan Retention Time qiymatiga qarab, ushbu qiymatni Ret. Time maydoniga kiriting.
Yuqoridagi jadvalda Ret. Time uchun 3.121 qiymati kiritilgan.
Har bir pikni quyida ko‘rsatilganidek sozlab, pik nomi va RT (Retention Time) ni belgilang..

3) Peak lar o'rnatilgandan so'ng, daraja(level)ni sozlang.

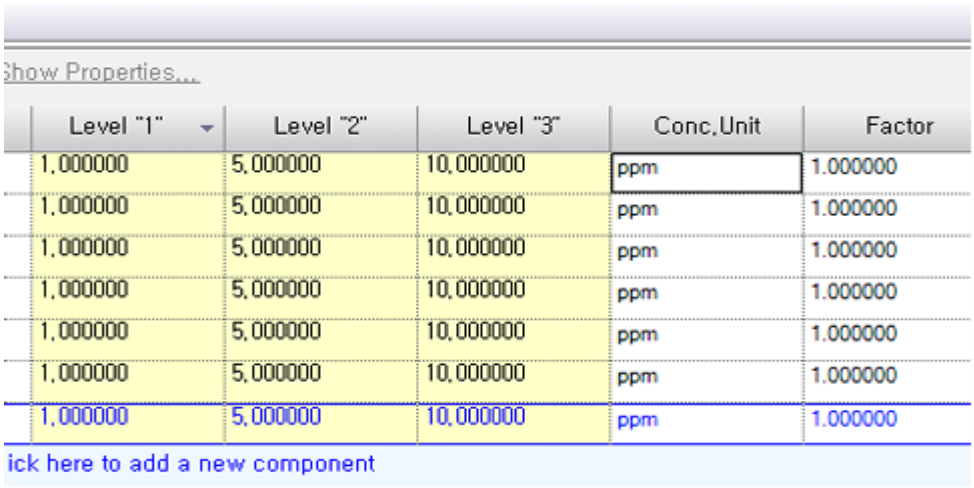


Level ustunida sichqonchaning o‘ng tugmasini bosing va Add Level ni tanlang.
Kerakli miqdordagi darajalar (ya’ni standartlar) qo‘shish uchun ushbu jarayonni takrorlang..

Keyin, har bir darajaga mos keladigan konsentratsiyani kiriting.

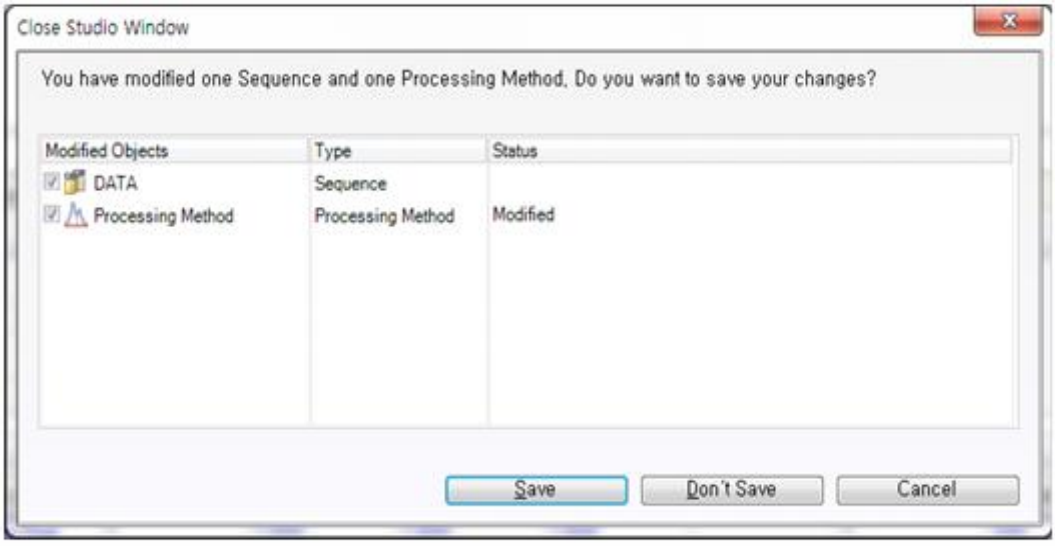


4) Konsentratsiya birligini kiriting.



Konsentratsiya birligini "Conc.Unit " ustuniga kiriting.

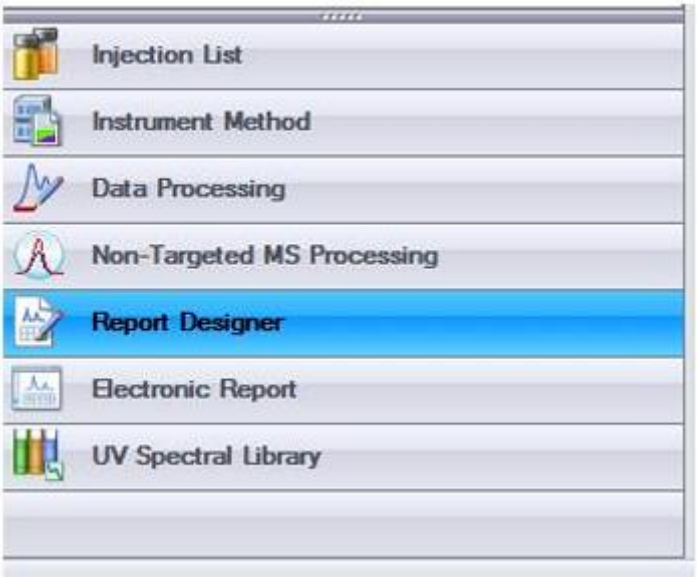
5) Tahrirlangan Processing Method ni yopayotganda, quyida ko‘rsatilganidek pop-up oyna paydo bo‘ladi.



DATA va Processing Method ni tanlang, so‘ng Save tugmasini bosing.

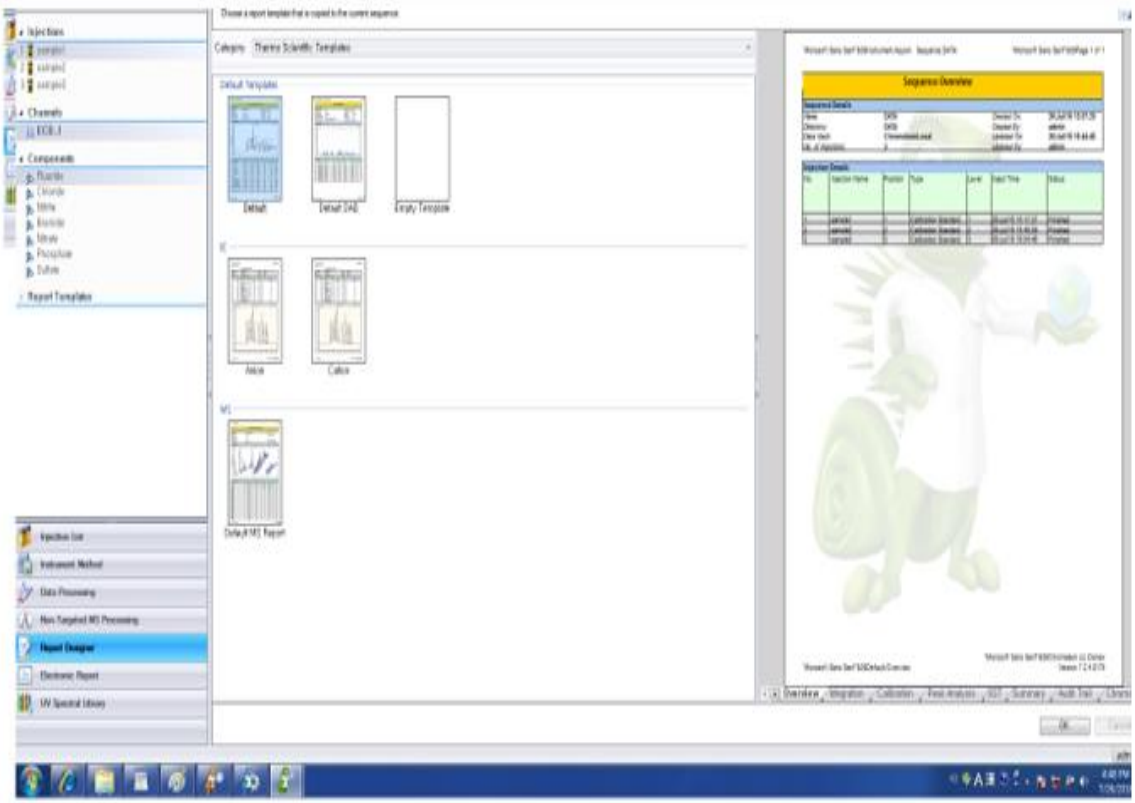
3. Ma'lumotlar chiqishi

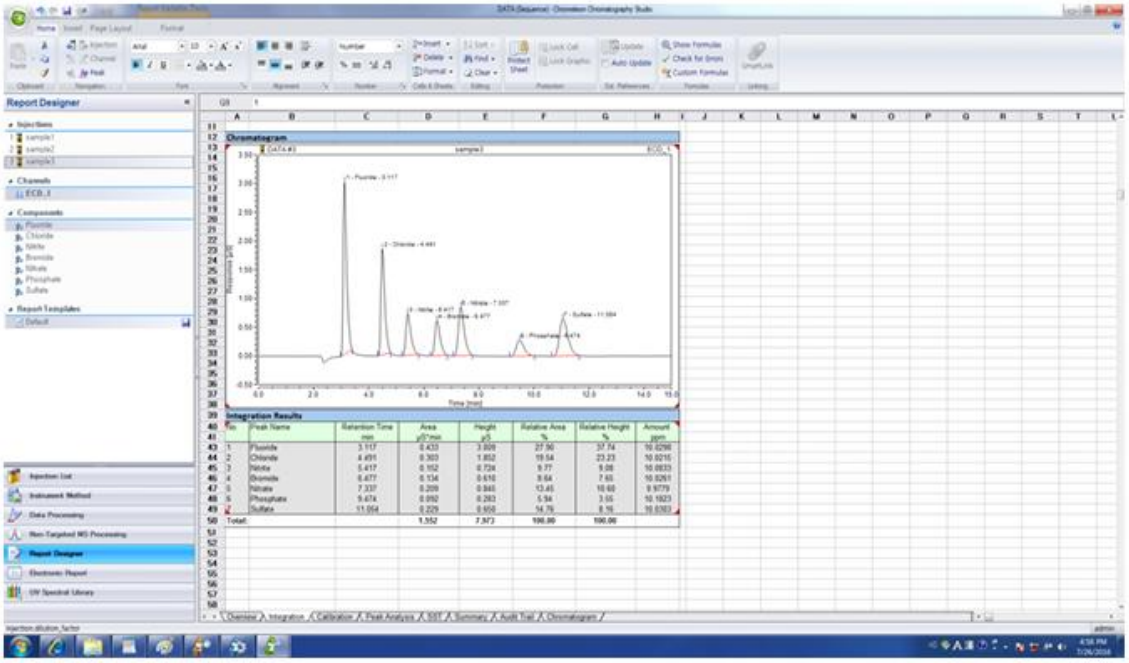
1) Kerakli peakni ikki marta bosing.



2) Report Designer tugmasini bosing.

3) Quyidagi ekran paydo bo'ladi. Kerakli shablonni tanlang, so'ngra quyida ko'rsatilgandek hisobot shaklini ko'rsatish uchun OK tugmasini bosing.

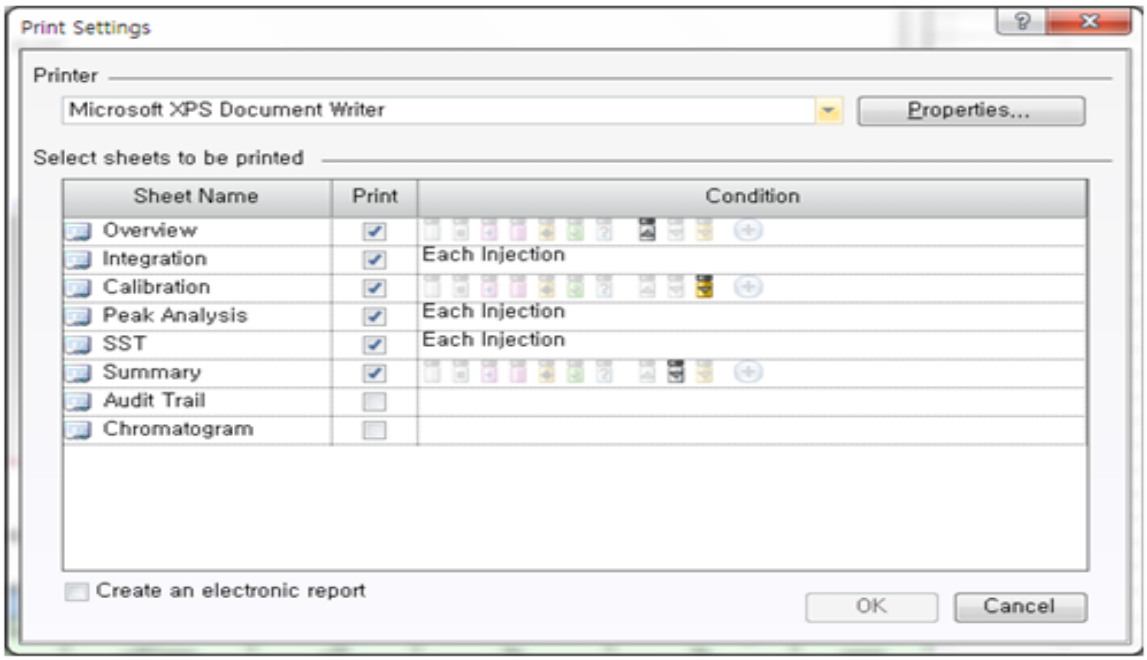




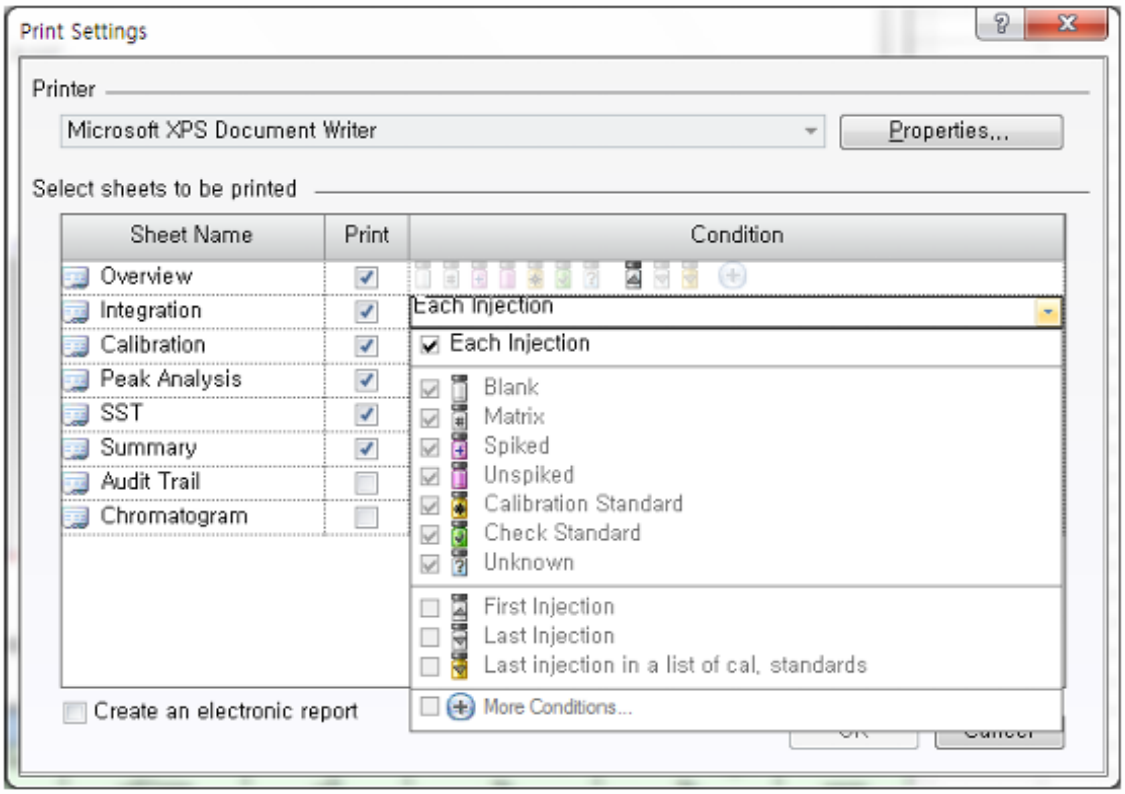
4) Hisobotni chop etish uchun Print settings tugmasini bosing.



5) Kerakli varaqni tanlang.



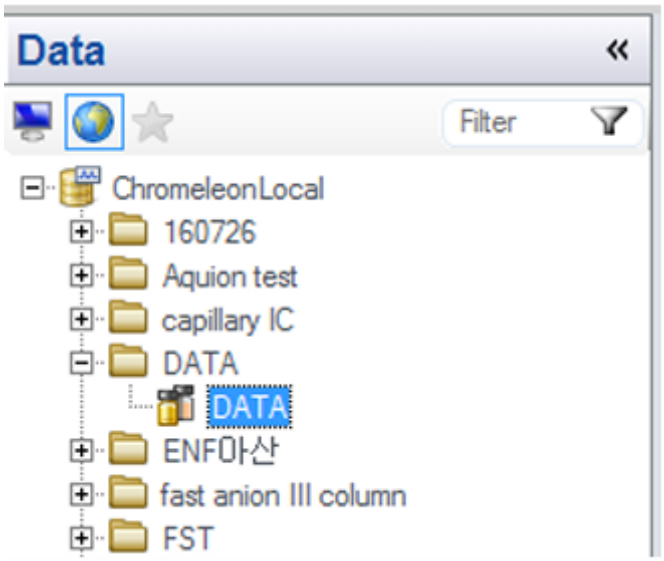
6) Kerakli shartni tanlang. Chiqariladigan namunalarni tanlagandan so'ng, chop etish uchun OK tugmasini bosing.



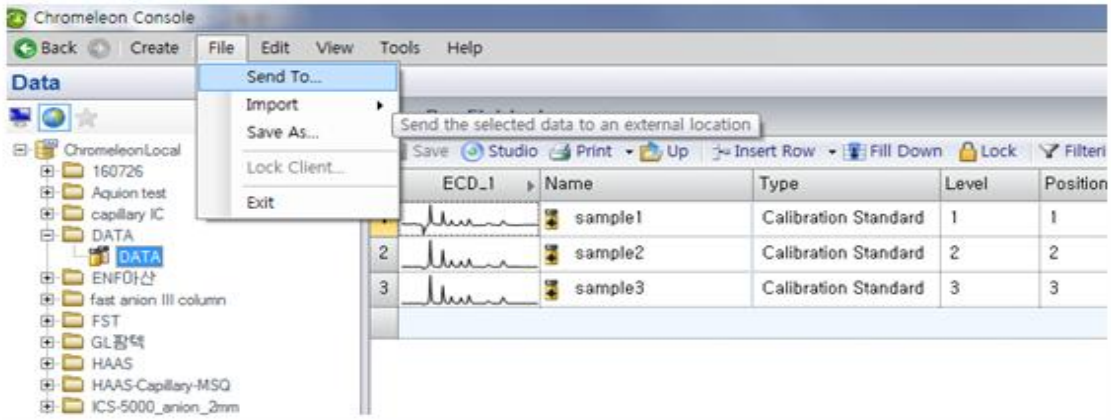
Namunalarni chiqarishni tanlagandan so'ng, chop etish uchun OK tugmasini bosing.

4. DATA Backup → Ma'lumotlarni zaxiralash

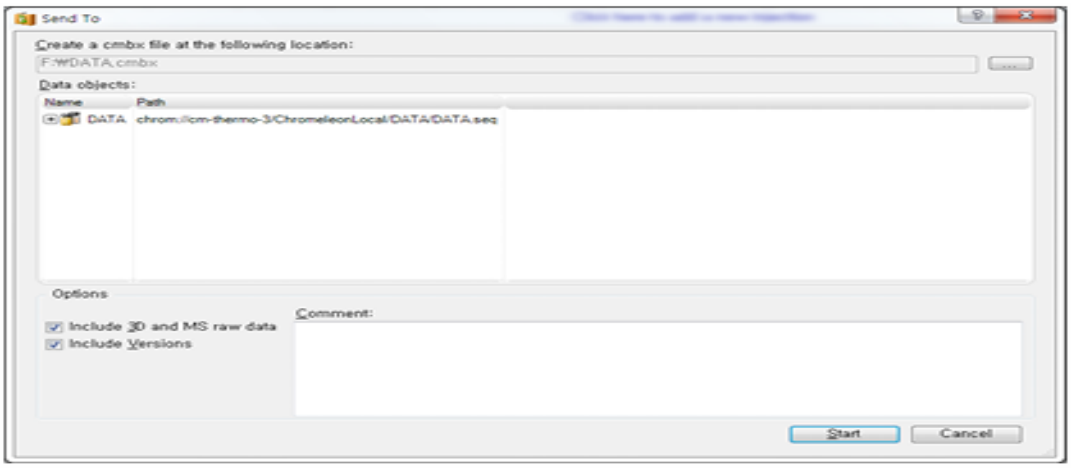
1) Kerakli Sequenceni bosing.



2) Fayl / Yuborish(Send To)... ni bosing.

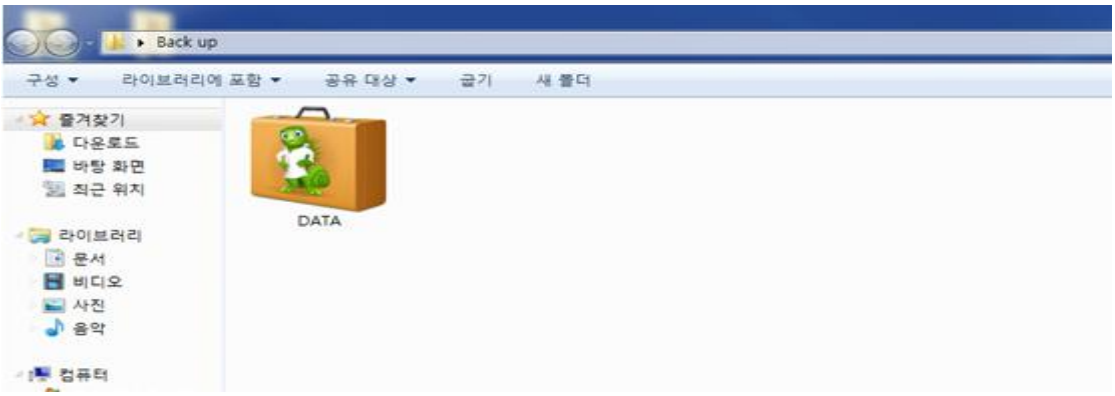


3) Quyidagi ekran paydo bo'lganda, "Start" tugmasini bosing.



4) Keyingi oynada zaxira fayl saqlanadigan joyni belgilang va fayl nomini kiriting.

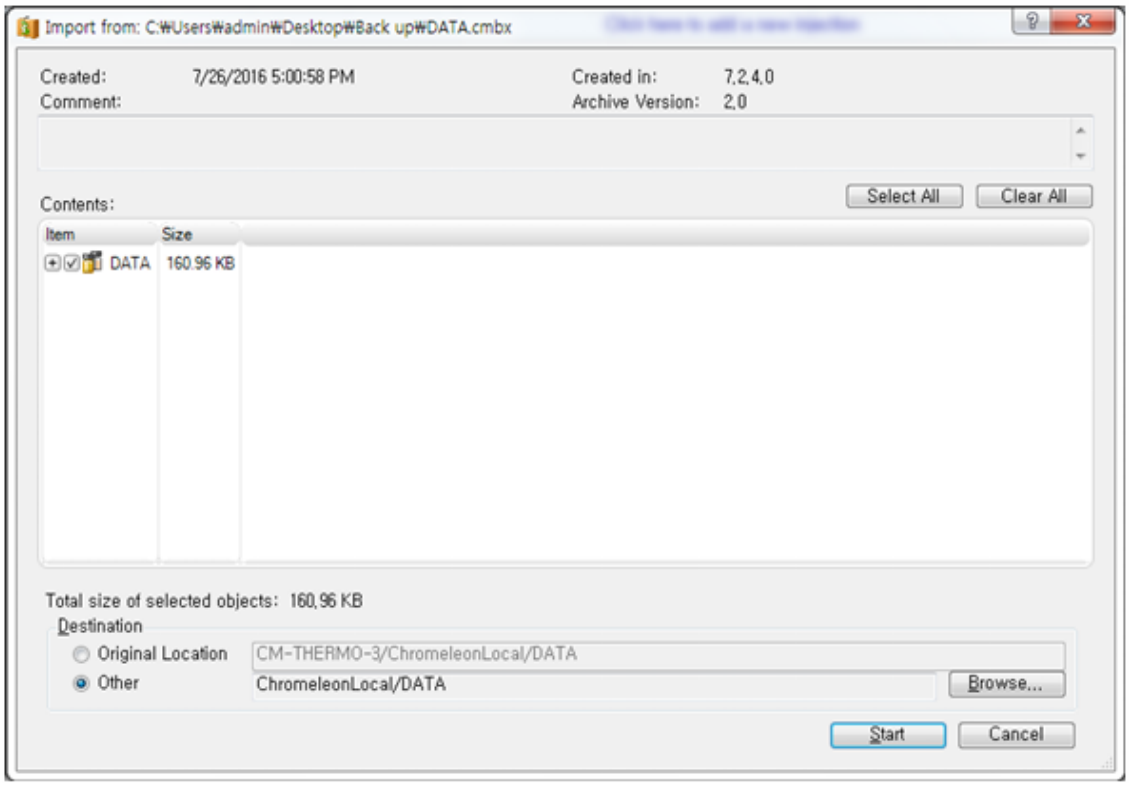
5) Zaxira fayli saqlangan joyda quyidagicha ko'rsatilishini tasdiqlashingiz mumkin.



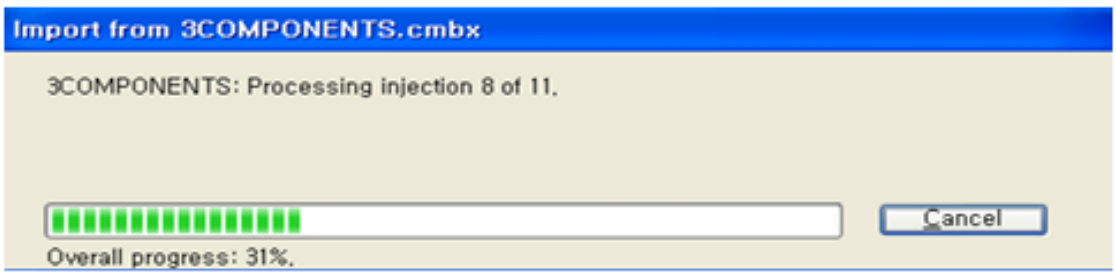
5. Zaxiralangan DATA faylini tiklash

1) Zaxira faylini ikki marta bosing.

2) Keyingi oynada Other bo‘limida Browse.. tugmasidan foydalanib, zaxira fayl tiklanadigan joyni belgilang, so‘ng Start tugmasini bosing.

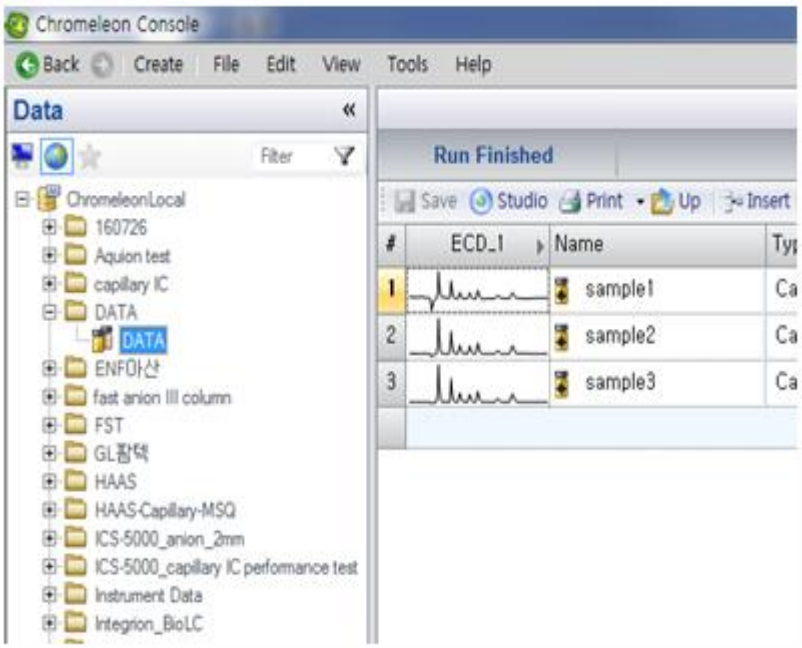


3) Quyida ko'rsatilgandek davom etgandan so'ng, fayl belgilangan joyga tiklanadi.

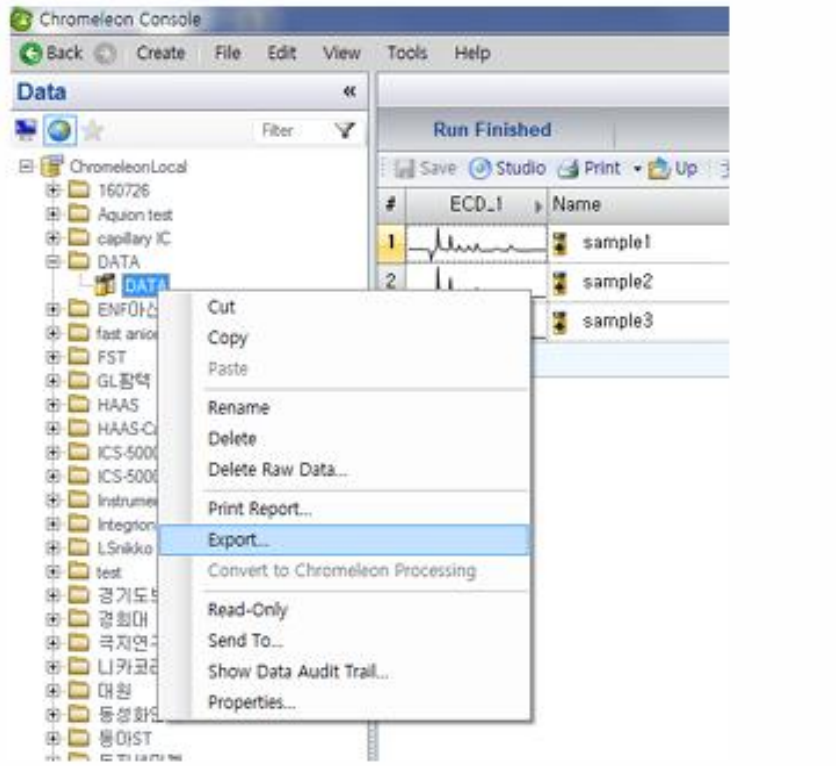


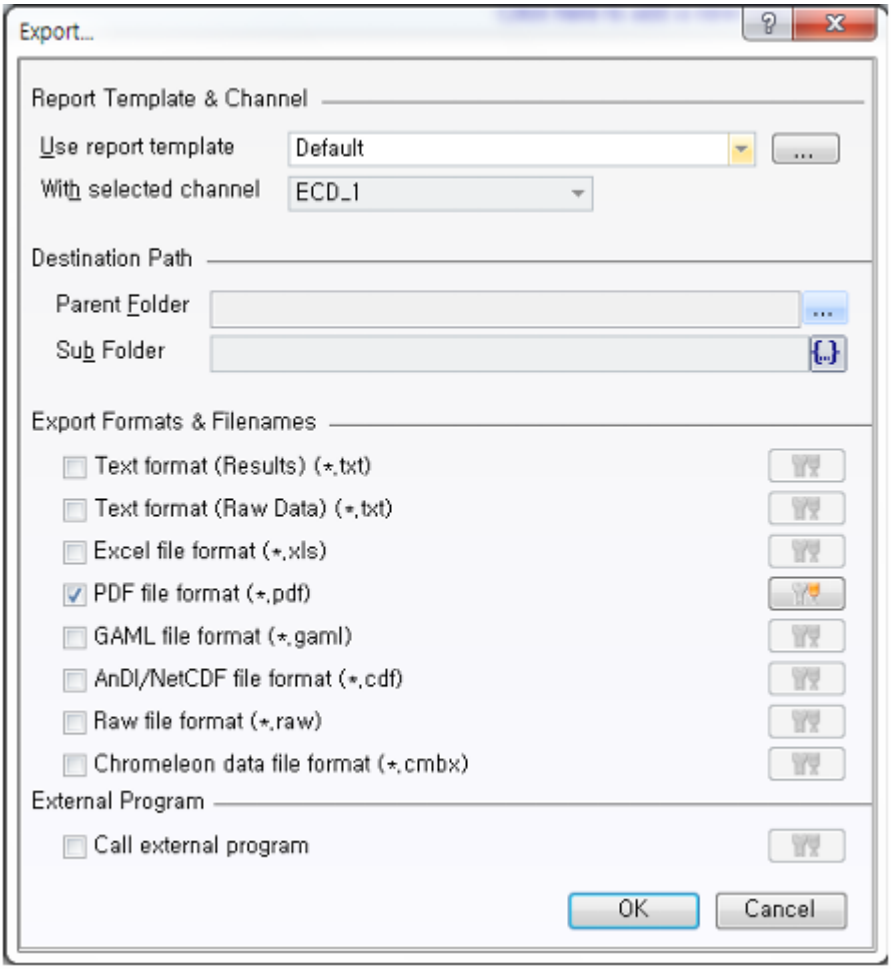
6. Excel yoki PDF fayliga aylantirish

1) Kerakli Sequence ni bosing.



2) Sichqonchaning o'ng tugmachasini bosing va Eksport-ni tanlang.





3) Yuqoridagi oynada:

(1) Use report template: kerakli hisobot formatini tanlang.

(2) With Selected Channel: kerakli kanalni tanlang.

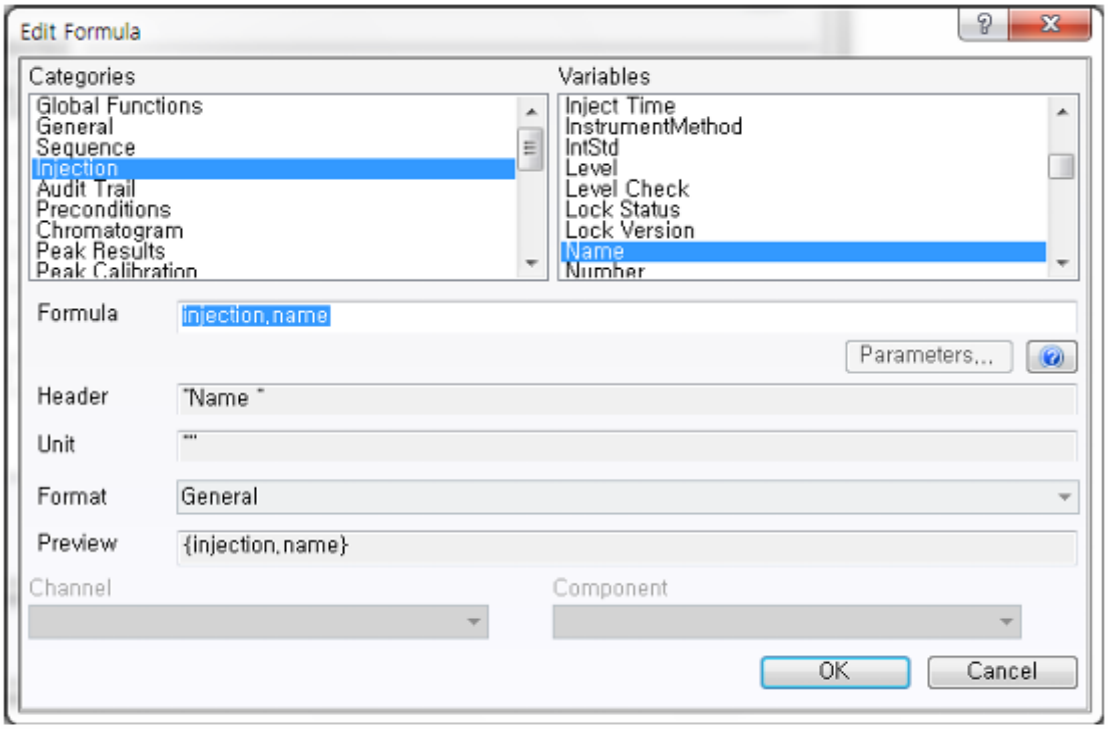
(3) Parent Folder: saqlanadigan joyni belgilang.

(4) Sub Folder: odatda avtomatik tarzda papka yaratiladi.

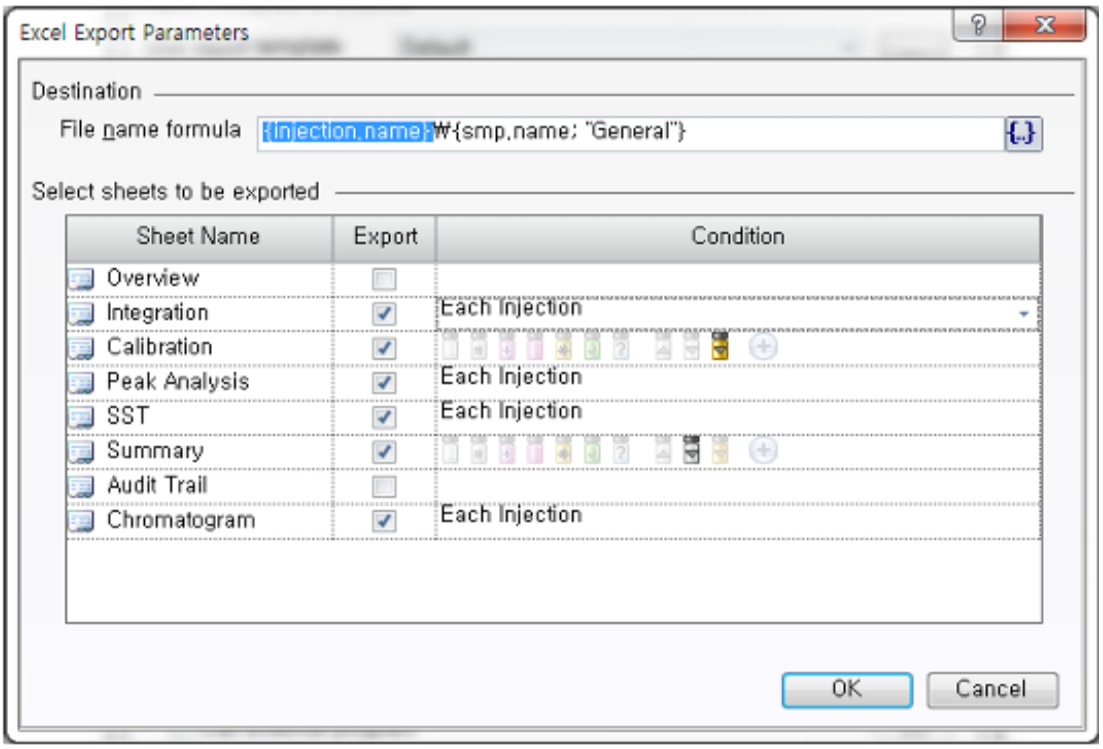
(5) Agar Excel formatiga o‘tkazmoqchi bo‘lsangiz, Excel File Format ni belgilang.

Agar PDF formatiga o‘tkazmoqchi bo‘lsangiz, PDF File Format ni belgilang.

Agar Excel fayli uchun konvertatsiya sozlamalarini o‘zgartirmoqchi bo‘lsangiz, quyidagi tugmani bosib sozlamalarni o‘zgartiring (xuddi shu tartib PDF fayl uchun ham amal qiladi).



Fayl nomi formulasini quyida ko'rsatilgandek o'zgartiring.



Barcha sozlamalarni bajargandan so'ng, kerakli joyga o'tkazishni yakunlash uchun OK tugmasini bosing.

VI. Texnik xizmat ko'rsatish

Ushbu bob tahlil jarayonida har bir modul yoki qismda yuzaga kelishi mumkin bo‘lgan turli muammolar ichida eng ko‘p uchraydigan muammolarni, ularning sabablarini va bartaraf etish choralarini bayon qiladi.

1. Kolonka ajratish samaradorligining pasayishi

1) Ifloslantiruvchi moddalar

(1) Analitlar (Analytes)

Namuna ortiqcha yuklanishi (overloading) va boshqa sabablar tufayli analitlarning kolonkada to‘planishi.

Kutiladigan belgilar: blank (bo‘sh) namunada piklarning paydo bo‘lishi, piklarning cho‘zilishi (tailing) va retensiya vaqtining qisqarishi.

(2) Og‘ir metallar (Heavy metal)

Namunadagi og‘ir metallar tufayli kolonkani ajratish samaradorligi pasayadi va shovqin ortadi.

(3) Organik birikmalar (Organic compounds)

Namunadagi turli organik moddalar to‘planishi natijasida kolonkadagi bosim oshadi, shovqin kuchayadi va “ghost peaks” (soxta piklar) paydo bo‘ladi.

2) Tozalash eritmasi (Cleaning Solution)

Tozalash eritmasi va uni ifloslantiruvchi moddalarga qarab tanlash bo‘yicha batafsil ma’lumot uchun har bir kolonkaga oid qo‘llanmaga murojaat qiling.

(1) Analitlar bilan ifloslanish

Anion/kation va boshqa umumiy tahlil kolonkalarida analit sababli yuzaga kelgan ifloslanishni bartaraf etish uchun tozalash eritmasi tahlil davomida ishlatilgan eluент konsentratsiyasidan 5–10 marta yuqori qilib tayyorlanadi va qo‘llaniladi.

(2) Organik birikma bilan ifloslanish

Organik moddalar bilan ifloslanish turli manbalardan kelib chiqishi mumkin. Bunday holatda odatda tozalash eritmasiga atsetonitril (yoki metanol) qo‘shilib qo‘llaniladi.

3) Kolonkani tozalash usuli

Kolonkani tozalash vaqtida ifloslantiruvchi moddalar tizimga qayta kirib ketmasligi uchun kolonkani nasosning Pressure Transducer Outlet qismidan to‘g‘ridan-to‘g‘ri ulash kerak. Shu bilan birga, tizimdagi barcha klapanlar, shuningdek suppressor va detektor yacheykasi olib tashlanishi va tozalash jarayonidan chiqarib tashlanadi.

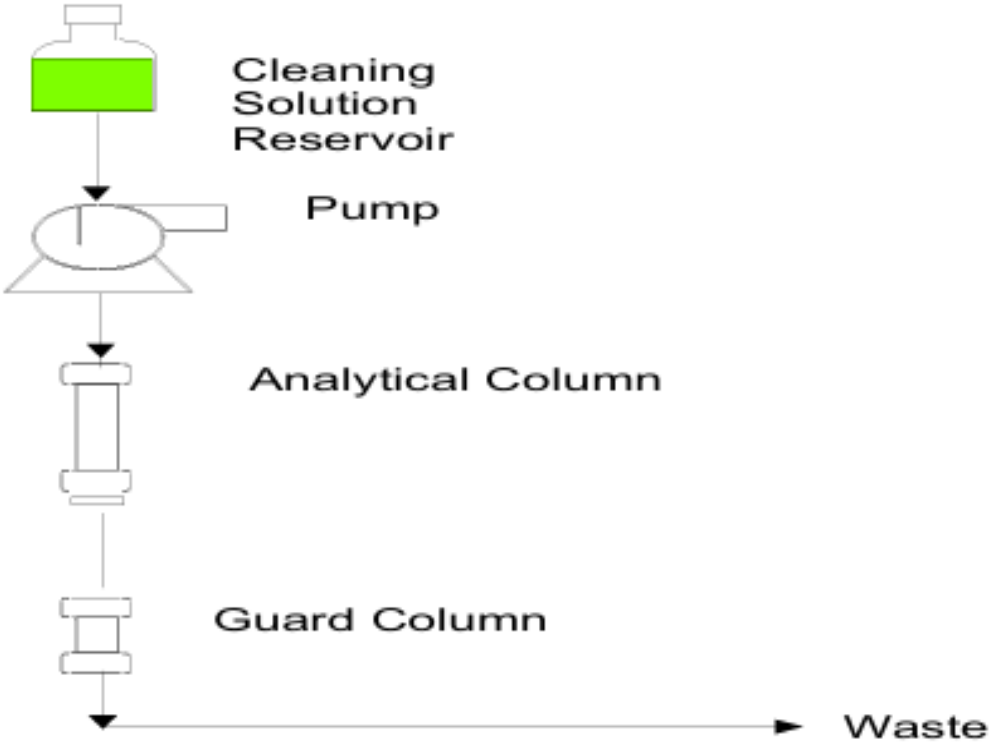
(1) Suppressor va detektor yacheykasini ajrating. (Kolonka chiqishini to‘g‘ridan-to‘g‘ri chiqindi (waste) ga yo‘naltiring.)

(2) Oqim yo‘lini Guard Column → Separator Column dan Separator Column → Guard Column ga o‘zgartiring. (Bunda kolonkadagi oqim yo‘nalishi o‘zgarmasligi kerak.)

(3) Tayyorlangan tozalash eritmasidan taxminan 60 mL yuboring (30–60 daqiqa davomida).

(4) Kamida 30 mL yuqori tozalikdagi (≥18 MΩ) deionizatsiyalangan suv bilan yuvib chiqing (10–30 daqiqa).

(5) Oqim yo‘lini yana normal holatga keltirgandan so‘ng (oldin ajratilgan suppressor va detektor yacheykasini qayta ulang), kolonkani muvozanatlash uchun analitik eluent bilan 30–60 daqiqa davomida yuvib chiqing.



2. Suppressor bilan bog‘liq muammo (Suppressor Trouble)

1) Baseline noise kuchayadi va peak maydoni anormal darajada kichiklashadi. Ushbu holat odatda namuna tarkibida organik moddalar bilan ifloslanish mavjud bo‘lganda yoki quyidagi hollarda yuzaga keladi:

(1) ERS holatida: eluент uzoq vaqt davomida elektrolizsiz ishlatilgan bo‘lsa.

(2) MMS holatida: eluent uzoq vaqt davomida regenerant ishlatilmasdan oqizilgan bo‘lsa.

* Tuzatish choralari

(1) (1) ELUENT IN portiga (ilgari analitik kolonkaga ulangan) ulangan eluent liniyasini ajratib, uni chiqindi (waste) trubkasiga ulang.

(2) ELUENT OUT portidagi (ilgari detektor yacheykasiga ulangan) eluent liniyasini yacheykadan ajrating, so'ngra Luer adapteri bilan birlashtiruvchini liniyaning oxiriga ulang ("Suppressor Hydrating" bo'limiga qarang).

(3) Anion uchun o‘z-o‘zini regeneratsiya qiluvchi suppressor (AERS) bo‘lsa, shprits yordamida tayyorlangan 0,5N H_2SO_4 dan 5 mL yuboring; kation suppressori (CERS) bo‘lsa, 0,5N NaOH dan 5 mL yuboring. Oqim yo‘nalishi

ELUENT OUT → ELUENT IN bo‘lishi kerak.

- AERS-300 uchun yuqoridagi protsedura bajarilmaydi.

(4) (4) Suppressorni yana analitik kolonkaga va detektorni tizimga qayta ulang (asl konfiguratsiyani tiklang), so‘ng taxminan 30 daqiqa davomida deionizatsiyalangan suv bilan yuvib chiqing. Tizim yana tahlil sharoitlarida barqarorlashgach, tahlilni davom ettiring.

2) High Background Conductivity → Yuqori fon o‘tkazuvchanligi

* Tuzatish choralari

(1) Ushbu holat ERS quvvat manbaida muammo bo‘lganda yoki eluent konsentratsiyasi noto‘g‘ri bo‘lganda yuzaga kelishi mumkin. Eluent konsentratsiyasini qayta tekshiring va tahlil sharoitlariga muvofiq regenerant oqim tezligini tasdiqlang.

(2) Oqim yo'lining noto’g’ri shakllanishi

Agar eluent ERS ning ELUENT IN portiga kirib, lekin ELUENT OUT orqali chiqmasa, oqim yo‘lida tiqilib qolish bor-yo‘qligini tekshiring. Suppressor yon tomoni yoki pastki qismidan eluent sizib chiqishi ko‘pincha oqim yo‘li tiqilib qolganini bildiradi. Agar trubkalar tiqilib qolganligi aniqlansa, ularni almashtiring. Agar suppressorning to‘rtta kirish/chiqish portlari tiqilib qolgan bo‘lsa, tozalash jarayonini amalga oshiring. Bu holat membrananing shikastlanishiga ham asosiy sabab bo‘lishi mumkin.

Shu sababli tizimni muntazam tekshirib turish zarur. Agar membrana shikastlanganligi aniqlansa, suppressorni almashtirish kerak.

3) Baseline drift (baseline asta-sekin ko‘tariladi)

*Tuzatish choralari:

(1) Tozalash tartibiga muvofiq tozalashni amalga oshiring.

(2) Fon o‘tkazuvchanligini kamaytirish va uni barqarorlashtirish uchun ERS oqim sozlamasini (tok berish darajasini) bir pog‘ona oshiring.

(100 mA dan yuqori oshirmaslik tavsiya etiladi.)

4) Sezgirlik pasayishi

Tuzatish chorasi

(1) Tizimda sizib chiqish (leak) ehtimoli mavjud. Sizib chiqish joyini aniqlang va uni bartaraf etish uchun fittinglarni mos ravishda mahkamlang.
(2) Injection valve (yuborish klapani) to‘g‘ri ishlayotganini tekshiring.
(3) Agar yuqoridagi choralar yordam bermasa, ERS Clean Up jarayonini amalga oshiring.

5) ERS Chemical Mode va AutoSuppression Mode o‘rtasida o‘tish usuli
Ushbu ikki rejimni birgalikda qo‘llaganda, odatda bazaviy chiziqning (baseline) beqarorligi muammosi yuzaga keladi. Agar rejimlar hech qanday tayyorgarliksiz darhol almashtirilsa, membrana muvozanat holatiga tez kira olmaydi. Shuning uchun ushbu ikki rejimni birgalikda ishlatishda, rejimni o‘zgartirishdan oldin quyidagi choralarni ko‘rish zarur..
(1) Chemical Mode → AutoSuppression Mode
: Shprits yordamida suppressorning Regen Out portiga taxminan 30 mL deionizatsiyalangan suv yuboring.
(2) AutoSuppression Mode → Chemical Mode
: Shprits yordamida suppressorning Regen Out portiga taxminan 30 mL 0,5N sulfat kislota yuboring. (Oqim tezligi: 5–10 mL/min)
※ Eslatma (Caution)
ERS ni tozalash vaqtida suppressor quvvatini albatta o‘chirib qo‘ying.

6) ERS namlantirish (ERS Hydrating)
(1) Reagent tayyorlash
- Anion: 200 mN H_2SO_4
- Kation: 200 mM NaOH
SRS birinchi marta o‘rnatilganda: deionizatsiyalangan suv (DI water)
(2) ERS ni qurilmadan ajrating va rasmda ko‘rsatilgan qismlarni tayyorlang.
(3) Eluent Out va Regen In portlariga Luer adapterini o‘rnating.
(4) Quyidagi rasmda ko‘rsatilganidek, Eluent Out porti orqali 3 mL namlantiruvchi reagentni yuboring..
※ Eslatma (Caution): Sekin yuboring va albatta 1 cc shpritsdan foydalaning. Yuborish jarayonida reagent qarama-qarshi Eluent In portidan chiqadi, shuning uchun yuborish vaqtida uni salfetka bilan artib turing..

((5) Regen In porti orqali 5 mL namlantiruvchi reagent yuboring.

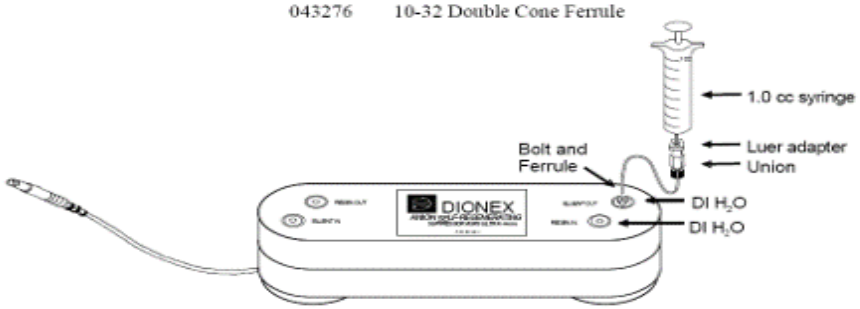
→ ERS birinchi marta o‘rnatilganda, deionizatsiyalangan suv bilan yuboring, so‘ng 20 daqiqa o‘tgach o‘rnatib foydalaning.

(6) 20 daqiqadan so‘ng, D va E bosqichlarini deionizatsiyalangan suv bilan takrorlang.

→ Eslatma: reagentdan foydalanmang.

(7) ERS ni o‘rnatgandan so‘ng, qurilmani barqaror holatga keltiring..

P/N	Description
016388	1 cc plastic syringe
024305	1/4-28 Luer adapter
046888	10-32 Luer adapter
042627	10-32 Union
043275	10-32 Bolt
043276	10-32 Double Cone Ferrule



3. Suppressorni tozalash tartibi (Suppressor Clean Up Procedure)

1) Anion suppressor

(1) AMMS:

Organik ifloslanish holatida: 90% ACN yoki 90% MeOH (30–60 mL)

→ so‘ng deionizatsiyalangan suv (10–20 mL)

Metall ifloslanish holatida: 0,1M KCl + 1M HCl (30–60 mL)

→ so‘ng deionizatsiyalangan suv (10–20 mL)

(2) ASRS:

Metall ifloslanish holatida: 0,1M KCl + 1M HCl (30–60 mL)

→ so‘ng deionizatsiyalangan suv (10–20 mL)

Organik ifloslanish holatida: 90% ACN yoki 90% MeOH + 10% 1M HCl (V/V) (30–60 mL) → so‘ng deionizatsiyalangan suv (10–20 mL)

2) Kation suppressor (Cation Suppressor)

(1) CMMS:

Kislotada eruvchan ifloslanish holatida: 0,1M KCl + 1M HCl (30–60 mL)

→ so‘ng deionizatsiyalangan suv (10–20 mL)

Ishqorda eruvchan ifloslanish holatida: 0,1M NaOH (30–60 mL)

→ so‘ng deionizatsiyalangan suv (10–20 mL)

Organik ifloslanish holatida: 90% ACN yoki 90% MeOH (30–60 mL)

→ so‘ng deionizatsiyalangan suv (10–20 mL)

CERS: Metall ifloslanish holatida: 1M MSA (30–60 mL)

→ so‘ng deionizatsiyalangan suv (10–20 mL)

Organik ifloslanish holatida:90% ACN yoki 90% MeOH + 10% 1M MSA (V/V) (30–60 mL)

→ so‘ng deionizatsiyalangan suv (10–20 mL))

4. O‘tkazuvchanlik yacheykasini tozalash (Conductivity Cell Cleaning)

1) O‘tkazuvchanlik detektori yacheykasini (CDM) tozalash usuli

(1) Tozalash eritmasini tayyorlang: 3M HNO_3 eritmasini tayyorlab, 10 cc shprits tayyorlang.

Yuvish eritmasi sifatida 10 mL deionizatsiyalangan suv (DI Water) tayyorlang va 10 cc shprits tayyorlang..

(2) O‘tkazuvchanlik yacheykasini IC tizimidan ajrating.

(3) Suppressorning Eluent Out porti bilan yacheyka kirish qismi o‘rtasidagi ulanishni uzing.

(4) Yacheyka kirish qismiga Union ulang va unga Female Luer adapter ni o‘rnating. Yacheyka chiqishini to‘g‘ridan-to‘g‘ri chiqindi (waste) liniyasiga ulang.

(5) O‘tkazuvchanlik yacheykasini o‘chiring.

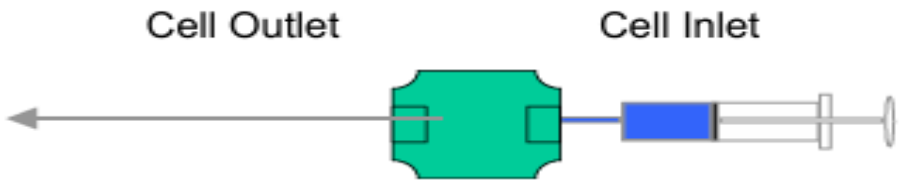
(6) Avval shprits yordamida yacheykaga 5 mL 3M HNO_3 yuboring.

(7) (6)-bosqichdan so‘ng 2 daqiqa kuting, keyin qolgan 5 mL HNO_3 ni yuboring.

(8) (7)-bosqichdan so‘ng yana 2 daqiqa kuting, so‘ng yacheykani yuvish uchun deionizatsiyalangan suv yuboring.

(9) O‘tkazuvchanlik yacheykasini yana IC tizimiga ulang.

(Agar Cell Calibration Constant o‘zgargan bo‘lsa, (7)-bosqichdan keyin 0,001 KCl yordamida yacheykani kalibrlang.)



5. Injektorni tozalash

Injektorlarning har xil turlari mavjud, ammo tozalash usuli bir xil.

Agar ichkarida tuz yoki zarrachalar qolsa, tiqilib qolish kabi muammolar paydo bo'lishi mumkin. Shuning uchun, namuna loop (sample loop) ga deionizatsiyalangan suv yuborib, Load va Inject amallarini bir necha marta takrorlash orqali injektorni tozalang.

Injektordan har safar foydalanilganda uni tozalash tavsiya etiladi.

Vaqti-vaqti bilan texnik xizmat ko'rsatish uchun injektorni qismlarga ajratib oling va yaxshilab tozalang.

Hajmi (µL)	Naycha(tubing) hajmi	
	0.02 ID (sm) – To'q sariq rangli naycha	0.01 ID (sm) – Qora naycha
10	4.93	19.72
20	9.87	39.48
25	17.27	69.08
50	24.67	98.68
100	49.34	197.36
150 ta	74.01	296.04
200	98.68	394.72
250	103.35	413.40
500	246.69	986.76
1000	493.38	1973.52

*. Injektor halqasi hajmi

6. Nasos bilan bog'liq muammolar va ularni bartaraf etish choralari

Nasos - bu eluentni doimiy va aniq oqim tezligida yetkazib berishi kerak bo'lgan qurilma.

Agar bu funksiya to'g'ri saqlanmasa, quyidagi holatlar yuzaga kelishi mumkin. Bunday hollarda, nasosni saqlash va unga xizmat ko'rsatish uchun tegishli tuzatish choralari ko'rilishi kerak.

1) Oqish sodir bo'ladi

Nasoslash jarayonida nasos yig'ilishida oqish paydo bo'lishi mumkin.

Ko‘p hollarda bunga nasos bosh qismidagi porshen muhrlari (piston seal) va qo‘shimcha muhrlar (back-up seal) ning eskirishi sabab bo‘ladi.

Bunday holatda muhrlarni almashtirish tavsiya etiladi. Almashtirish jarayonida yangi muhrlarga zarar yetkazmaslikka e’tibor berish kerak, va odatda ikkala muhrni bir vaqtda almashtirish standart hisoblanadi.

2) Bosim qo'llanilmaganda

· CAM / Motor to'g'ri ishlayotganini tekshiring (nasosni ishga tushirish va nasosni yoqish holatini tasdiqlang).

· Ion xromatografiyasi oqim yo'lidagi kirish va chiqish klapanlarini tekshiring.

· Nasos bosh qismidagi porshen shikastlangan-yo‘qligini tekshiring.

· Nasos bosh qismidan havo chiqarilganligini (air removal) tekshiring..

* Havoni chiqarish tartibi (Air Removal Procedure)

(1) Mos keladigan eluent oqim tezligini tanlang.

(2) Transducer Knob Valve ni oching.

(3) Priming blokidagi Luer adapterga 10 cc shprits ulang.

(4) Priming klapanini oching.

(5) Nasosni ishga tushiring.

(Batafsilroq havo chiqarish tartibi uchun har bir tizimning ishlash bo‘limiga murojaat qiling.)

Ⅶ. Nosozliklarni bartaraf etish

Alomat	Tavsif	Tuzatish choralari
Oqish	Oqim yo'lidan oqish	Oqish joyini tekshiring. Fitingni mahkam torting yoki eskirgan yoki shikastlangan bo'lsa, uni almashtiring.
	Tiqilib qolgan yoki noto‘g‘ri o‘rnatilgan trubka (tubing)	Trubka bukilib qolmagan yoki tiqilib qolmaganligini tekshiring. Agar chiqindi liniyasi tizimdan yuqoriroqda o‘rnatilgan bo‘lsa, trubkani almashtiring yoki uning joylashuvini o‘zgartiring..
	Nasosni tekshirish klapani va boshqalar / Nasos bosh qismi noto‘g‘ri o‘rnatilishi	Nasos bosh qismidagi har bir detal to‘g‘ri o‘rnatilganligini tekshiring. Oqish to‘xtaguncha ehtiyotkorlik bilan mahkamlang. Haddan tashqari kuch ishlatish shikastlanishga olib kelishi mumkin.
	Nasos porshen muhrining shikastlanishi	Porshen muhrini almashtiring. Agar oqish davom etsa, porshenning o‘zini ham almashtiring.

Alomat	Tavsif	Tuzatish choralari
Oqish	Transduser oqishi	(a) Tekshirish klapani va o'tkazgich orasidagi trubka mahkam ulanganligini tekshiring. Agar u eskirgan yoki shikastlangan bo‘lsa, almashtiring. (b) Transduser tugmasi (knob) to‘g‘ri mahkamlanganligini tekshiring. Haddan tashqari mahkamlash transduser tugmasi va korpusiga zarar yetkazishi mumkin, shuning uchun ehtiyot bo‘ling. (c) O'tkazgichni tekshiring. Agar oqish sababi o'tkazgich tugmasi bo'lsa, o'tkazgich tugmachasining O-halqasini almashtiring yoki agar u shikastlangan bo'lsa, o'tkazgich tugmachasini almashtiring. Agar oqish o'tkazgichning pastki qismida sodir bo'lsa, bosim o'tkazgichining O-halqasini almashtiring.
	ERS oqishi	ERS Eluent In, Out va Regenerant In, Out portlarida biron bir tiqilib qolish bor-yo'qligini tekshiring. Agar tiqilib qolish mavjud bo'lsa, ERS tozalashni amalga oshiring. Agar tiqilib qolish bartaraf etilmasa va tozalashdan keyin ham oqish davom etsa, ERSni almashtiring. Batafsil ma'lumot uchun SRS qo'llanmasiga qarang.
Peak Aniqlash Anormal	Namuna kiritilgandan so'ng, analit yetarlicha elutsiyalanmagan va keyingi namuna yuborilgan.	Keyingi namunani yuborishdan oldin, oldingi namuna to'liq elutsiya qilinmaguncha kuting.

Alomat	Tavsif	Tuzatish choralari
Peak Reproducibility Poor → Pik takrorlanuvchanligi past	Namuna yuborishdan oldin sample loop ifloslangan	Namuna yuborishdan oldin sample loop ni deionizatsiyalangan suv yoki namuna bilan taxminan uning hajmidan 10 baravar ko‘p miqdorda yuvib, yaxshilab tozalang.
	Kolonkaning ortiqcha yuklanishi (sig‘imidan ortiq analitli namunalarni bir necha marta yuborish)	(a) Namunani suyultiring. (b) Namuna halqasini kichikroq halqa bilan almashtiring.
	Eritmaning oqishi	Kolonkada oqish bor-yo'qligini tekshiring.
	Namuna halqasi hajmidan kamroq namunani yuborish yoki kuchli ifloslangan namunalarni yuborish	(a) Yetarli miqdorda namuna yuboring (kamida sample loop hajmidan 5–10 baravar ko‘p). (b) Agar namuna to‘g‘ridan-to‘g‘ri yuborib bo‘lmasa, mos oldindan tayyorlash (pretreatment) jarayonini bajaring.

Alomat	Tavsif	Tuzatish choralari
Retensiya vaqti va selektivlikning buzilishi	Eluent almashtirilgandan keyin tizim barqarorlashmagan	Taxminan 10 daqiqa davomida deionizatsiyalangan suvni yuborgandan so‘ng, tizimni mos eluent yordamida barqarorlashtiring.
	Tizim oqim tezligi noto'g'ri	(a) Belgilangan oqim tezligining to'g'riligini tekshiring. (b) Agar haqiqiy oqim tezligi belgilangan qiymatdan farq qilsa, nasos oqimini kalibrlashni amalga oshiring.
	Eluent ifloslangan yoki noto'g'ri tayyorlangan	Yuqori tozalikdagi reagentlar yordamida eluentni tayyorlang.
	Standart ifloslangan yoki yomonlashgan	Standart tayyorlash va saqlash paytida ifloslanish yoki o'zgartirishning oldini olishga e'tibor bering. Agar ifloslanishga shubha qilinsa yoki yaroqlilik muddati o'tgan bo'lsa, standartni yangidan tayyorlang yoki sotib oling.
	Kolonka ifloslangan	(a) Kolonkani tozalang. (b) Agar tozalashdan keyin ham xuddi shu muammo davom etsa, kolonkani almashtiring.

Alomat	Tavsif	Tuzatish choralari
Detektor chiqish signali past	Yetarli darajada namuna yuborilmagan	(a) Namuna yuborish hajmini yoki konsentratsiyasini oshiring. (b) Sample loop hajmini kattaroq qilib almashtiring.
	Kalibrlash yacheykasida nosozlik	Yacheykani kalibrlashni amalga oshiring.
	Auto Offset bajarilmagan	Namuna yuborishdan oldin offset holatini albatta tekshiring.
	ERS nosozligi	(a) ERS quvvati yoqilganligini tekshiring. (b) Agar supressor ifloslangan bo'lsa, uni yuvish (regeneratsiya) jarayonini amalga oshiring.
	Namuna konsentratsiyasi juda yuqori	(a) Namunani suyultiring (b)Namuna halqasi (loop) hajmini kamaytiring.
	Noto'g'ri eluent tanlangan	To'g'ri eluent ishlatilayotganligini va konsentratsiyasini tekshiring

Alomat	Tavsif	Tuzatish choralari
Detektordan signal yo‘q	Yacheyka ulanmagan	Yacheyka kabelining ulanishini tekshiring.
	Oqim yo'q	(a) Oqim normal hosil bo‘layotganini tekshiring. (b) Nasos quvvatini tekshiring. (c) Nasos oqim tezligi “0” ga o‘rnatilmaganligini tekshiring.
Bazaviy shovqin yoki drift (siljish)	Yacheyka kalibrlashi noto‘g‘ri	Yacheykani kalibrlashni amalga oshiring.
	Yacheyka kirishidan oldingi oqim yo‘lida oqish	Barcha trubkalar, kolonkalar va suppressorni sizib chiqish bor-yo‘qligiga tekshiring. Oqish joyini mahkamlang va eskirgan yoki shikastlangan qismlarni almashtiring.
	Nasos primlanmagan	Nasosni primlash (priming) jarayonini bajaring.
	Atrof-muhit haroratining tez o‘zgarishi	Laboratoriya haroratini doimiy darajada saqlang.